# Probabilistic Machine Learning

## Lecture 02: Probability and Statistics

Prof. Dr. Michael Granitzer

Speaker notes

The lecture opens with an introductory example, Probability by Counting: one table of student exam results, read for joint, marginal and conditional probabilities, the sum and product rules and Bayes' theorem, before any definition. Nine modules then make it formal, in two strands that meet at the data. The first strand is about events. Random Experiments and Events sets up the language: sample space, outcomes, events and their algebra. The Probability Measure assigns numbers to events, starting from probability as area, and fixes the Kolmogorov axioms. Conditional Probability defines conditioning, the product rule and total probability; Bayes' Theorem inverts a conditional and resolves the base-rate trap; Independence names the case in which conditioning changes nothing. The second strand is about quantities. Random Variables turns outcomes into numbers with distributions; Joint Distributions describes several at once. Data as a Sample reads a table as draws from a joint distribution, using independence for the i.i.d. assumption, and names what else enters recorded values: noise, artefacts, outliers, missing values. Statistics and Estimation computes statistics and estimators with their standard error and confidence interval. Conditioning a parameter on data, the Beta–Bernoulli model, follows in Lecture 03, where naive Bayes needs it.

Combinatorics and the axiomatic construction of probability (σ-algebras, measurability) are left out: the course needs the calculus of probability — sum rule, product rule, Bayes' theorem — rather than its measure-theoretic foundation.

# Where We Left Off

Lecture 01 ended on an ideal — model the joint, condition on the data, read off answers — and on the shortcuts most methods take. Today builds the language all of them are written in:

**Model the joint**
Random variables and their joint distribution

→

**Condition on the data**
For events (Conditional Probability) — for a parameter in L03

→

**Read off answers**
An estimate and its standard error (Statistics and Estimation)

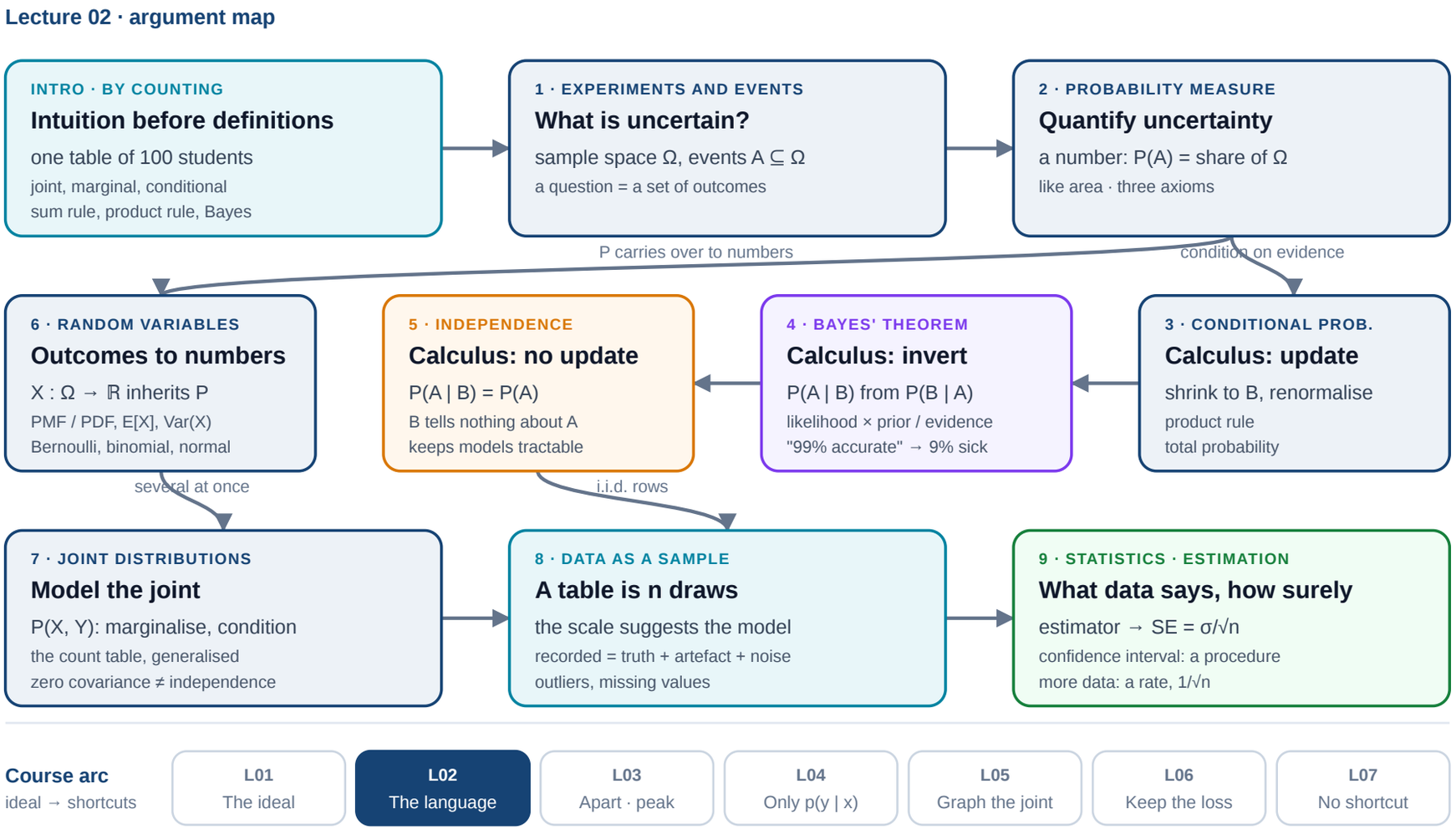
> It also left a claim unpaid: that **epistemic** uncertainty is the kind more data removes. Statistics and Estimation turns that slogan into a rate.

Speaker notes

The modules follow the order the mathematics requires, not the order in which the three moves are performed: conditioning is defined for events (Conditional Probability and Bayes) before random variables and joint distributions exist (Random Variables, Joint Distributions), and only then can a joint over data and parameters be conditioned (Lecture 03).

The estimation module gives the rate: the standard error of an estimate from $n$ i.i.d. observations shrinks like $\sigma/\sqrt{n}$, so quadrupling the data halves the uncertainty about the estimated quantity. The same rate reappears for the posterior over a parameter in Lecture 03, whose standard deviation shrinks roughly like $1/\sqrt{n}$; keeping only that posterior's peak is the first shortcut (Lecture 03), and the full treatment for many parameters returns in Lecture 07.

# Argument Map



Speaker notes

One box per module, numbered in lecture order; the path snakes through the rows. The introduction computes everything on one count table; modules 1–2 make the table's ingredients formal (what is uncertain, how likely). From the measure two strands leave. The event strand: conditioning turns evidence into a new probability measure; Bayes' theorem inverts a conditional and resolves the base-rate trap, i.e. the confusion of $P(\text{evidence} \mid \text{hypothesis})$ with $P(\text{hypothesis} \mid \text{evidence})$; independence is the special case in which evidence changes nothing. The quantity strand: random variables carry the measure over to numbers, and a joint distribution describes several of them at once — the object "model the joint" in Lecture 01 refers to, and the count table of the introduction in general form. The strands meet at the data: a table is read as $n$ draws from a joint, and independence supplies the i.i.d. assumption; recorded values additionally carry noise, artefacts and outliers. Statistics and estimation then quantifies what a sample says about the population and how precisely; Bayes' theorem returns for a parameter in Lecture 03. There is no arrow from independence to random variables: random variables need only the measure.

# Probability by Counting

> ***Guiding Question:*** *A hundred students, two courses, one exam each. How far do we get with nothing but counting — before any definition?*

Speaker notes

An introductory example rather than a formal module: every quantity is a count divided by a count. Joint, marginal and conditional probability, the sum and product rules and Bayes' theorem all appear here informally; the following modules define each of them and remove the restriction to finite tables.

# Example: Students, Courses and Exams

**Probabilistic ML** — 40 students
a quarter of them fail the exam

**Databases** — 60 students
one in ten fails the exam

Each of 100 students sat exactly one of the two exams.

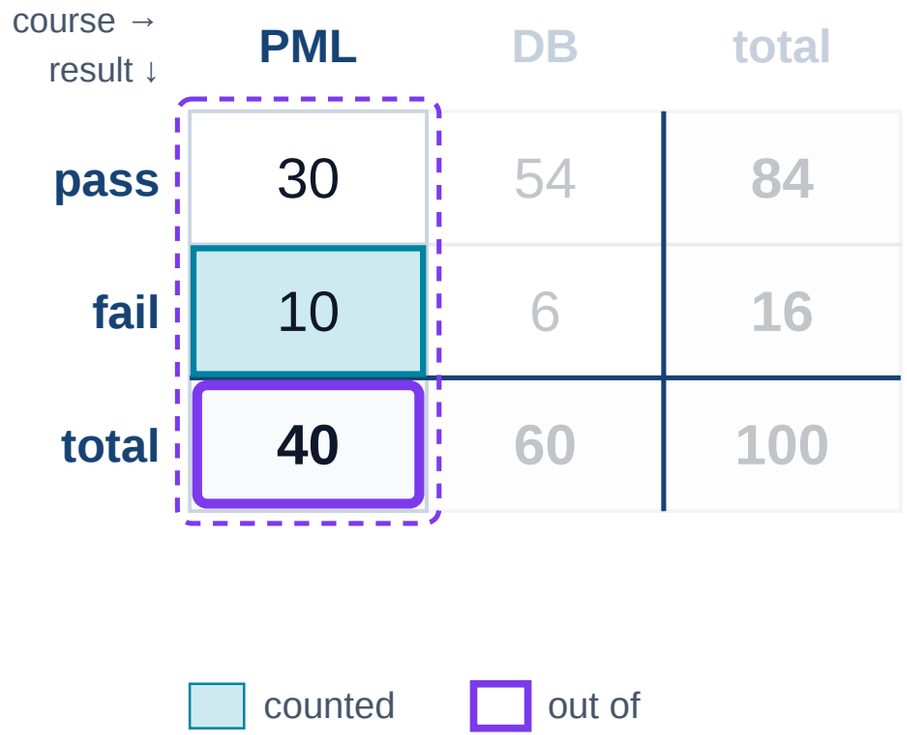
- **Forward:** how many students fail overall?
- **Backward:** a student failed — how probable is it that they took PML?

Speaker notes

The example is Bishop's two-box experiment (*Pattern Recognition and Machine Learning*, §1.2) with a course in place of a box and an exam result in place of a fruit. The experiment has two stages: a course $C \in \{\mathrm{PML}, \mathrm{DB}\}$ with $P(C = \mathrm{PML}) = 0.4$, then a result $R \in \{\mathrm{pass}, \mathrm{fail}\}$ with $P(R = \mathrm{fail} \mid \mathrm{PML}) = 0.25$ and $P(R = \mathrm{fail} \mid \mathrm{DB}) = 0.1$. The forward question asks for a marginal, $P(\mathrm{fail}) = 0.4 \cdot 0.25 + 0.6 \cdot 0.1 = 0.16$; it runs in the direction of the experiment. The backward question asks for $P(\mathrm{PML} \mid \mathrm{fail})$; it runs from the observed result back to its cause and needs Bayes' theorem. Both are answerable by counting alone.

# Joint, Marginal, Conditional — One Table

**? Of the PML students, what share failed?**




**P(fail | PML)**

= **10** / **40** = 0.25

*conditional: a cell, out of its own column*

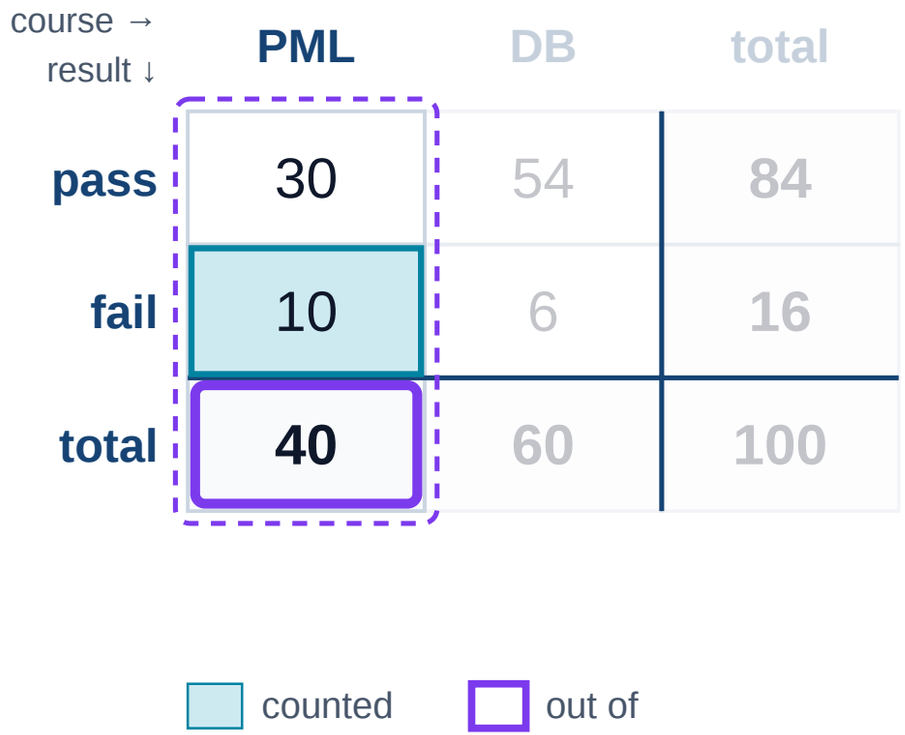
Speaker notes

Each question is shown first; the room answers before the next click marks the counted cells (teal) and the count they are taken out of (violet). Joint: $P(\mathrm{PML}, \mathrm{fail}) = 10/100$, one cell over the grand total. Marginal: $P(\mathrm{fail}) = 16/100$, a row total over the grand total. Conditional: $P(\mathrm{fail} \mid \mathrm{PML}) = 10/40$, a cell over its own column total; the DB column is not taken into account at all.

The joint probabilities sum to 1 over the four cells; the conditionals sum to 1 only within their own column, $P(\mathrm{pass} \mid \mathrm{PML}) + P(\mathrm{fail} \mid \mathrm{PML}) = 30/40 + 10/40 = 1$. The formal definitions follow later; on a finite table they reduce to exactly these divisions.

# Sum Rule and Product Rule — On the Table

**? 40% took PML, and a quarter of them failed. What share failed PML?**




… of those, failed: P(fail | PML) = **10** / **40**

**Product rule**

P(PML, fail) = P(PML) · P(fail | PML)

= 40/100 · 10/40 = 10/100

*a share of a share — the 40s cancel*

Speaker notes

Sum rule: the row total is the sum of the row's cells, so $P(\text{fail}) = P(\text{PML}, \text{fail}) + P(\text{DB}, \text{fail}) = 0.10 + 0.06 = 0.16$; summing over the course removes it. Product rule: a cell is its column's share of the table times the cell's share of that column, $P(\text{PML}, \text{fail}) = P(\text{PML})\,P(\text{fail} \mid \text{PML}) = \frac{40}{100} \cdot \frac{10}{40}$. The column total appears once as a numerator and once as a denominator and cancels.

# The Two Rules

**Sum rule**

$$P(X) = \sum_{Y} P(X, Y)$$

Add along a row or column to remove a variable.

**Product rule**

$$P(X, Y) = P(Y \mid X) P(X)$$

A cell is its column total times its share of that column.

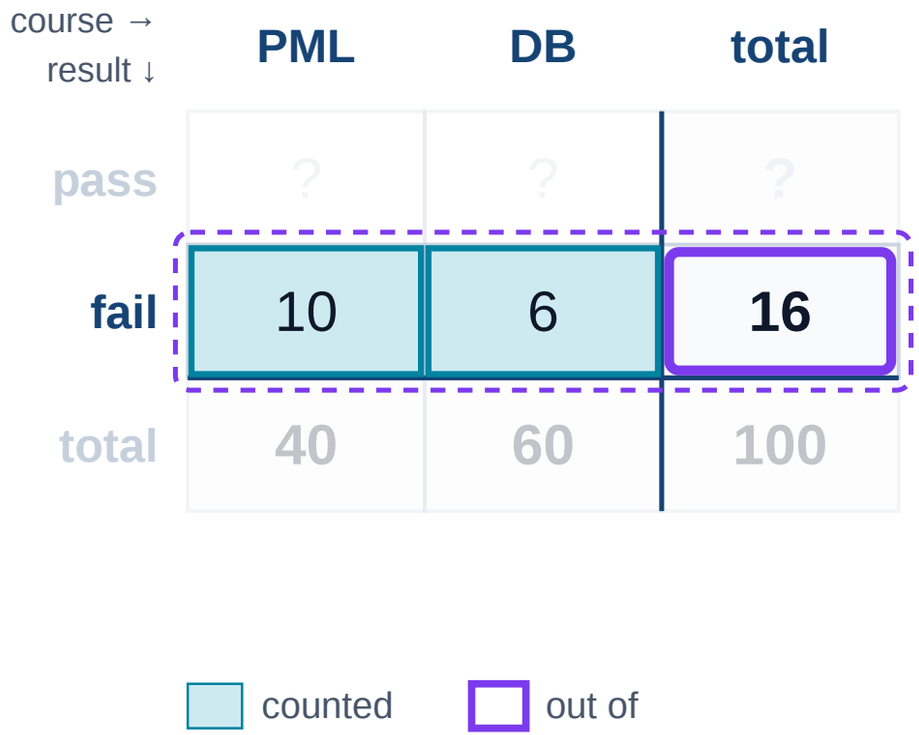
Almost all of probability theory is these two rules applied repeatedly.

Speaker notes

$X$ and $Y$ stand for the course and the result. $P(X)$ is shorthand for the whole distribution of $X$, so each rule is one equation per value. For continuous variables the sum becomes an integral, $p(x) = \int p(x, y)\, dy$. The product rule holds in either order, $P(X, Y) = P(Y \mid X)P(X) = P(X \mid Y)P(Y)$; equating the two forms gives Bayes' theorem. The formal treatment justifies both rules from the axioms and the definition of conditional probability and extends them beyond finite tables.

# Bayes' Theorem — On the Table

**? And which course did they most probably take?**




P(PML | fail) = **10** / **16** ≈ 0.63

P(DB | fail) = **6** / **16** ≈ 0.37

PML — odds 10 : 6

*same normaliser: comparing needs only the cells*

Speaker notes

Steps 1–3 answer the question by counting: the evidence "failed" removes the pass row, 16 students remain, 10 of them from PML, so $P(\mathrm{PML} \mid \mathrm{fail}) = 10/16 = 0.625$, up from the prior $P(\mathrm{PML}) = 0.4$ because PML students fail more often (a quarter against a tenth).

Steps 4–8 rebuild the same answer from what is usually known instead of a table: the course sizes (the prior) and the fail rate per course (the likelihoods). The cell counts are hidden and appear only once they can be inferred. Prior times likelihood gives the cell, $0.4 \cdot 0.25 = 0.10$ (product rule); summing that over both courses gives the normaliser $P(\mathrm{fail}) = 0.10 + 0.06 = 0.16$ (sum rule); their ratio is the posterior. The pass row is never needed. Step 9 asks the comparative question, which course is more probable: both posteriors share the normaliser, so the comparison needs only the cells, $10 : 6$ in favour of PML.

# Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

Speaker notes

With equal fail rates, $P(\text{fail} \mid \text{PML}) = P(\text{fail} \mid \text{DB}) = 0.25$, the likelihood cancels and the posterior equals the prior: $0.10/(0.10 + 0.15) = 0.4$. Course and result are then independent. In general, evidence changes a belief only to the extent that the hypotheses assign it different probabilities; the ratio $P(\text{fail} \mid \text{PML})/P(\text{fail} \mid \text{DB})$ (2.5 originally, 1 in the variant) measures how much.

# Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

And if one in four DB students failed too?

# Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_Y P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

And if one in four DB students failed too?

Then $P(\text{PML} \mid \text{fail}) = 0.4$: the result says nothing about the course.

# Takeaway

> Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. What follows makes them precise and extends them beyond counting.

Speaker notes

The terms used informally here — joint, marginal, conditional probability, Bayes' theorem — are all derived later from three axioms plus the definition $P(A \mid B) = P(A \cap B)/P(B)$. The sum rule is additivity (the third axiom) applied to the disjoint events $\{Y = y\}$; the product rule is that definition rearranged; combined, they give the theorem of total probability. The formal treatment does not contradict the counting picture; it removes the restriction to finite tables of counts.

# Random Experiments and Events

> ***Guiding Question:*** *Before any number: what exactly is uncertain, and which questions about it can be asked at all?*

# What Is a Random Experiment?

A coin flip, tomorrow's rainfall, whether a user clicks a link. What makes each of them a *random experiment* — and what would not be one?

Speaker notes

The definition does not require the world to be indeterministic; it requires only that the outcome cannot be predicted from what is specified. Repeatability "in theory" means the configuration and procedure can be re-created, not that the experiment is actually repeated. Many random experiments in data analysis are not designed by anyone — daily rainfall, whether a user clicks a link, the length of a queue at noon. There the configuration is whatever system produced the data, and treating it as repeatable is a modelling assumption, typically that the observations come from the same unchanged process.

# What Is a Random Experiment?

A coin flip, tomorrow's rainfall, whether a user clicks a link. What makes each of them a *random experiment* — and what would not be one?

> **Definition (Random experiment).** A procedure that can — at least in theory — be repeated infinitely often, characterised by:
>
> 1. **Configuration** — a precisely specified system that can be reconstructed.
> 2. **Procedure** — an instruction for executing the experiment.
> 3. **Unpredictability** of the outcome.

**Coin flip:** a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

# Randomness as Missing Information

> A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.
>
> Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

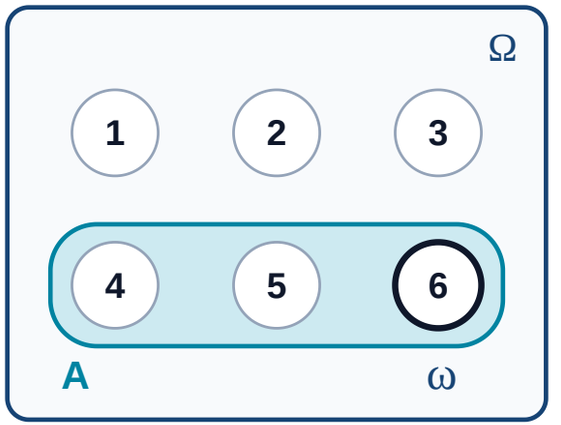
Speaker notes

Calling something random is a statement about the observer's knowledge, not only about the system. A coin flip is governed by classical mechanics; given initial velocity, spin and height, the outcome is determined, and Diaconis, Holmes and Montgomery (2007) built a coin-tossing machine that produces the same face every time. The flip is random only because these quantities are not measured.

For a model the implication is that its "noise" term collects everything it does not represent: genuine indeterminacy, measurement error, and unmodelled signal. These cannot be told apart from the residuals alone; adding a relevant input variable can turn part of what looked like noise into explained variation.

# Sample Space and Events

**Definition (Sample space, event space).** A set $\Omega = \{\omega_1, \ldots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$, an **outcome**.
For finite $\Omega$, every subset $A \subseteq \Omega$ is an **event**; $A$ **occurs** iff the observed $\omega \in A$. All events together, $\mathcal{P}(\Omega)$, form the **event space**.




| | |
|---|---|
| **Ω** | sample space: all outcomes |
| **ω = 6** | an outcome: an element |
| **A = {4, 5, 6}** | an event: a subset |

ω ∈ A: the event "at least 4" occurs

An outcome is an element; an event is a subset — a *question you can ask* about the result.

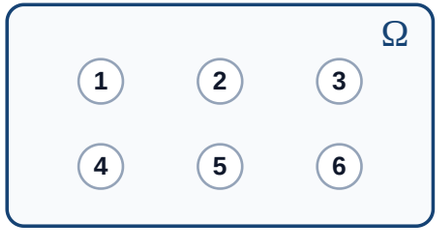
Speaker notes

"Exactly one" means the outcomes are mutually exclusive (no result maps to two elements) and exhaustive (every possible result maps to some element). Which results are distinguished is a modelling decision: for a die, $\{1, \ldots, 6\}$ and $\{\mathrm{even}, \mathrm{odd}\}$ are both valid sample spaces, but the second cannot express the event "a six".

An outcome $\omega$ is an element of $\Omega$; an event is a subset. The elementary event $\{\omega\}$ is a set containing one outcome and is not the same object as $\omega$. Probabilities are assigned to events, so one writes $P(\{\omega\})$, often abbreviated $P(\omega)$.

The figure is the die panel used for all event types that follow: the roll showed 6, and since $6 \in \{4, 5, 6\}$ the event "at least 4" occurred. For infinite $\Omega$ — the drug dosage below — the event space is a suitable family of subsets rather than the full power set; this returns with continuous distributions.

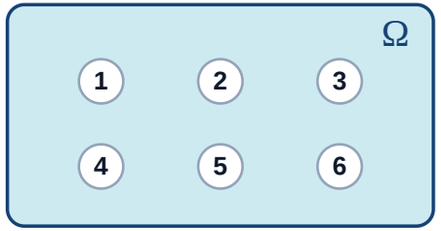
# Important Event Types




$\varnothing$

impossible event

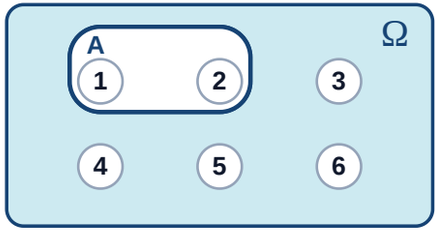
no outcome belongs to it




$\Omega$

certain event

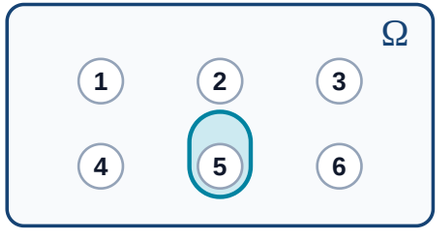
every outcome belongs to it




$\bar{A} = \Omega \setminus A$

complementary event

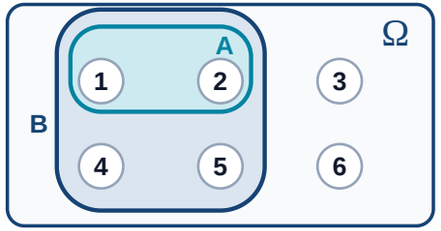
A = {1, 2}, $\bar{A}$ = {3, 4, 5, 6}




$|A| = 1$

elementary event

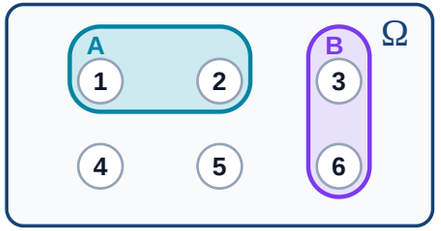
A = {5}: one outcome




$A \subseteq B$

A entails B

A = {1, 2}, B = {1, 2, 4, 5}




$A \cap B = \varnothing$

incompatible events

A = {1, 2}, B = {3, 6}

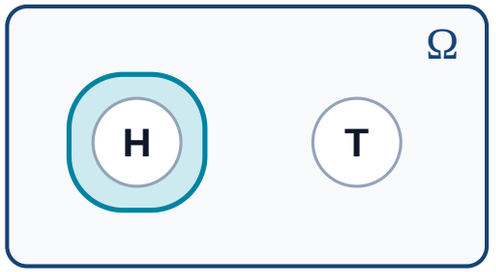
Speaker notes

All six panels use the die, $\Omega = \{1, \ldots, 6\}$. The impossible event $\emptyset$ contains no outcome and never occurs; the certain event $\Omega$ always occurs. The complement $\bar{A} = \Omega \setminus A$ occurs exactly when $A$ does not. An elementary event contains one outcome. $A \subseteq B$ means $A$ entails $B$: whenever $A$ occurs, $B$ occurs too. Incompatible (disjoint) events, $A \cap B = \emptyset$, cannot occur together.

Set operations correspond to logical connectives between the statements the events express: inclusion is implication, intersection is "and", union is "or", complement is "not". This correspondence is why identities between events can be read off Venn diagrams.

# Examples

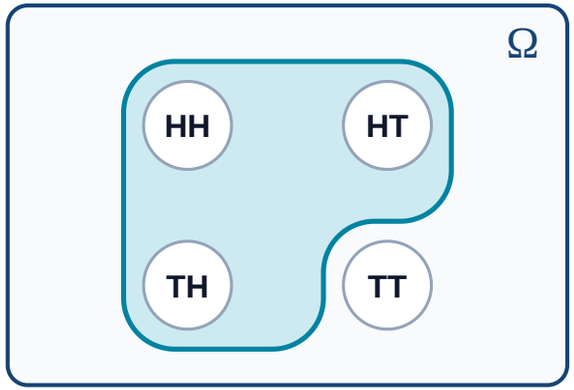
**One coin flip**




E = {H}: "heads"

events: ∅, {H}, {T}, {H, T}

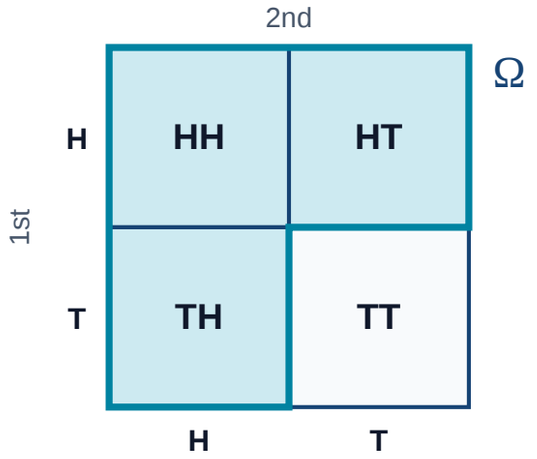
**Two coin flips: the set**




E = {HH, HT, TH}: "at least one head"

4 outcomes, $2^4 = 16$ events

**… as a grid: 1st × 2nd**



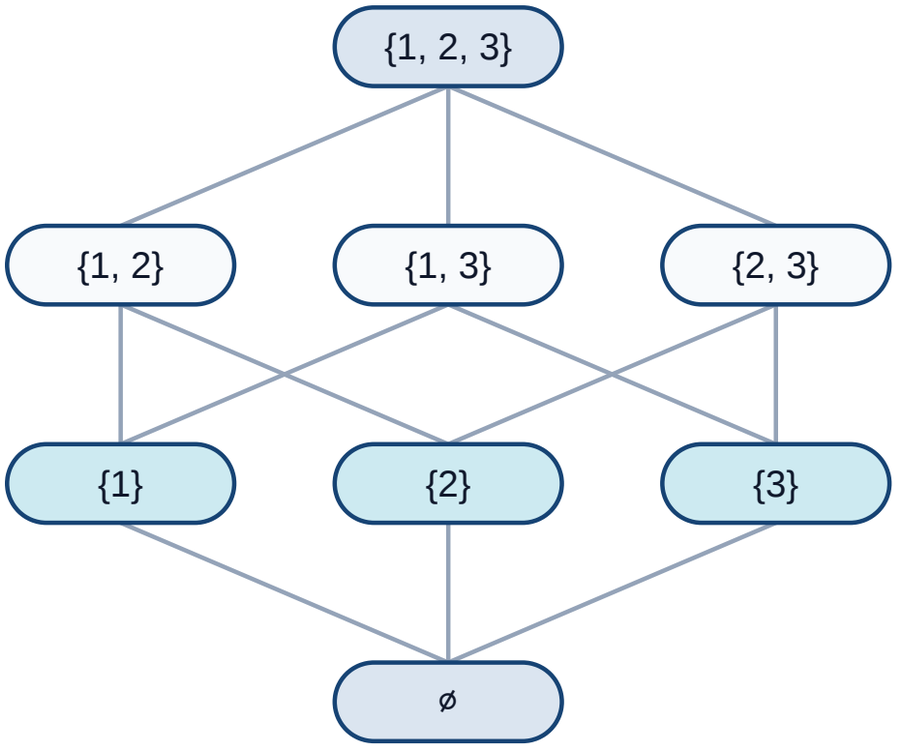

same outcomes, arranged by flip

The same event, twice: as a subset of Ω, and as cells of the 1st × 2nd grid. The sample space fixes which questions can be asked.

Speaker notes

For two flips, $\Omega = \{HH, HT, TH, TT\}$. "At least one head" is $\{HH, HT, TH\}$, "exactly one head" is $\{HT, TH\}$, and "the first flip is heads" is $\{HH, HT\}$: same sample space, different questions. If the sample space recorded only the number of heads, $\{0, 1, 2\}$, the event "the first flip is heads" could not be expressed at all, and the three outcomes would not be equally likely ($\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$).

# The Event Space Grows Fast




**$|\Omega| = n \rightarrow 2^n$ events**

| | |
|---|---|
| 1 | 2 |
| 2 | 4 |
| 3 | 8 |
| 6 | 64 |
| 10 | 1,024 |
| 20 | 1,048,576 |

lines: one element added

For $|\Omega| = n$ the event space has $2^n$ elements — far too many to list, which is why probability is done by counting, not by enumeration.

Speaker notes

Each event is determined by an independent in/out choice for each of the $n$ outcomes, so there are $2^n$ events. For three outcomes these are $\emptyset$, three singletons, three pairs and $\Omega$, ordered by inclusion in the figure. Already a modest $n = 300$ gives more events than the estimated $10^{80}$ atoms in the observable universe.

A probability measure on a finite $\Omega$ is nevertheless fixed by only $n$ numbers, the probabilities of the elementary events; additivity gives every other event. For infinite $\Omega$, in particular $\Omega = \mathbb{R}$, the full power set contains sets to which no consistent "length" can be assigned (Vitali sets), which is why a smaller family of events, a $\sigma$-algebra, is used there.

# Not Every Sample Space Is Finite

**Dosage of a drug: Ω = (0, ∞)**




uncountably many outcomes — probability can only live on intervals, not on points

**Race with 7 teams: all orderings**




7! = 5,040 outcomes; shaded: the event "team 3 wins"

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

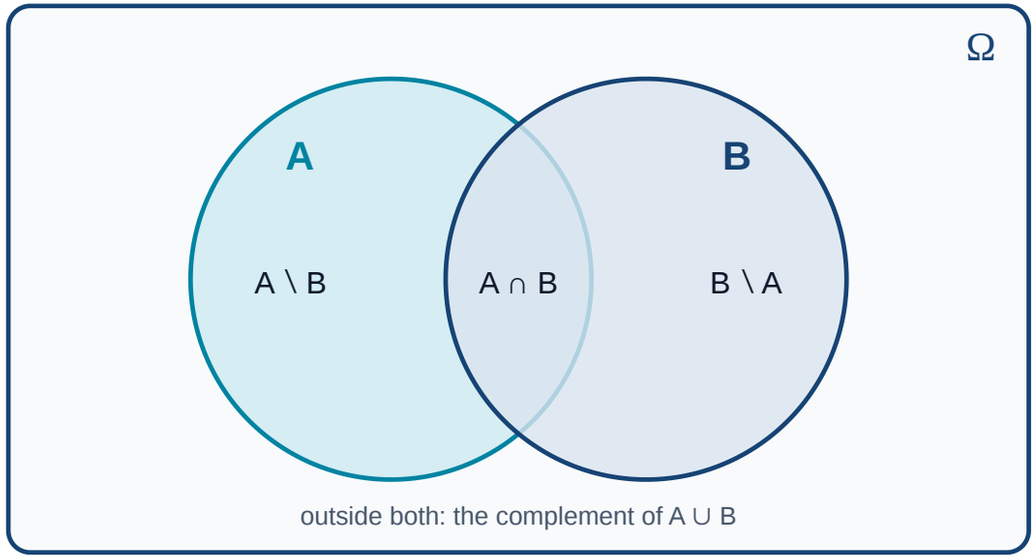
Speaker notes

A drug dosage modelled as a positive real number gives an uncountable sample space $\Omega = [0, \infty)$; events are intervals such as $[5, 10]$ mg, and any single exact value has probability zero under a continuous model. The model ignores that real dosages are measured with finite precision and administered in fixed steps; the idealisation is chosen because intervals and densities are easier to work with than a fine discrete grid.

The race example is finite but large: seven teams can finish in $7! = 5{,}040$ orders, so the sample space of full rankings has 5,040 outcomes and $2^{5040}$ events. The same modelling choice — what counts as an outcome — arises for every attribute of a dataset.

# Venn Diagrams

**Venn diagrams** are the graphical representation of events:




- the **rectangle** is the sample space $\Omega$
- a **circle** is an event $A \subseteq \Omega$
- every **region** is a set — and every set operation is a region you can point at

Speaker notes

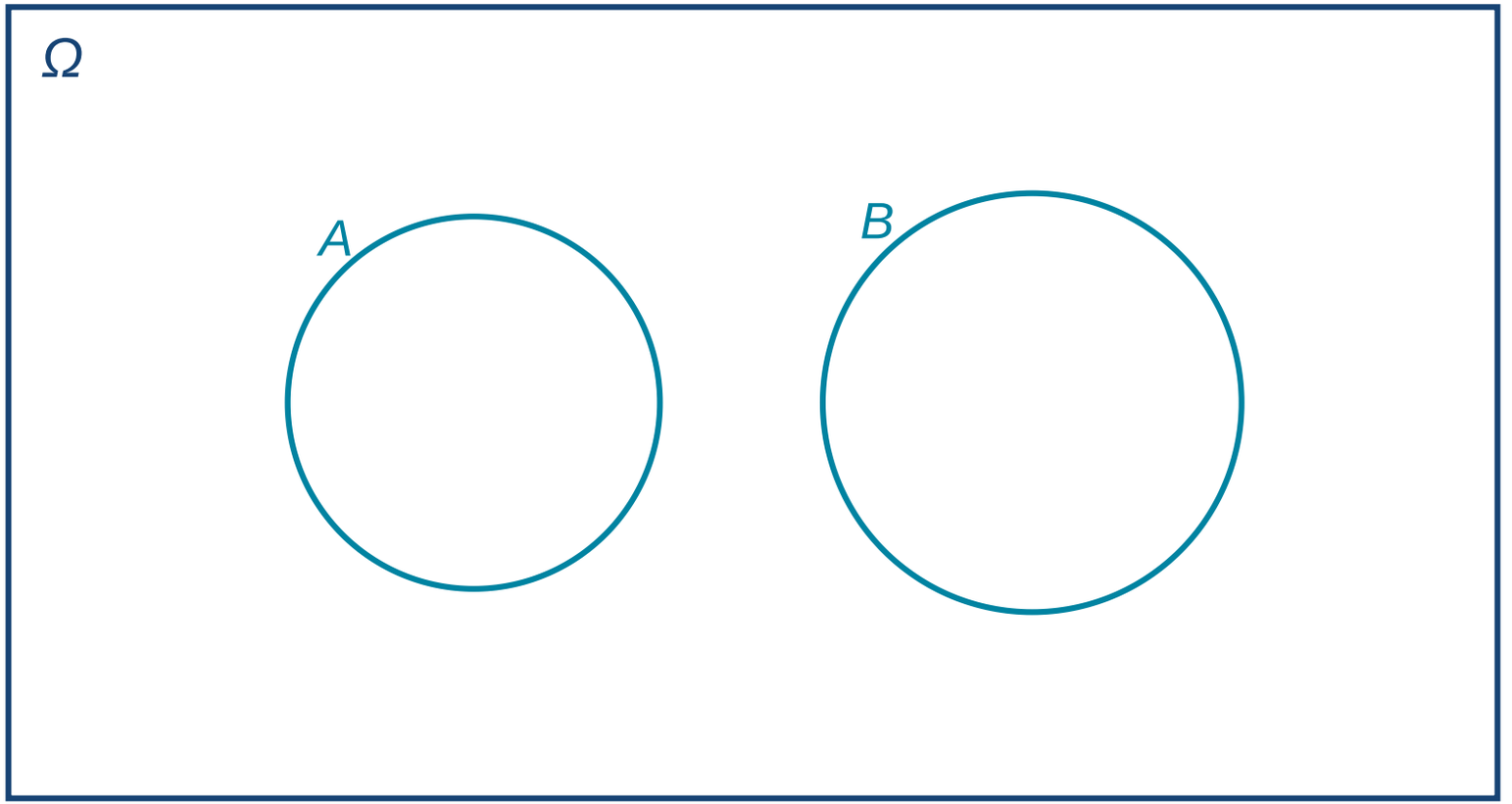
Two circles divide the rectangle into four regions: $A \setminus B$, $A \cap B$, $B \setminus A$ and $\overline{A \cup B}$. These four are pairwise disjoint and together make up $\Omega$, so every event built from $A$ and $B$ by set operations is a union of some of them; there are $2^4 = 16$ such events. Three circles give eight regions.

The sizes of the circles carry no meaning in a plain Venn diagram. Once probabilities are attached and areas are drawn proportional to them, laws such as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ become statements about areas.

# Important Event Types

**Before each step: which region will light up?**

$$A \cap B = \emptyset$$




A and B are disjoint

Speaker notes

The four steps highlight: the event $A$; its complement $\bar{A} = \Omega \setminus A$, everything outside the circle; a sub-event $B \subseteq A$, drawn as a circle inside $A$; and two disjoint events, $A \cap B = \emptyset$, drawn as non-overlapping circles.

Disjointness is a property of the sets themselves, fixed before any probability is assigned. It is often confused with independence, which is a property of the probabilities; two disjoint events with positive probability are never independent, since $P(A \cap B) = 0 < P(A)P(B)$.

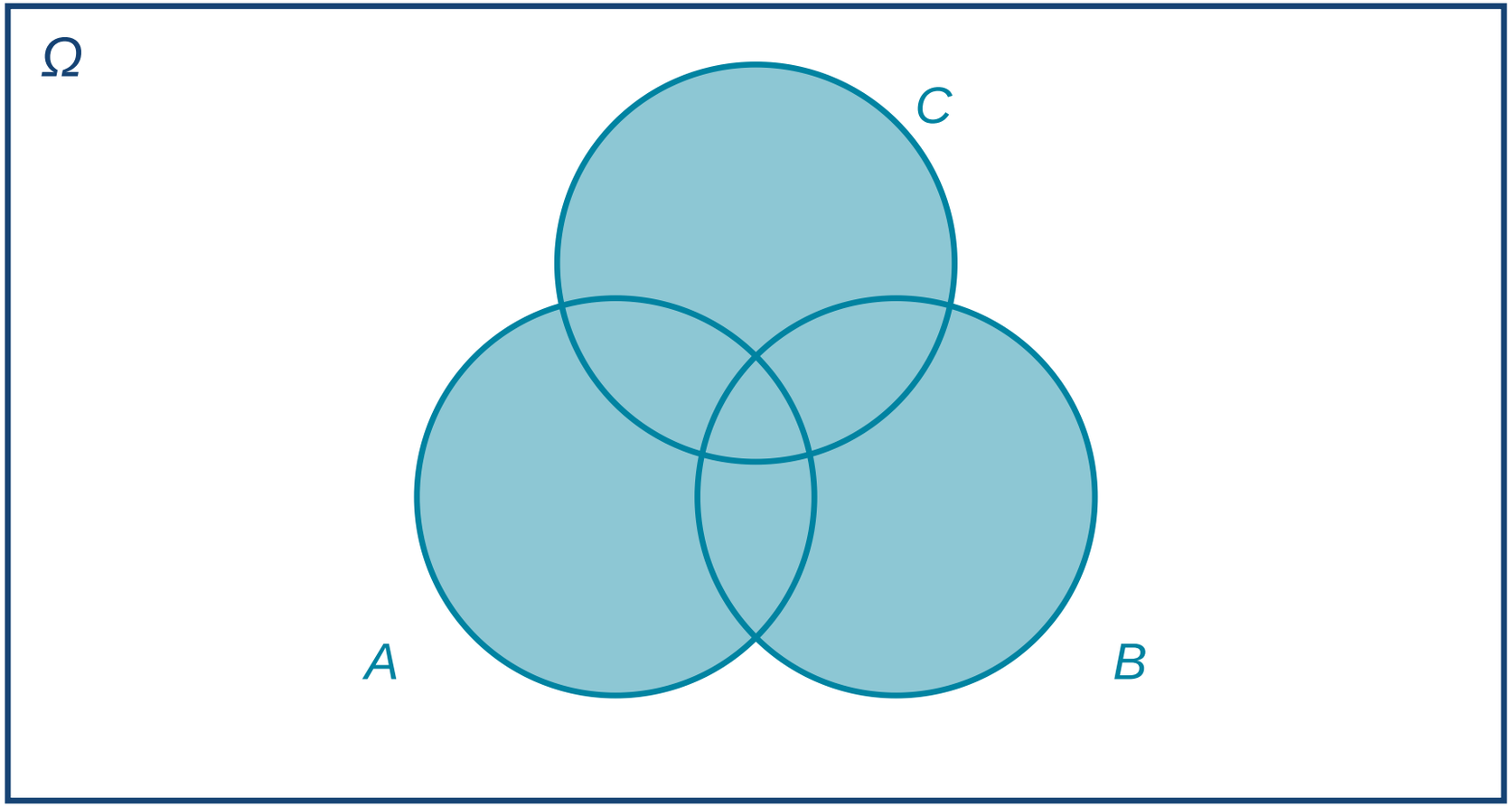
# Algebra of Events

| Law | Union form | Intersection form |
|---|---|---|
| Commutative | $A \cup B = B \cup A$ | $A \cap B = B \cap A$ |
| Associative | $(A \cup B) \cup C = A \cup (B \cup C)$ | $(A \cap B) \cap C = A \cap (B \cap C)$ |
| Distributive | $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ | $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$ |

Events form a set algebra — the same laws you know, but now every identity is a picture.

# Set Operations as Regions

**A ∪ B ∪ C**




Union of three events

Speaker notes

The four steps highlight, with three sets: $A \cup B$ (everything in either circle), $A \cap B$ (the lens shared by $A$ and $B$, including the part also in $C$), $A \cap B \cap C$ (the single central region), and $A \cup B \cup C$ (all seven regions inside the circles). Union enlarges and intersection shrinks: $A \cap B \cap C \subseteq A \cap B \subseteq A \cup B \subseteq A \cup B \cup C$.

De Morgan's laws complete the algebra: $\overline{A \cup B} = \bar{A} \cap \bar{B}$ and $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

# Mini-Exercise

Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

Speaker notes

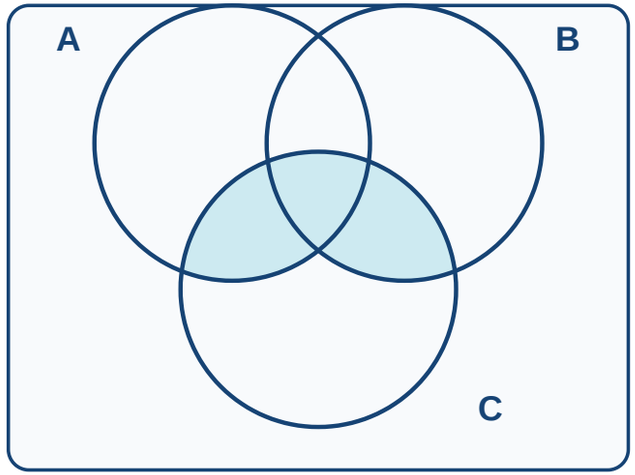
The outer operation decides. $(A \cup B) \cap C$ is an intersection with $C$, so it lies inside $C$. $(A \cap B) \cup C$ is a union with $C$, so it contains $C$. Expanded with the distributive laws: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$.

Since one is a subset of $C$ and the other a superset, they are equal exactly when both equal $C$, i.e. when $A \cap B \subseteq C \subseteq A \cup B$. A frequent error is to move the brackets freely; the associative law allows this only when both operations are the same.

# Mini-Exercise

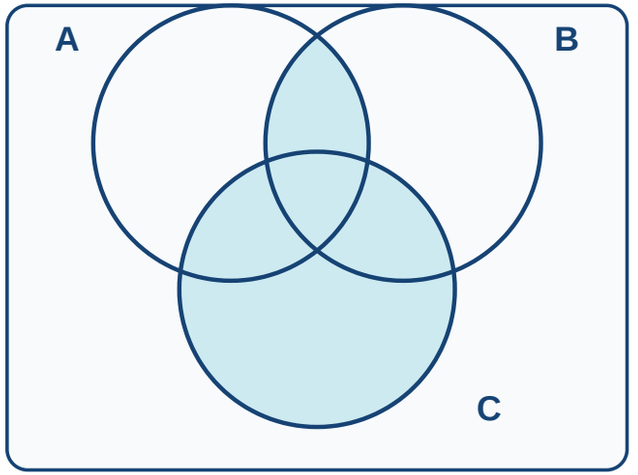
Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

**(A ∪ B) ∩ C**




inside C only

**(A ∩ B) ∪ C**




all of C, plus the lens A ∩ B

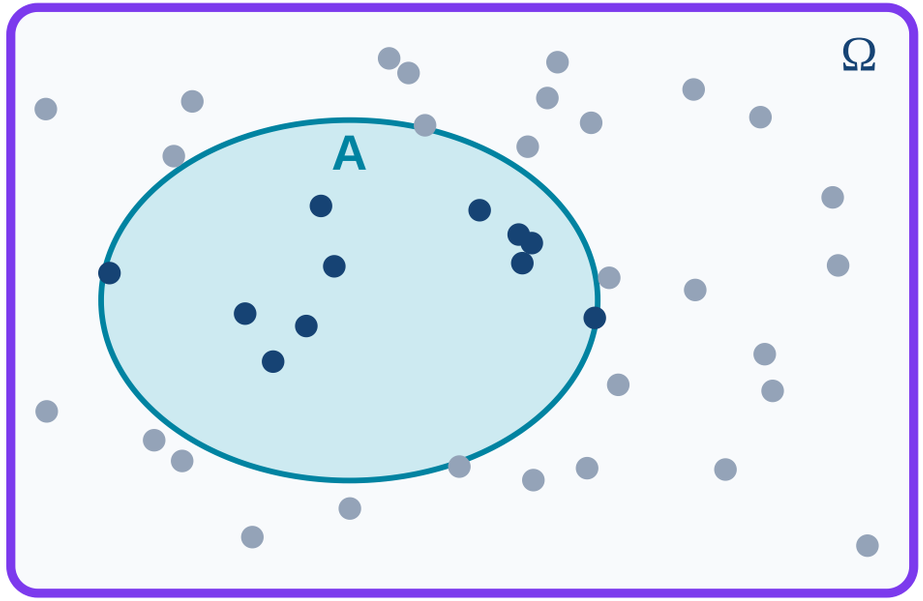
**No.** The first lies entirely inside $C$; the second is all of $C$ plus the lens $A \cap B$. The distributive laws expand them into different expressions.

# The Probability Measure

> ***Guiding Question:*** *We need a language that can say "how strongly", not just "true or false". What must such a language satisfy — and who decides the numbers?*

# An Intuition: Probability as Area

**Throw 40 stones at random into Ω**




P(A) = area of A / area of Ω ≈ 11 of 40 stones

counted out of

Draw the sample space $\Omega$ on the ground, with each event as a region. Throw a stone at random.

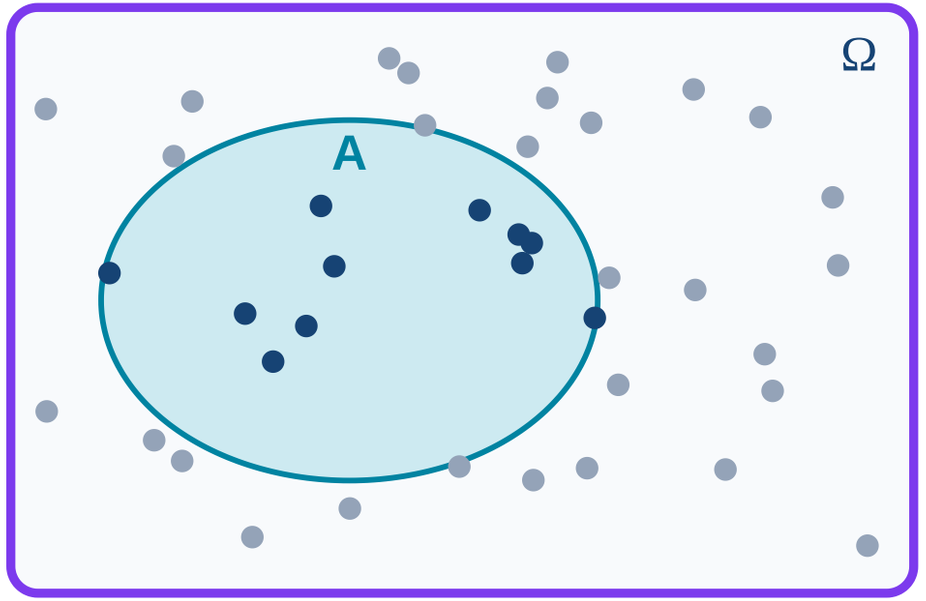
How likely does it land in $A$?

Speaker notes

In the figure $A$ covers about 27% of $\Omega$, and 11 of 40 stones (27.5%) land in it; with more stones the share converges to the area ratio, which is the frequentist reading on the next slide. Under the two assumptions, $P(A)$ is the area of $A$ divided by the area of $\Omega$. Assumption 1 corresponds to normalisation, $P(\Omega) = 1$. Assumption 2 makes all elementary events equally likely, which is the Laplace setting. Dropping assumption 2 means that some parts of $\Omega$ carry more weight than their area: the stone is thrown with a non-uniform aim. Area is then replaced by weighted area, and the axioms keep exactly the properties that any such weighting shares with plain area — non-negativity, total one, and additivity over disjoint regions.

# An Intuition: Probability as Area

**Throw 40 stones at random into Ω**




P(A) = area of A / area of Ω ≈ 11 of 40 stones

counted out of

Draw the sample space $\Omega$ on the ground, with each event as a region. Throw a stone at random.

How likely does it land in $A$?

$$P(A) = \frac{\text{area of } A}{\text{area of } \Omega}$$

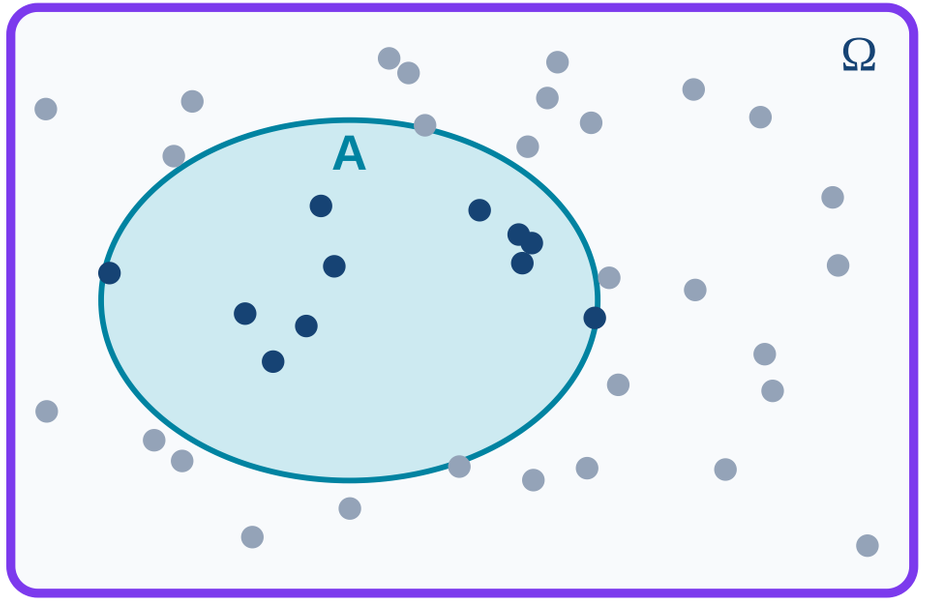
provided that

1. the stone always lands in $\Omega$, and
2. every part of $\Omega$ is equally likely.

Probability measures **how much of** $\Omega$ an event occupies.

# An Intuition: Probability as Area

**Throw 40 stones at random into Ω**




P(A) = area of A / area of Ω ≈ 11 of 40 stones

counted out of

Draw the sample space $\Omega$ on the ground, with each event as a region. Throw a stone at random.

How likely does it land in $A$?

$$P(A) = \frac{\text{area of } A}{\text{area of } \Omega}$$

provided that

1. the stone always lands in $\Omega$, and
2. every part of $\Omega$ is equally likely.

Probability measures **how much of** $\Omega$ an event occupies.

> Assumption 2 is the one we will have to give up.

# Two Ways to Read a Probability

**Frequentist**

The probability of an elementary event $a$ is the fraction of repetitions of the experiment that yield $a$.

*Requires* the experiment to be repeatable.

**Bayesian**

The probability of $a$ is the degree of certainty that $a$ will happen — even without conducting the experiment.

*Requires* a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

Speaker notes

"The probability that this machine part fails within a year" has a natural frequentist reading: the fraction of parts of the same type that fail within a year. "The probability that this patient has the disease" refers to one person in one state; it has no repetition, so it is read as a degree of belief, or frequentistically only by placing the patient in a reference class (all patients with these symptoms), whose choice changes the number.

The frequentist reading assigns probabilities to repeatable events, not to fixed but unknown quantities such as a model parameter. The Bayesian reading assigns probabilities to both, which is what makes a prior distribution over a parameter meaningful. Both readings obey the same axioms, so every theorem holds under either.

# The Classical (Laplace) Definition

**Definition (Classical / Laplace probability).** If every elementary event in $\Omega$ is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment**; its probabilities need not be estimated, they are implied by the setup.

Speaker notes

Counting (combinatorics) exists to evaluate $|A|$ and $|\Omega|$ when they are too large to list. The formula is valid only if the elementary events really are equally likely, and that depends on the choice of $\Omega$. For the sum of two dice, $\Omega = \{2, \ldots, 12\}$ has 11 outcomes but they are not equiprobable; $P(\text{sum} = 7) = 6/36$, not $1/11$. The Laplace formula applies to the 36 ordered pairs, not to the 11 sums.

Where no symmetry of the setup justifies equal probabilities — a loaded die, a click-through rate — the probabilities have to be estimated from data or specified by a model.

# Why That Definition Is Not Enough

> **The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

**So what do we do?** Stop trying to *define* what probability is, and instead specify what it must *do*.

Speaker notes

"Equally likely" already presupposes a notion of probability, so the Laplace formula cannot serve as its definition. The frequentist definition $P(A) = \lim_{n\to\infty} n_A/n$ has two problems: no experiment is repeated infinitely often, and the limit is not guaranteed to exist for an arbitrary sequence of results. The law of large numbers states that relative frequencies converge, but only with probability one — it presupposes probability rather than defining it.

The axiomatic approach, due to Kolmogorov (1933), avoids both problems: it specifies the rules every probability assignment must obey and leaves the origin of the numbers — symmetry, frequencies, or belief — to the application.

# The Kolmogorov Axioms

**Definition (Probability measure, Kolmogorov 1933).** Let $\Omega$ be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \to \mathbb{R}$ is a **probability measure** if

| | | |
|---|---|---|
| **I** | $P(A) \geq 0$ | non-negative |
| **II** | $P(\Omega) = 1$ | normalised |
| **III** | $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$ | additive |

**INTUITION**

Probability is **one kilogram of sand** spread over $\Omega$; $P(A)$ is the sand lying in $A$.

- **I** — no region holds negative sand.
- **II** — all the sand lies somewhere in $\Omega$.
- **III** — two regions that do not overlap hold their sand together.

The axioms say how *any* assignment must behave — not which probabilities the events get. How the sand is spread is the modeller's choice.

*For infinite $\Omega$ the event space is a $\sigma$-algebra and Axiom III holds for countably many disjoint events — measure theory; nothing in this course depends on it.*

Speaker notes

The sand picture is the area picture of the first slide with the uniformity assumption dropped: area is replaced by mass, and the mass may pile up anywhere. All three axioms hold for any such distribution of mass — it is never negative, the total is fixed at one, and the mass in two non-overlapping regions is the sum of their masses. What the axioms deliberately leave open is the spreading itself; symmetry (Laplace), frequencies and degrees of belief are three ways to choose it.

Upper bounds such as $P(A) \leq 1$ are not axioms; they follow from them (since $P(A) + P(\bar{A}) = 1$ and $P(\bar{A}) \geq 0$). On a finite $\Omega$, a probability measure is completely specified by the values on elementary events, $p_\omega \geq 0$ with $\sum_\omega p_\omega = 1$; additivity then gives $P(A) = \sum_{\omega \in A} p_\omega$. Countable additivity — Axiom III for countably many disjoint events — is needed, for example, for the geometric distribution, whose infinitely many probabilities must sum to 1.

# Probability Space

**Definition.** Let $\Omega$ be a sample space, $\mathcal{P}(\Omega)$ an event space, and $P$ a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written $(\Omega, P)$ — is a **probability space**.

INTUITION

Three separate modelling choices: $\Omega$ — **what can happen** (the ground); $\mathcal{P}(\Omega)$ — **what may be asked** (the regions one may draw); $P$ — **how likely each answer is** (how the sand is spread).

### Same $\Omega$, different $P$

A fair and a loaded die share $\Omega = \{1, \ldots, 6\}$ but are two probability spaces.

### Finite $\Omega$

$P$ is fixed by the elementary events: $p_\omega \geq 0$, $\sum_\omega p_\omega = 1$, and $P(A) = \sum_{\omega \in A} p_\omega$.

Speaker notes

The three components state three separate modelling choices: $\Omega$ fixes what counts as an outcome, the event space fixes which questions may be asked, and $P$ fixes how likely each answer is. A theorem that begins with "let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space" holds for every such choice.

In practice the underlying probability space is often left implicit and only the distributions of random variables are written down; it is still assumed to exist.

# What the Axioms Already Give Us

| Property | Statement | From | Intuitively |
|---|---|---|---|
| **Complement** | $P(\overline{A}) = 1 - P(A)$ | II, III | *the sand not in $A$ lies in $\overline{A}$* |
| **Empty event** | $P(\emptyset) = 0$ | complement, $A = \Omega$ | *no region, no sand* |
| **Bounds** | $0 \leq P(A) \leq 1$ | I, complement | *no region holds more than all the sand* |
| **Monotonicity** | $A \subseteq B \Rightarrow P(A) \leq P(B)$ | I, III | *a larger region holds at least as much* |
| **Inclusion–exclusion** | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ | III | *the overlap was weighed twice* |
| **Union bound** | $P(A \cup B) \leq P(A) + P(B)$ | I, inclusion–exclusion | *weighing twice never weighs less* |
| **Finite additivity** | $P(A_1 \cup \ldots \cup A_p) = \sum_i P(A_i)$ for disjoint $A_i$ | III, repeated | *disjoint regions add up* |

Speaker notes

Derivations. Complement: $A$ and $\bar{A}$ are disjoint with union $\Omega$, so $P(A) + P(\bar{A}) = P(\Omega) = 1$. Bounds: $P(A) \geq 0$ by Axiom I and $P(A) = 1 - P(\bar{A}) \leq 1$. Empty set: $\bar{\Omega} = \emptyset$, so $P(\emptyset) = 1 - P(\Omega) = 0$. Monotonicity: for $A \subseteq B$, $B = A \cup (B \setminus A)$ disjointly, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$. Inclusion–exclusion: write $A \cup B = A \cup (B \setminus A)$ and $B = (B \setminus A) \cup (A \cap B)$, both disjoint; then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. On the Venn picture, $P(A) + P(B)$ counts the overlap twice. The union bound (Boole's inequality) drops the non-negative term $P(A \cap B)$; it extends to any finite or countable union and is often the only usable estimate when the overlaps are unknown. Finite additivity follows from Axiom III by induction over $p$.

Monotonicity transfers logic to numbers: if $A$ entails $B$, $A$ cannot be more probable than $B$. The conjunction fallacy (Tversky and Kahneman's "Linda problem") is the empirical finding that people often judge $P(A \cap B) > P(A)$, which violates it.

# Takeaway

> Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Speaker notes

An axiomatic definition fixes how probabilities must behave and leaves their source to the modeller: symmetry (Laplace), long-run frequency, or a stated degree of belief. Any assignment that is non-negative, normalised and additive is a probability measure, whichever source it comes from, and every theorem derived from the axioms — total probability, Bayes' theorem, the law of large numbers — applies to it. This is why frequentist and Bayesian statistics share one mathematics and differ only in which quantities they treat as random.

# Conditional Probability

> ***Guiding Question:*** *We learn that $B$ happened. What becomes of every other probability — and which rules follow from that one step?*

# Motivating Example

You roll a 6-sided die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

Speaker notes

Without the information, "sum is 8" is $\{(2,6),(3,5),(4,4),(5,3),(6,2)\}$, so $P = 5/36 \approx 0.14$. Knowing the first roll is 3 leaves six equally likely outcomes, of which only $(3,5)$ sums to 8: $P = 1/6 \approx 0.17$. The information raised the probability slightly; other observations lower it — after a first roll of 1, a sum of 8 is impossible.

Conditioning is not a new kind of probability. It is the old probability restricted to a smaller sample space, with the outcomes that remain keeping their relative weights and being rescaled to sum to 1.

# Motivating Example

You roll a 6-sided die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

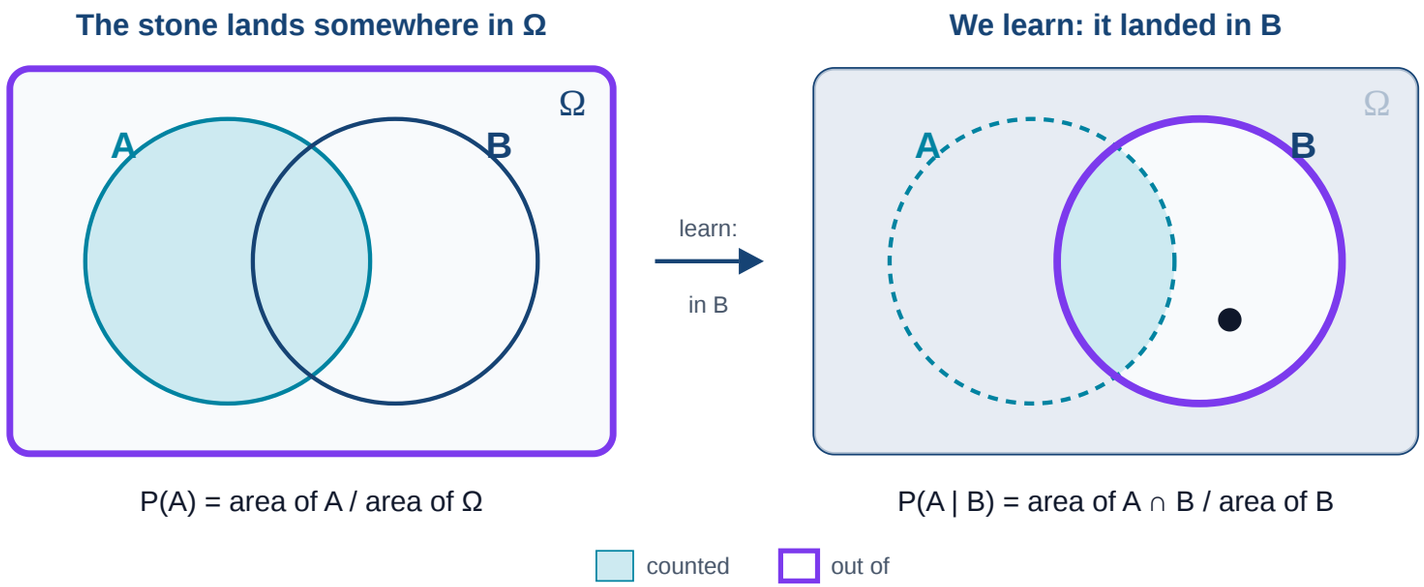
- Before the first roll: $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3,1),\ldots,(3,6)\}$ remains
- Outcomes like $(2,4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3,5)\}$

$$P = \frac{1}{6}$$

# Definition

**Definition (Conditional probability of $A$ under the condition $B$).**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{if } P(B) > 0$$




Back to the stone: once it is known to lie in $B$, $B$ **is the whole world**.

Speaker notes

The figure carries the colour code used from here on: teal is what is counted, the violet frame is what it is counted out of. Before the information, $A$ is counted out of $\Omega$; after it, only the part of $A$ inside $B$ counts, and it is measured against $B$.

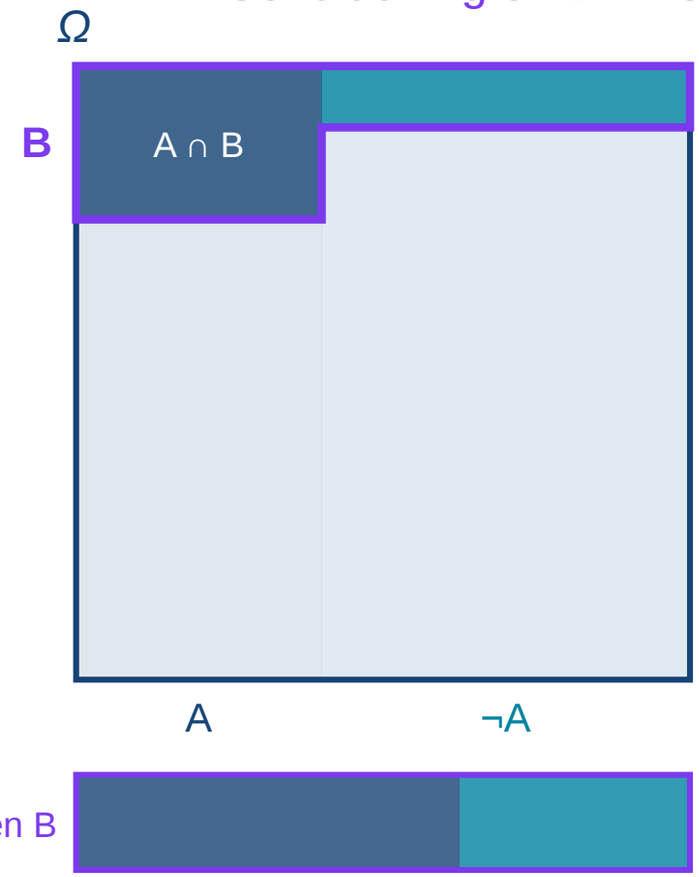
In the dice example $P(A \cap B) = 1/36$ and $P(B) = 6/36$, so $P(A \mid B) = 1/6$; dividing by $P(B)$ is the rescaling of the six remaining outcomes to total probability 1.

The condition $P(B) > 0$ is necessary because the ratio is otherwise undefined. Conditioning on events of probability zero, such as a continuous variable taking an exact value, requires densities instead and is treated with continuous distributions.

$P(A \mid B)$ and $P(A \cap B)$ are different quantities: $P(A \cap B) \leq P(A \mid B)$ always, with equality only when $P(B) = 1$ or $P(A \cap B) = 0$.

# Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1




P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

P(A | B) = 0.63

**Before the last click:** what share of $B$ is $A$?

**Then:** set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

Speaker notes

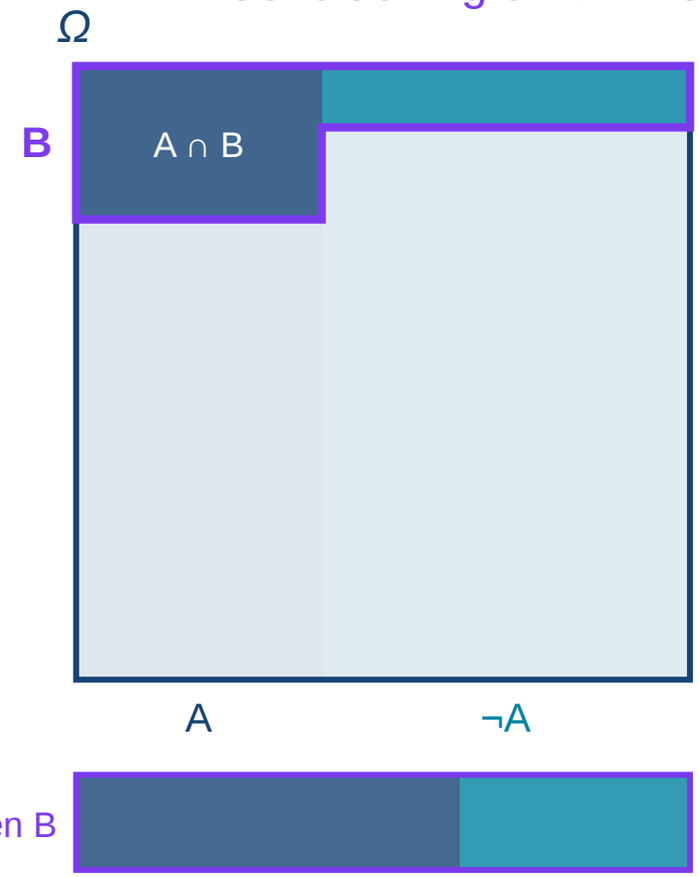
The four steps: the unit square split into a column $A$ of width $P(A) = 0.4$; the event $B$, outlined in violet, made of two cells, $A \cap B$ of area $0.10$ inside the column and $\bar{A} \cap B$ of area $0.06$ outside; everything outside $B$ removed; and the remainder stretched to total 1, giving $P(A \mid B) = 0.10/0.16 = 0.625$. The stretch factor is $1/P(B)$. These are the numbers of the students-and-exams example with $A$ = took PML and $B$ = failed.

The three sliders are areas, the quantities the definition is written in: $P(A)$, $P(B)$ and $P(A \cap B)$. They are free except that the overlap can be no larger than either event and no smaller than $P(A) + P(B) - 1$ (otherwise $A \cup B$ would exceed $\Omega$); outside that range the overlap slider snaps back. Everything else is derived: $P(\bar{A} \cap B) = P(B) - P(A \cap B)$ and $P(A \mid B) = P(A \cap B)/P(B)$. The heights of the two cells are the conditionals $P(B \mid A)$ and $P(B \mid \bar{A})$; the picture contains them, but they are not needed to define $P(A \mid B)$.

Two settings worth showing. $P(A \cap B) = P(B)$ puts all of $B$ inside $A$: $P(A \mid B) = 1$. $P(A \cap B) = P(A) \cdot P(B)$ (e.g. $0.4 \cdot 0.16 \approx 0.06$) gives both cells the same height: $A$ takes the same share of $B$ as of $\Omega$, $P(A \mid B) = P(A)$, and the caption reports that learning $B$ changes nothing — the case called independence.

# Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1




P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

P(A | B) = 0.63

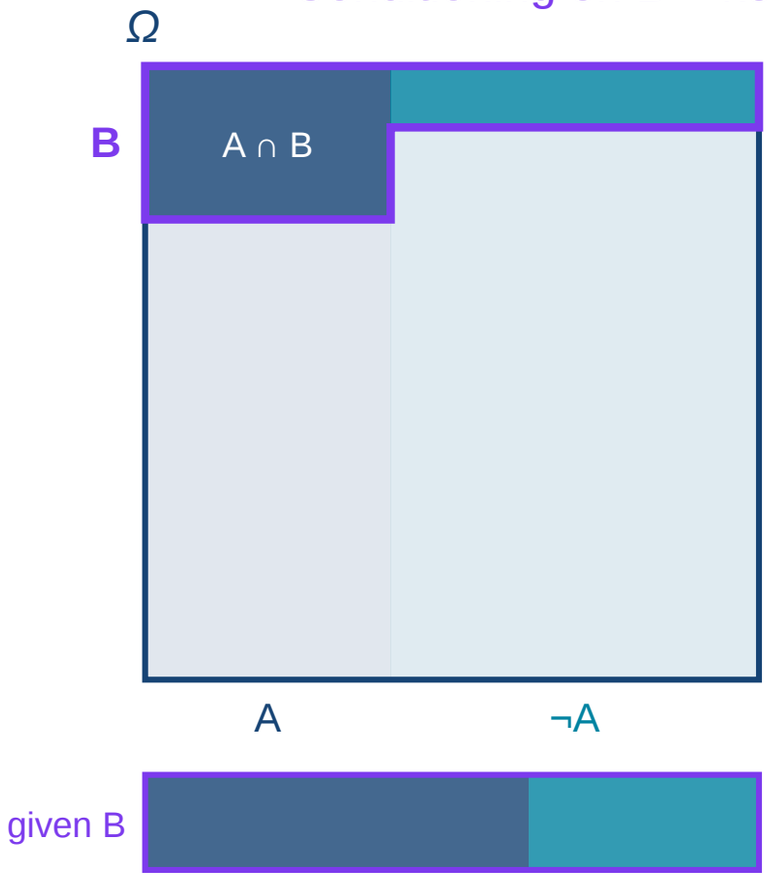
**Before the last click:** what share of $B$ is $A$?

$P(A \cap B)/P(B) \approx 0.63$

**Then:** set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

# Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1




P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

P(A | B) = 0.63

**Before the last click:** what share of $B$ is $A$?

$P(A \cap B)/P(B) \approx 0.63$

**Then:** set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

1: all of $B$ lies inside $A$.

# Example: Two Coins

Two fair coins are flipped. How likely is $B$ = **both heads**, if we learn …

**(a) "The first coin is heads"**

$F = \{hh, ht\}$

**(b) "At least one coin is heads"**

$A = \{hh, ht, th\}$

Speaker notes

Let the room estimate both before the answers appear; (b) is most often answered with $\frac{1}{2}$.

# Example: Two Coins

Two fair coins are flipped. How likely is $B$ = **both heads**, if we learn …

### (a) "The first coin is heads"

$F = \{hh, ht\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

### (b) "At least one coin is heads"

$A = \{hh, ht, th\}$

# Example: Two Coins

Two fair coins are flipped. How likely is $B$ = **both heads**, if we learn …

### (a) "The first coin is heads"

$F = \{hh, ht\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

### (b) "At least one coin is heads"

$A = \{hh, ht, th\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

# Example: Two Coins

Two fair coins are flipped. How likely is $B$ = **both heads**, if we learn …

**(a) "The first coin is heads"**

$F = \{hh, ht\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

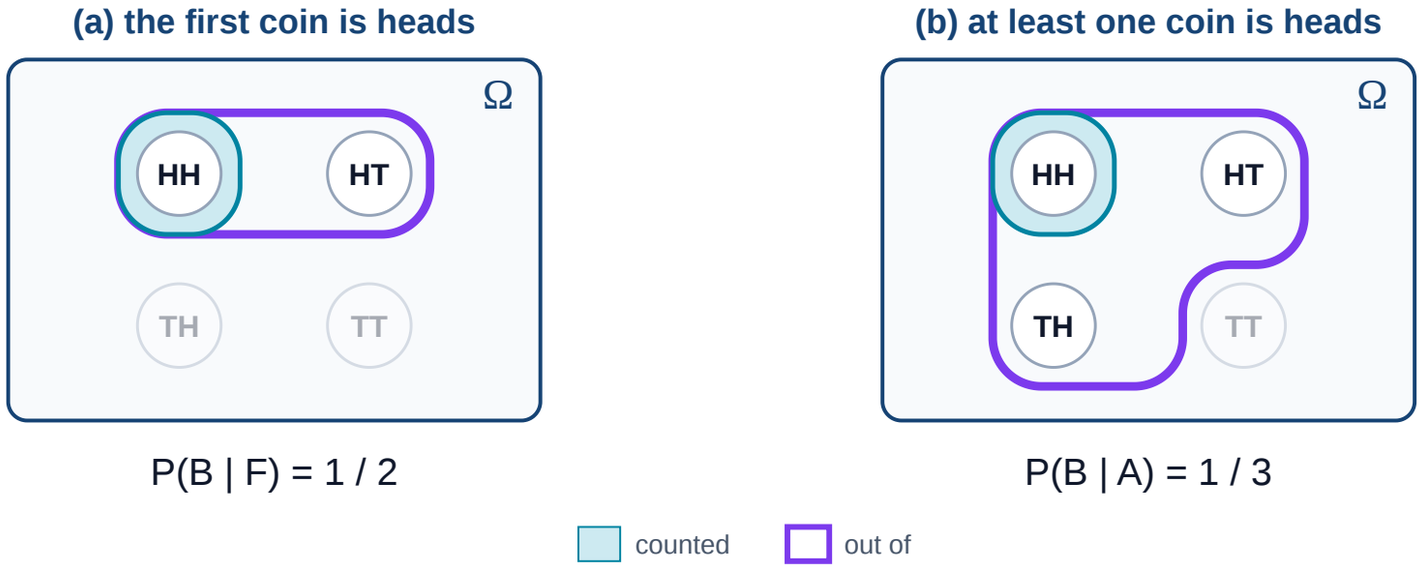
**(b) "At least one coin is heads"**

$A = \{hh, ht, th\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event, similar news. **Why is (b) smaller?**

# Two Coins, Drawn




Both count the same outcome $hh$. News (a) leaves **two** possible outcomes, news (b) leaves **three**: weaker news rules out less, so $hh$ gets a smaller share.

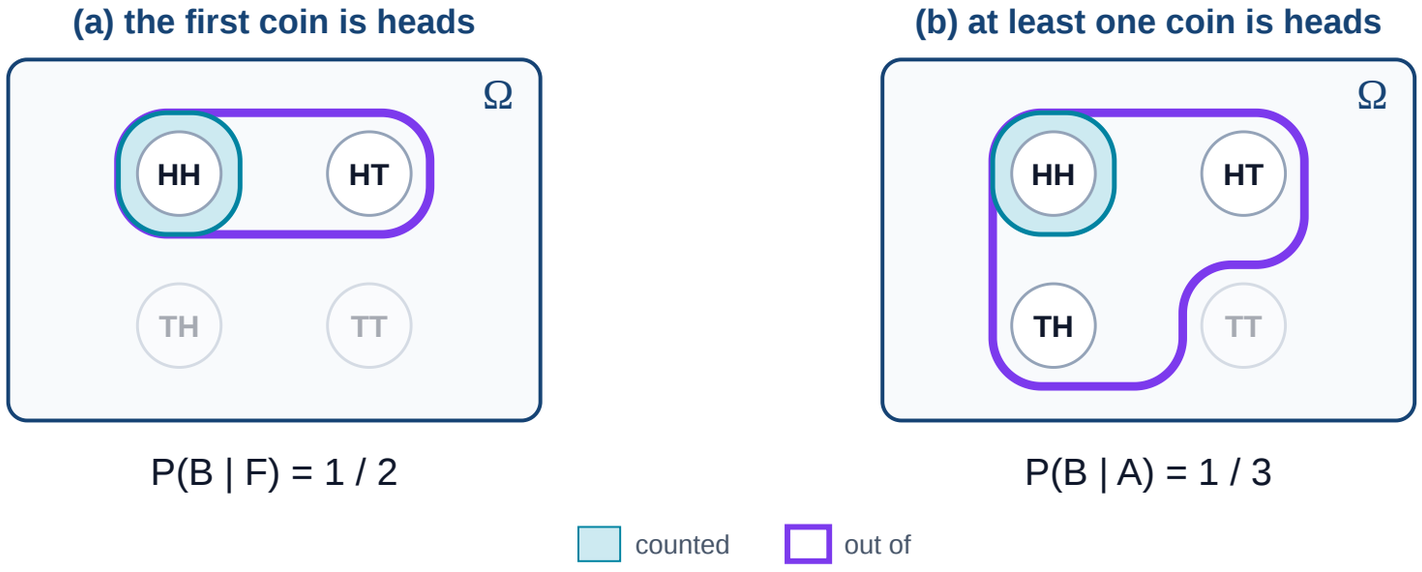
Speaker notes

(b) is smaller because the information in (a) is stronger: naming *which* coin is heads rules out two of the four outcomes, leaving $\{hh, ht\}$, whereas "at least one is heads" rules out only $tt$, leaving three equally likely outcomes of which one is $hh$.

A frequent error is to answer $\frac{1}{2}$ in (b) by reasoning "one coin is heads, the other is heads with probability $\frac{1}{2}$". That argument silently fixes which coin is known to be heads, which is the information of (a), not of (b). The figure makes the difference countable: the violet frame holds two outcomes in (a) and three in (b), and the counted outcome $hh$ is the same.

Venn diagrams and outcome sets are drawable for two, at most three events; a model with ten binary variables has $2^{10} = 1{,}024$ joint outcomes. The definition of conditional probability applies unchanged to any number of events, which is why it, not the picture, is the tool. The same structure underlies the "boy or girl paradox" (Gardner, 1959): "a family with two children has at least one boy" gives probability $\frac{1}{3}$ for two boys, "the older child is a boy" gives $\frac{1}{2}$. The answer also depends on how the information was obtained; the $\frac{1}{3}$ assumes that "at least one" was the full content of what was learned.

# Two Coins, Drawn




Both count the same outcome $hh$. News (a) leaves **two** possible outcomes, news (b) leaves **three**: weaker news rules out less, so $hh$ gets a smaller share.

> Pictures stop at two or three events. **Learn the rule** — it does not run out.

# Total Probability

**Theorem (Total probability).** Let $E_1, \ldots, E_k$ be mutually exclusive with $\Omega = E_1 \cup \ldots \cup E_k$ and $P(E_i) > 0$. Then for any event $B$:

$$P(B) = \sum_{i=1}^{k} P(E_i) \cdot P(B \mid E_i)$$

**Proof.** Decompose $B$ along the partition and use additivity:

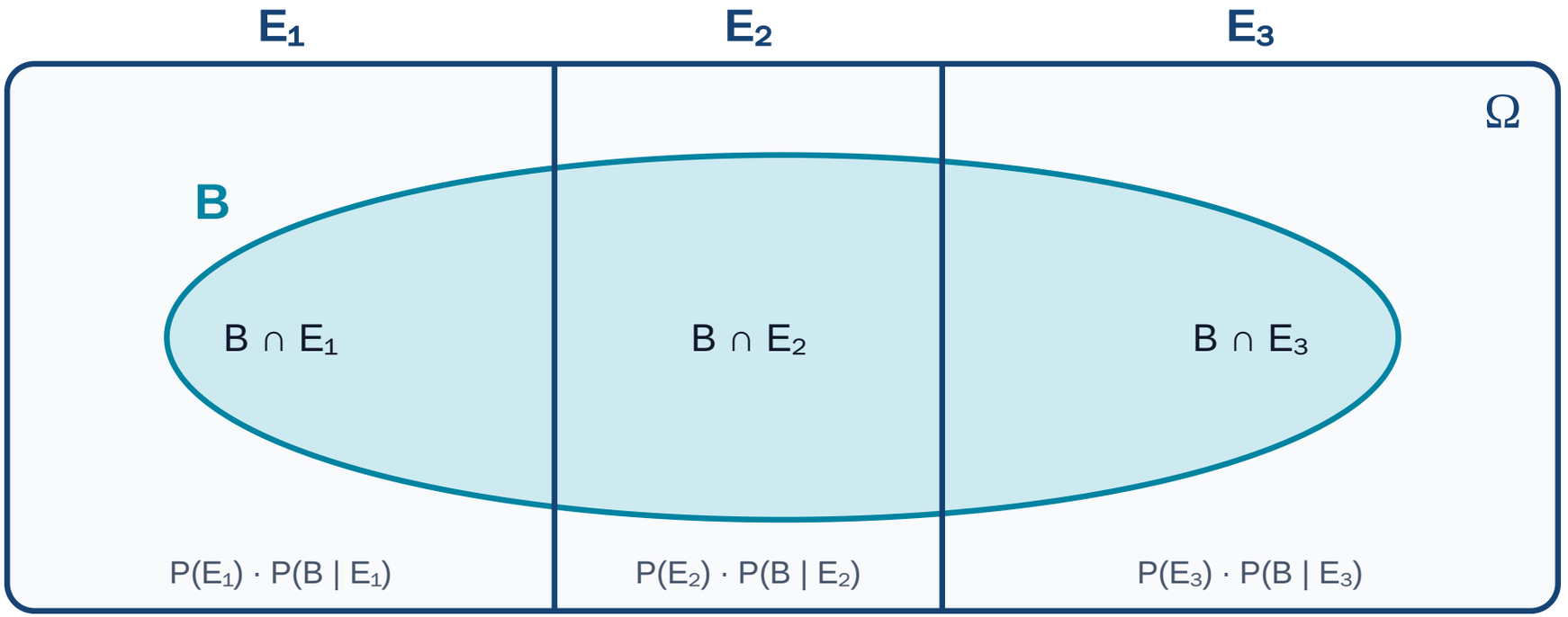
$$P(B) = P((E_1 \cup \ldots \cup E_k) \cap B) = \sum_{i=1}^{k} P(E_i \cap B) = \sum_{i=1}^{k} P(E_i) \cdot P(B \mid E_i)$$

Speaker notes

The proof uses three facts. First, $B = B \cap \Omega = \bigcup_i (E_i \cap B)$ because the $E_i$ cover $\Omega$. Second, the pieces $E_i \cap B$ are pairwise disjoint because the $E_i$ are, so additivity (Axiom III) turns the union into a sum. Third, $P(E_i \cap B) = P(E_i)\,P(B \mid E_i)$ by the definition of conditional probability, which needs $P(E_i) > 0$.

The theorem computes a probability by cases: in the students-and-exams example, $P(\text{fail}) = 0.4 \cdot 0.25 + 0.6 \cdot 0.1 = 0.16$. It supplies the denominator $P(B)$ of Bayes' theorem whenever $P(B)$ is not known directly. The simplest partition is $\{E, \bar{E}\}$: $P(B) = P(E)P(B \mid E) + P(\bar{E})P(B \mid \bar{E})$.

# Total Probability, Drawn




$$P(B) = P(B \cap E_1) + P(B \cap E_2) + P(B \cap E_3) = \Sigma_i P(E_i) \cdot P(B \mid E_i)$$

$B$ is cut into disjoint pieces, one per $E_i$; each piece is "first $E_i$, then $B$ within $E_i$" — the product rule.

Speaker notes

The strips $E_1, E_2, E_3$ are a partition: disjoint and together all of $\Omega$. Each piece $B \cap E_i$ has probability $P(E_i)\,P(B \mid E_i)$ by the product rule, and the pieces add up to $B$ by additivity. In the students-and-exams example the partition is the course $\{\mathrm{PML}, \mathrm{DB}\}$ and $B$ = failed: $P(\mathrm{fail}) = 0.4 \cdot 0.25 + 0.6 \cdot 0.1 = 0.16$ — the sum rule of the count table, written with conditionals.

# Rules That Follow

**Product rule**

$$P(A \cap B) = P(B) \cdot P(A \mid B) = P(A) \cdot P(B \mid A)$$

**Complement, same condition**

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

**But in general**

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at $B$, $P_B(E) = P(E \mid B)$ for every event $E$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** $P_B$.

Speaker notes

The product rule is the definition of conditional probability multiplied out; it also holds when $P(B) = 0$ if both sides are read as 0. Its two forms, set equal and divided by $P(A) > 0$, turn $P(A \mid B)$ into $P(B \mid A)$: that is Bayes' theorem, which gets its own unit.

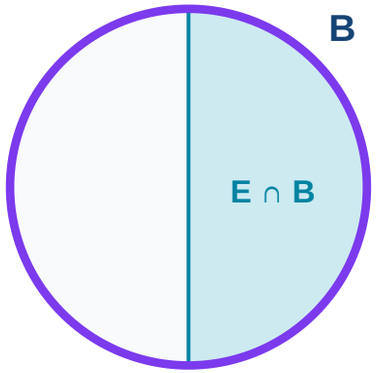
Complementing the event under a fixed condition is valid because $P_B$ is a probability measure, and $P_B(\bar{A}) = 1 - P_B(A)$ holds for every measure. Complementing the condition is not valid: $P(A \mid B)$ and $P(A \mid \bar{B})$ belong to two different measures. In the dice example, $P(\text{sum } 8 \mid \text{first is } 3) = 1/6$, whereas $P(\text{sum } 8 \mid \text{first is not } 3) = 4/30 = 2/15$, not $5/6$.

That $P_B$ satisfies the axioms is checked directly: $P(A \cap B)/P(B) \geq 0$; $P(\Omega \cap B)/P(B) = 1$; and disjoint $E_1, E_2$ give disjoint $E_1 \cap B, E_2 \cap B$. Every rule proved for $P$ therefore holds for $P(\cdot \mid B)$.

The letter $E$ is a placeholder for an arbitrary event; writing the map as $A \mapsto P(A \mid B)$ is equally correct, but $A$ already names a specific event in the product rule on this slide.

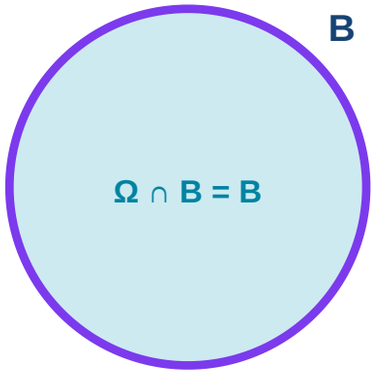
# Conditioning Gives a New Measure

$P_B(E) = P(E \cap B) / P(B)$: every event E is measured by its piece inside B



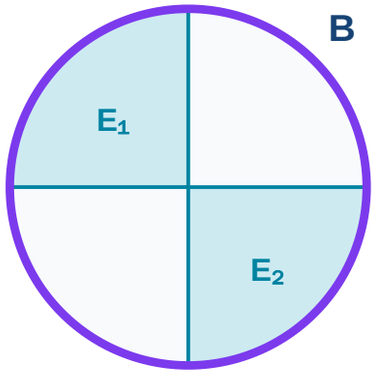

Axiom I: $P_B(E) \geq 0$

a piece of B is never negative




Axiom II: $P_B(\Omega) = 1$

all of B is the whole world




Axiom III: $P_B(E_1 \cup E_2) = P_B(E_1) + P_B(E_2)$

disjoint pieces of B add up

Inside $B$, the three axioms hold again — so **every rule proved for $P$ holds for $P_B$**.

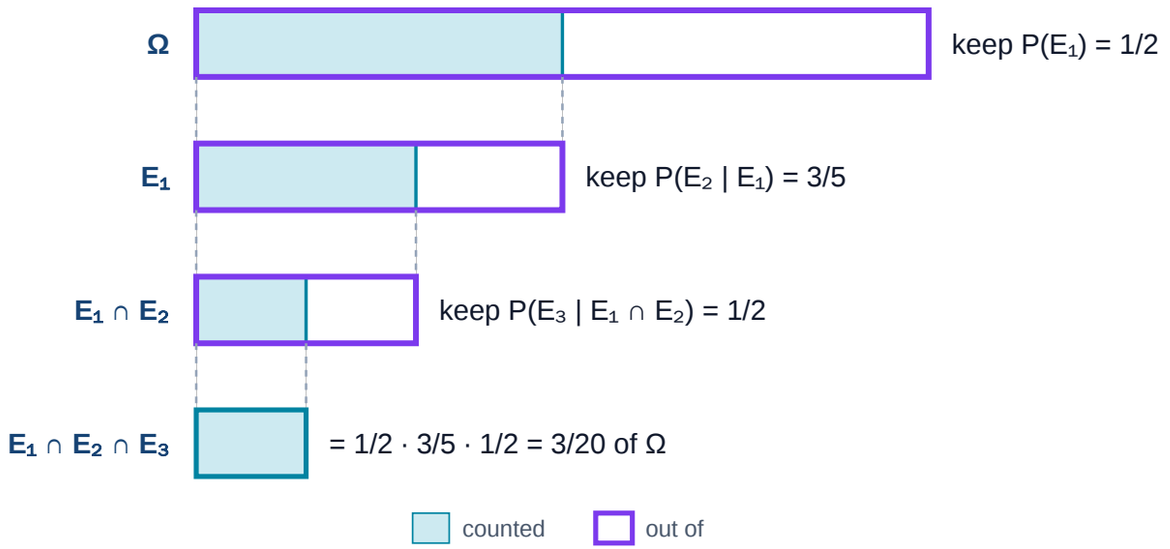
Speaker notes

Axiom I: $P_B(E) = P(E \cap B)/P(B)$ is a ratio of a non-negative number and a positive one. Axiom II: $P_B(\Omega) = P(\Omega \cap B)/P(B) = P(B)/P(B) = 1$; also $P_B(B) = 1$, so after conditioning $B$ is certain. Axiom III: if $E_1 \cap E_2 = \emptyset$, the pieces $E_1 \cap B$ and $E_2 \cap B$ are disjoint too, and additivity of $P$ carries over after dividing by $P(B)$. Events disjoint from $B$ get $P_B(E) = 0$.

This licenses the complement rule under a fixed condition, $P(\bar{A} \mid B) = 1 - P(A \mid B)$, and every other consequence of the axioms. It does not license changing the condition: $P_B$ and $P_{\bar{B}}$ are two different measures.

# Chain Rule

$$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \ldots \cap E_{n-1})$$




Each factor keeps a share of what is left. Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Speaker notes

Proof for three events: $P(E_1 \cap E_2 \cap E_3) = P(E_1 \cap E_2)\, P(E_3 \mid E_1 \cap E_2) = P(E_1)\, P(E_2 \mid E_1)\, P(E_3 \mid E_1 \cap E_2)$; induction gives the general case. It requires $P(E_1 \cap \ldots \cap E_{n-1}) > 0$.

Any order of the events gives a valid factorisation. The chain rule holds without assumptions; simplification comes only from conditional independence, when a factor such as $P(E_3 \mid E_1 \cap E_2)$ equals $P(E_3 \mid E_2)$. For $n$ binary variables the full joint needs $2^n - 1$ numbers; if each factor depends on at most $k$ earlier variables, at most $n \cdot 2^k$ suffice. Bayesian networks are built on exactly this combination of an ordering and dropped conditions.

The figure uses $P(E_1) = \frac{1}{2}$, $P(E_2 \mid E_1) = \frac{3}{5}$ and $P(E_3 \mid E_1 \cap E_2) = \frac{1}{2}$: each bar is the shaded part of the bar above it, so the last one is $\frac{1}{2} \cdot \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{20}$ of $\Omega$. Every conditional is measured against what is left, never against $\Omega$.

Example: drawing two aces in a row from a 52-card deck without replacement has probability $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.

# Takeaway

> Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

# Bayes' Theorem

> ***Guiding Question:*** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and how do we turn* $P(+ \mid \text{sick})$ *into* $P(\text{sick} \mid +)$*?*

# Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

**Your test comes back positive. How likely is it that you are actually sick?**

Speaker notes

The test has sensitivity $P(+ \mid \text{sick}) = 0.99$ and specificity $P(- \mid \text{healthy}) = 0.99$; the prevalence is $0.001$. The exact answer is $P(\text{sick} \mid +) = \dfrac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.01 \cdot 0.999} = \dfrac{0.00099}{0.01098} \approx 0.090$.

The common intuitive answer is 99%, the test's accuracy. Physicians make the same error on comparable screening problems (Eddy, 1982, for mammography). The error is called base-rate neglect: the prevalence, 1 in 1,000, is ignored. Errors of this size occur because false positives from the large healthy group outnumber true positives from the small sick group about ten to one.

# Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

**Your test comes back positive. How likely is it that you are actually sick?**

> Only about **9%** — the answer most people miss by a factor of ten.

# Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

**P(sick | positive) = 1/11 ≈ 9%**



Speaker notes

The steps show: a population of 1,000; the one person among them who is sick (prevalence 0.001); the healthy people the test flags, $1\% \cdot 999 \approx 10$; and only the positives kept, about 11 people of whom one is sick. Hence $P(\text{sick} \mid +) \approx 1/11 \approx 9\%$; exactly, $0.99/(0.99 + 9.99) \approx 0.090$.

Counting people in a fixed population ("natural frequencies") rather than multiplying percentages is known to reduce base-rate neglect substantially (Gigerenzer and Hoffrage, 1995). The computation is the same: restricting the population to the positives is conditioning on the event "positive".

# What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space $B$; the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

**The trap**

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Speaker notes

Confusing $P(B \mid A)$ with $P(A \mid B)$ is known as the confusion of the inverse; in court it is called the prosecutor's fallacy, where the probability of the evidence given innocence is presented as the probability of innocence given the evidence. When $P(A \cap B) > 0$, the two conditionals agree only if $P(A) = P(B)$, since $P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$.

Bayes' theorem converts one into the other, and the conversion requires the prior $P(A)$. A test's sensitivity and specificity are properties of the test; $P(\text{sick} \mid +)$, the positive predictive value, also depends on the population tested.

# Bayes' Theorem — Two Events

**Theorem (Bayes, 2 events).** Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

Speaker notes

The four steps highlight in turn the posterior $P(A \mid B)$, the likelihood $P(B \mid A)$, the prior $P(A)$ and the evidence $P(B)$. For the disease test, $A$ = sick and $B$ = positive: "99% accurate" is the likelihood $P(B \mid A) = 0.99$, and the term ignored by intuition is the prior $P(A) = 0.001$.

The likelihood is a probability of $B$, but it is read as a function of the hypothesis $A$: it need not sum to 1 over hypotheses ($P(+ \mid \text{sick}) + P(+ \mid \text{healthy}) = 1.00$ here only by coincidence of the chosen rates). The evidence $P(B)$ does not depend on $A$; it normalises the posterior. Equivalent odds form: $\frac{P(A|B)}{P(\bar{A}|B)} = \frac{P(B|A)}{P(B|\bar{A})} \cdot \frac{P(A)}{P(\bar{A})}$, i.e. posterior odds = likelihood ratio × prior odds. Reference: (Sheldon & others 2002).

# Bayes' Theorem — Two Events

**Theorem (Bayes, 2 events).** Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

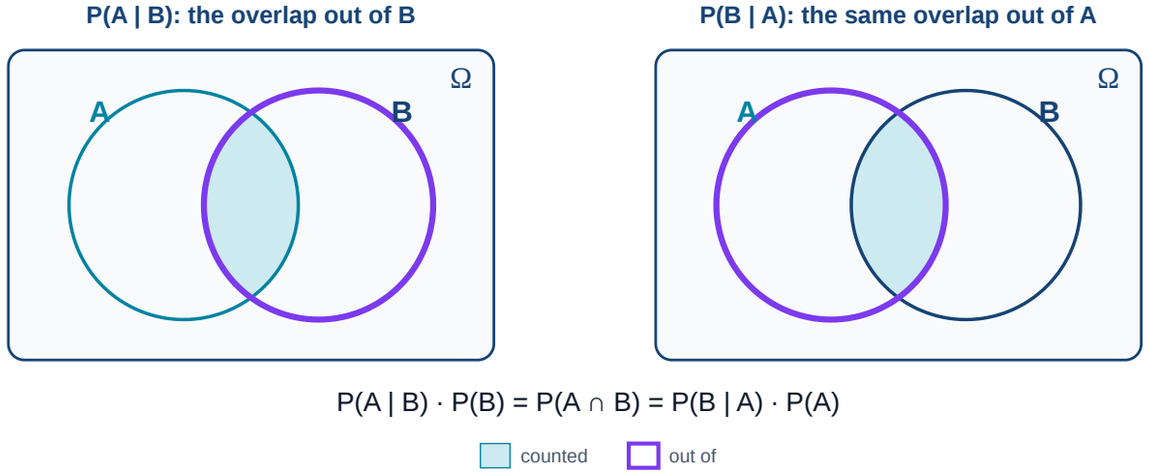
prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

"99% accurate" is the **likelihood** $P(+ \mid \text{sick})$; the intuition ignored the **prior** $P(\text{sick}) = 0.001$.

# Proof — Two Events

## Intuition




One overlap $A \cap B$, measured against two wholes: against $B$ it is $P(A \mid B)$, against $A$ it is $P(B \mid A)$.

## Proof

| step | because |
|---|---|
| $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ | definition |
| $= \frac{P(A \cap B)}{P(A)} \cdot \frac{P(A)}{P(B)}$ | multiply and divide by $P(A)$ |
| $= \frac{P(B \mid A)\, P(A)}{P(B)}$ | definition, roles swapped |

Needs $P(A), P(B) > 0$. ■

Speaker notes

The figure shows the same region $A \cap B$ measured against two different wholes: against $B$ it is $P(A \mid B)$, against $A$ it is $P(B \mid A)$. Both products with the size of the whole give back the overlap, $P(A \mid B)P(B) = P(A \cap B) = P(B \mid A)P(A)$; dividing by $P(B)$ is Bayes' theorem.

The first equality is the definition of $P(A \mid B)$ and needs $P(B) > 0$. The second multiplies and divides by $P(A)$ and needs $P(A) > 0$; the third recognises $P(A \cap B)/P(A) = P(B \mid A)$, the definition with the roles swapped. If $P(A) = 0$, $P(B \mid A)$ is undefined, but $P(A \mid B) = 0$ directly from the definition: a hypothesis with prior probability zero cannot gain probability from any evidence. This is why a prior should not assign zero to anything that may be true (Lindley's "Cromwell's rule").

# Mini-Exercise

**A high-risk group.** Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

Speaker notes

2,000 is chosen so that both rates give whole people: 1 in 20 of 2,000 is 100, and 1% of the 1,900 healthy is 19. Only the positives count once the result is known — 118 people, the new sample space — and 99 of them are sick. With prevalence 1 in 1,000 the same column held about 1 true positive against 10 false alarms; now true positives outnumber false alarms five to one.

Neither result corrects the other. The posterior depends on the test (sensitivity, false-alarm rate) and on the population tested (prevalence); here the test is the same and only the prevalence differs, raising the positive predictive value from about 9% to about 84%. For the same reason a positive screening result is commonly confirmed by a second, independent test: the first positive raises the prior for the second.

# Mini-Exercise

**A high-risk group.** Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

# Mini-Exercise

**A high-risk group.** Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

| | everyone | high-risk group |
|---|---|---|
| the test, $P(+ \mid \text{sick})$ / $P(+ \mid \text{healthy})$ | 99% / 1% | 99% / 1% |
| who is tested, $P(\text{sick})$ | 1 in 1,000 | 1 in 20 |
| $P(\text{sick} \mid +)$ | $\approx 9\%$ | $\approx 84\%$ |

# Mini-Exercise

**A high-risk group.** Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

| | everyone | high-risk group |
|---|---|---|
| the test, $P(+ \mid \text{sick})$ / $P(+ \mid \text{healthy})$ | 99% / 1% | 99% / 1% |
| who is tested, $P(\text{sick})$ | 1 in 1,000 | 1 in 20 |
| $P(\text{sick} \mid +)$ | $\approx 9\%$ | $\approx 84\%$ |

> Both answers are right. Same test, same rule — **different people tested**.

# The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

Speaker notes

The denominator, the evidence $P(+)$, is total probability over the partition $\{\text{sick}, \text{healthy}\}$: a positive comes either from a sick person, $P(+ \mid \text{sick})\,P(\text{sick}) = 0.0495$, or from a healthy one, $P(+ \mid \text{healthy})\,P(\text{healthy}) = 0.01 \cdot 0.95 = 0.0095$. The posterior is the first share of the total. In odds form: prior odds $1 : 19$, likelihood ratio $0.99/0.01 = 99$, posterior odds $99 : 19$, i.e. $99/118 \approx 0.84$.

# The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

$$= \frac{0.0495}{0.0495 + 0.0095} = \frac{0.0495}{0.0590} \approx 0.84$$

The denominator is the **evidence** $P(+)$: every way to test positive.

# The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

$$= \frac{0.0495}{0.0495 + 0.0095} = \frac{0.0495}{0.0590} \approx 0.84$$

The denominator is the **evidence** $P(+)$: every way to test positive.

> Times 2,000, the fraction is **99 / 118**. Counting people and the formula are **one calculation**, written two ways.

# Bayes' Theorem — Multiple Events

**Theorem (Bayes, multiple events).** Let $E_1, \ldots, E_k$ be mutually exclusive with $\Omega = E_1 \cup \ldots \cup E_k$ and $P(E_i) > 0$, and let $B$ be an event with $P(B) > 0$. Then

$$P(E_i \mid B) = \frac{P(B \mid E_i) \cdot P(E_i)}{P(B)} = \frac{P(B \mid E_i) \cdot P(E_i)}{\sum_{j=1}^{k} P(B \mid E_j) \cdot P(E_j)}$$

posterior of $E_i$

evidence

every $E_i$ explains B, weighted by its prior

likelihood — how well $E_i$ explains B

prior of $E_i$

The denominator is the **total probability** of the evidence.

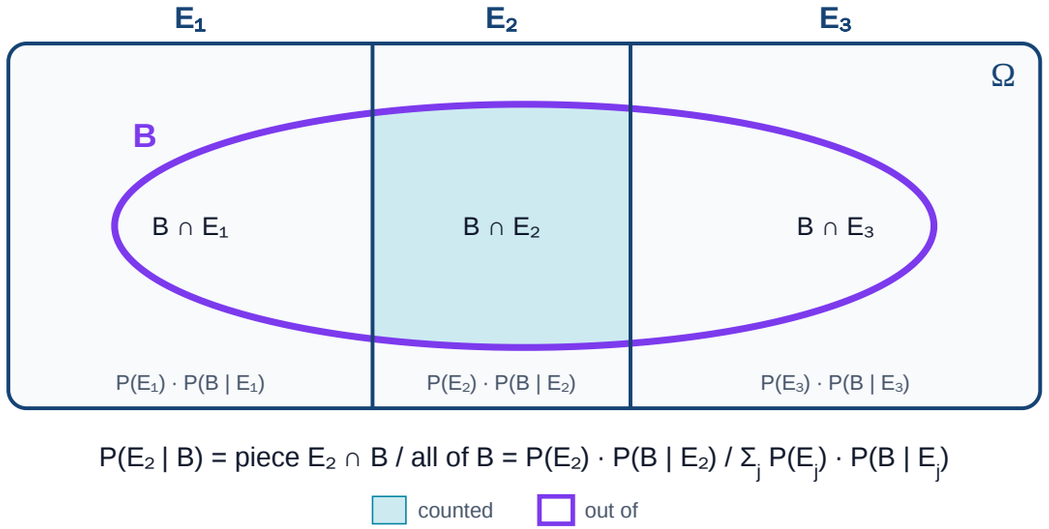
Speaker notes

The events $E_1, \ldots, E_k$ partition $\Omega$: exactly one of them occurs. Read as competing explanations of $B$, they are called hypotheses. "Sick vs. healthy" is the case $k = 2$, and the students-and-exams example (PML vs. DB) is another. The posteriors $P(E_i \mid B)$ sum to 1 because the denominator is the sum of all numerators.

The procedure is: compute likelihood × prior for every $E_i$, then divide by the total. Since the denominator is the same for all $i$, ranking the events — choosing the most probable one — needs only the numerators. A Bayes classifier does exactly this, with the classes as the $E_i$ and the observed features as $B$.

# Proof — Multiple Events

## Intuition




The events $E_i$ cut $B$ into pieces. The posterior of $E_i$ is **its piece out of all pieces**.

## Proof

| step | because |
|---|---|
| $P(E_i \mid B) = \frac{P(B\|E_i)\,P(E_i)}{P(B)}$ | Bayes, two events |
| $P(B) = \sum_j P(B \cap E_j)$ | additivity |
| $= \sum_j P(B \mid E_j)\,P(E_j)$ | product rule |

Additivity holds because the $E_j$ are disjoint and cover $\Omega$. Insert $P(B)$ into the first line. ■

Speaker notes

The events $E_j$ are the strips; the evidence $B$ is the ellipse. Each piece $B \cap E_j$ has probability $P(E_j)\,P(B \mid E_j)$ (product rule), and the pieces add up to $P(B)$ (total probability). The posterior of $E_2$ is its piece out of the whole ellipse. An event $E_j$ wins when its piece is large — because it was plausible beforehand (wide strip), or because it explains $B$ well (much of its strip lies inside $B$), or both.

# Takeaway

> Bayes' theorem is the arithmetic of changing your mind: **posterior ∝ likelihood × prior**. Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Speaker notes

$\propto$ means "proportional to, as a function of the hypothesis": the posterior is the product likelihood × prior, rescaled so that it sums to 1 over all hypotheses. Two consequences follow. Ignoring the prior gives the base-rate error (99% instead of 9% for the disease test). The same evidence gives very different posteriors under different priors (9% at prevalence 1/1,000, 84% at 1/20).

Sequential updating is consistent: the posterior after one piece of evidence serves as the prior for the next, and if the pieces of evidence are conditionally independent given the hypothesis, the result equals a single update on all of them at once.

# Back to Lecture 01: the Model Says "Fraud"

A fraud model is **99% accurate** in both directions. One transaction in **1,000** is fraudulent. Per 100,000 transactions:

| | flagged | not flagged | total |
|---|---|---|---|
| **fraud** | 99 | 1 | 100 |
| **legitimate** | 999 | 98,901 | 99,900 |
| **total** | **1,098** | 98,902 | 100,000 |

Speaker notes

The table is the disease-test computation with fraud in place of illness, for the fraud model of Lecture 01. Its cells: 100 frauds, of which 99% (99) are flagged; 99,900 legitimate transactions, of which 1% (999) are flagged. Hence $P(\text{fraud} \mid \text{flagged}) = 99/(99 + 999) = 99/1{,}098 \approx 0.090$. In odds form: prior odds $1 : 999$, times the likelihood ratio $0.99/0.01 = 99$, give posterior odds $99 : 999$, the same 9%.

The same holds for every classifier of a rare class. Accuracy is misleading there: a model that never flags anything is 99.9% accurate on this population. Precision ($99/1{,}098$) and recall ($99/100$) separate the two error types; Lecture 06 evaluates this model in those terms. A calibrated score of about 0.09 for a flagged transaction carries the prior; the bare label "fraud" does not.

# Back to Lecture 01: the Model Says "Fraud"

A fraud model is **99% accurate** in both directions. One transaction in **1,000** is fraudulent. Per 100,000 transactions:

| | flagged | not flagged | total |
|---|---|---|---|
| **fraud** | 99 | 1 | 100 |
| **legitimate** | 999 | 98,901 | 99,900 |
| **total** | **1,098** | 98,902 | 100,000 |

$$P(\text{fraud} \mid \text{flagged}) = \frac{99}{1{,}098} \approx 9\%$$

> Nine in ten blocked cards belong to honest customers — from a "99% accurate" model. The label "fraud" hid the **prior**; a score would at least have carried it.

# Independence

> ***Guiding Question:*** *When does learning one thing tell us nothing about another — and why is that the assumption every tractable model leans on?*

# Independence

**Definition (Statistical independence).** Two events $A$ and $B$ are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(A \mid B) = P(A \mid \overline{B})$$

Independence means: **learning $B$ changes nothing about $A$.**

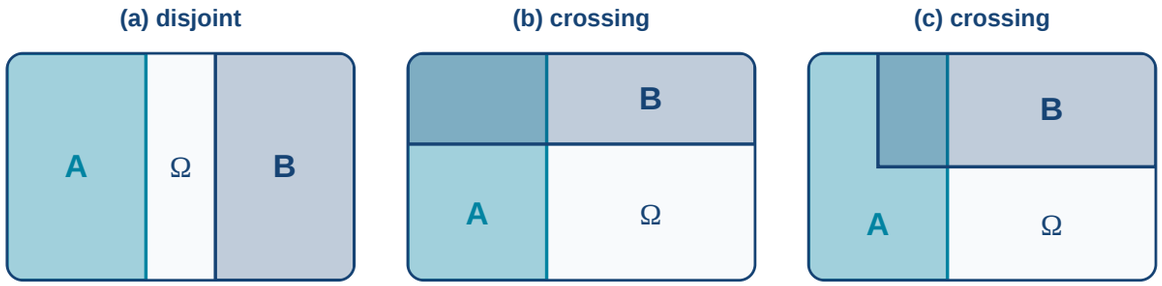
Speaker notes

The conditional form carries the meaning: learning that $B$ occurred leaves the probability of $A$ unchanged. The product form is the standard definition because it is symmetric in $A$ and $B$ and remains defined when $P(B) = 0$; an event of probability 0 or 1 is independent of every event.

Equivalence for $0 < P(B) < 1$: dividing the product form by $P(B)$ gives $P(A \mid B) = P(A)$. Total probability gives $P(A) = P(B)P(A \mid B) + P(\bar{B})P(A \mid \bar{B})$, so $P(A)$ is a weighted average of the two conditionals, and it equals $P(A \mid B)$ exactly when $P(A \mid B) = P(A \mid \bar{B})$. If $A$ and $B$ are independent, so are $A$ and $\bar{B}$, $\bar{A}$ and $B$, and $\bar{A}$ and $\bar{B}$.

Example: for two fair dice, "the first shows 6" and "the second shows 6" are independent, $\frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6}$.

# The Classic Confusion




In all three: P(A) = P(B) = 0.4

Which pair is **independent** — and why?

Speaker notes

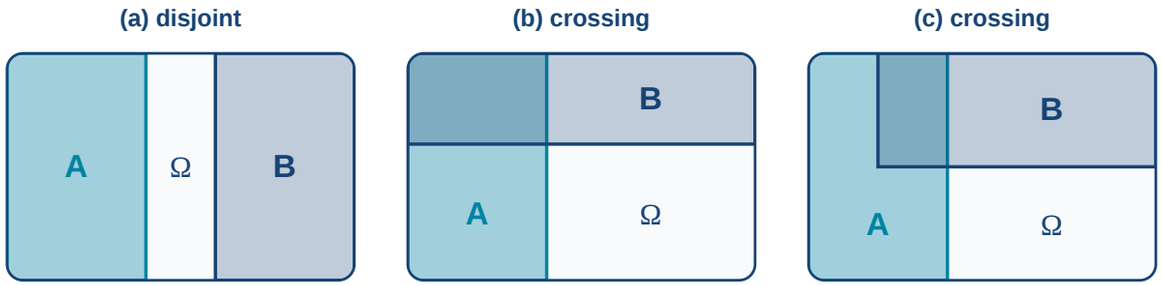
In the rectangle the areas are probabilities. (a): $P(A \cap B) = 0$ while $P(A)P(B) = 0.16$. (b): a vertical strip of width 0.4 and a horizontal strip of height 0.4 cross in a rectangle of area $0.4 \cdot 0.4 = 0.16 = P(A)P(B)$, so the product form holds exactly. Strips that cross the whole of $\Omega$ are always independent, whatever their widths; this is the picture behind the next slide. (c): $B$ is a block of width 0.8 and height 0.5, still area 0.4, starting inside $A$; the overlap is $0.2 \cdot 0.5 = 0.1 < 0.16$, so $A$ and $B$ are negatively dependent. Overlap alone decides nothing; only its size compared with $P(A)P(B)$ does.

The overlap is bounded: $A \cap B \subseteq B$, so $P(A \cap B) \leq \min(P(A), P(B))$ always, with $P(A \cap B) = P(B)$ only when $B$ lies inside $A$ (up to a set of probability 0). Below that, every value down to $\max(0, P(A) + P(B) - 1)$ is possible; here $[0, 0.4]$, with 0.16 the single independent value.

Disjoint events with positive probability exclude each other: the occurrence of one rules out the other, which is maximal information about it. Formally, $P(A \cap B) = 0$ while $P(A)P(B) > 0$, so the product form fails. Learning $B$ moves $P(A)$ to 0, the largest possible change downwards.

The Venn diagram misleads here because non-overlapping circles look "unrelated". Disjointness is a property of the sets; independence is a property of the probability measure and can change when the measure changes while the sets stay the same. The only case in which disjoint events are independent is when at least one of them has probability 0.

# The Classic Confusion




In all three: P(A) = P(B) = 0.4

Which pair is **independent** — and why?

**(a) dependent.** If $B$ occurs, $A$ certainly did not: $P(A \mid B) = 0 \neq 0.4$.

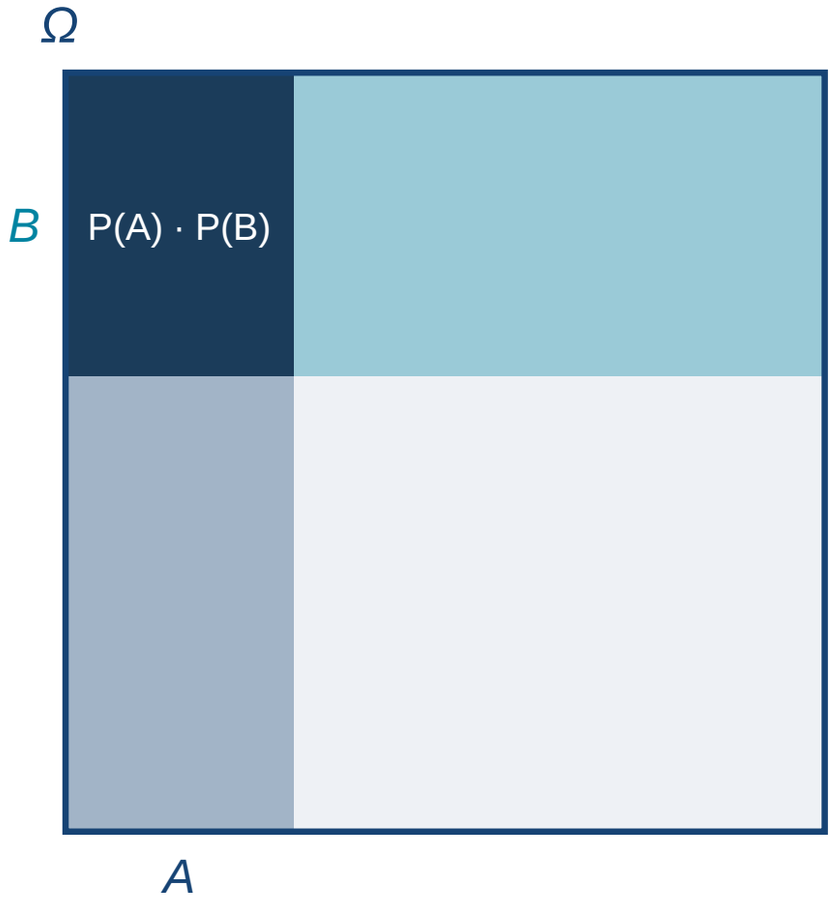
**(b) independent.** $A$ takes the same share of $B$ as of $\Omega$: $P(A \mid B) = 0.16/0.4 = 0.4$.

**(c) dependent.** They overlap, but too little: $P(A \mid B) = 0.1/0.4 = 0.25 \neq 0.4$.

> **Statistical independence does not mean the events are disjoint.** Disjoint events with positive probability are about as **dependent** as events can be.

# Independence Is About Areas, Not Overlap

P(A) · P(B) = 0.30 · 0.40 = 0.12 = P(A ∩ B) → independent




P(A) = 0.30

P(B) = 0.40

P(A ∩ B) = 0.12

P(A | B) = 0.12 / 0.40 = 0.30 = P(A)

Independent: **knowing $B$ tells nothing about $A$.** $A$ takes the same share of $B$ as of all of $\Omega$, so its probability stays the same.

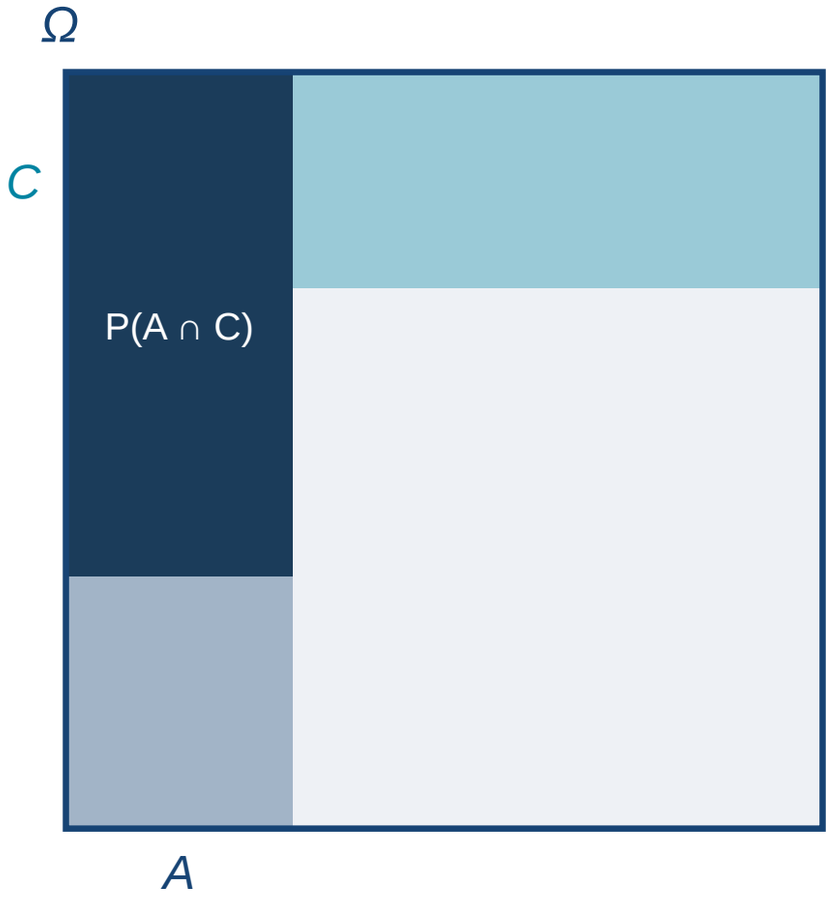
Speaker notes

The steps show the band $A$ with area $P(A) = 0.3$, the band $B$ with area $P(B) = 0.4$, their intersection with area $P(A \cap B) = 0.12$, and the comparison with $P(A)P(B) = 0.3 \cdot 0.4 = 0.12$: the events are independent. Equivalently $P(A \mid B) = 0.12/0.4 = 0.3 = P(A)$: $A$ occupies the same fraction of $B$ as of the whole of $\Omega$ .

Independence is an equality between numbers, not a property of the drawing: the shapes and positions of the regions are irrelevant, only their areas count. Overlapping regions can be independent, and so can regions that look strongly entangled.

# The Same Picture, Dependent

P(A) · P(C) = 0.30 · 0.40 = 0.12 ≠ 0.20 = P(A ∩ C) → dependent




P(A) = 0.30

P(C) = 0.40

P(A ∩ C) = 0.20

P(A | C) = 0.20 / 0.40 = 0.50 ≠ P(A)

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — $A$ and $C$ occur together more often than chance.

Speaker notes

Here $P(A \mid C) = 0.20/0.4 = 0.5 > P(A) = 0.3$: learning $C$ raises the probability of $A$, so $A$ and $C$ are positively dependent. A smaller intersection, below 0.12, would give negative dependence.

The marginals $P(A) = 0.3$ and $P(C) = 0.4$ are the same as on the previous slide; only the joint probability $P(A \cap C)$ differs. Marginals therefore never determine whether events are independent. Given these marginals, $P(A \cap C)$ can take any value between $\max(0, P(A) + P(C) - 1) = 0$ and $\min(P(A), P(C)) = 0.3$ (the Fréchet bounds); independence is the single value $0.12$ in that range. Answering questions about dependence requires the joint distribution.

# More Than Two Events

Events $A_1, \ldots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \ldots, i_l$:

$$P(A_{i_1} \cap \ldots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Speaker notes

Standard counterexample: flip two fair coins; let $A$ = "first is heads", $B$ = "second is heads", $C$ = "the coins agree". Each has probability $\frac{1}{2}$, and each pairwise intersection is $\{HH\}$ with probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, so the three are pairwise independent. But $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$: any two of them determine the third.

For $n$ events the definition imposes $2^n - n - 1$ equations, one for every subset of at least two events. The converse failure also exists: the triple product can factorise while a pair does not, so checking only the full intersection is not enough either. Models that assume features to be independent given a class (naive Bayes) assume mutual, not merely pairwise, conditional independence.

# Takeaway

> Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

# Random Variables

> ***Guiding Question:*** *We rarely care which outcome occurred, only what it was worth. How do outcomes become numbers — and what is left of the probability measure?*

Speaker notes

A random variable is a function on the sample space; its distribution is inherited from the probability measure. The module covers one variable at a time: discrete and continuous distributions, expectation and variance, and the families the rest of the course uses (Bernoulli, binomial, uniform, normal). It closes with what a family is: a formula with a few parameters instead of a table with one number per value, which needs less data but assumes a shape.

# Why Random Variables?

Three fair coins: what matters is rarely *which* sequence occurred, but the **number of heads** $Y$.

| $Y$ | **outcomes $\omega$ with $Y(\omega) = k$** | **count** | $P(Y = k)$ |
|---|---|---|---|
| $0$ | TTT | 1 | $\frac{1}{8}$ |
| $1$ | HTT · THT · TTH | 3 | $\frac{3}{8}$ |
| $2$ | HHT · HTH · THH | 3 | $\frac{3}{8}$ |
| $3$ | HHH | 1 | $\frac{1}{8}$ |

> $Y$ compresses eight outcomes into four numbers — the probabilities come along.

Speaker notes

The eight outcomes are equally likely, $\frac{1}{8}$ each. $Y$ maps them to $\{0, 1, 2, 3\}$, and $P(Y = k)$ is the number of sequences with $k$ heads divided by 8, i.e. $\binom{3}{k}/8$: 1, 3, 3, 1 sequences. This is the binomial distribution with $n = 3$, $p = \frac{1}{2}$.

The compression loses information: which coins showed heads, and in which order. It gains a numerical quantity that can be averaged, compared and plotted; for example $E[Y] = (0 + 3 + 6 + 3)/8 = 1.5$. Whether the lost information matters depends on the question: for the number of heads it does not, for "did the first coin show heads" it does.

# Definition

**Definition (Random variable).** A random variable $X$ maps outcomes to a measurable range $E$: $X : \Omega \to E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in $\omega$, not in $X$.

Speaker notes

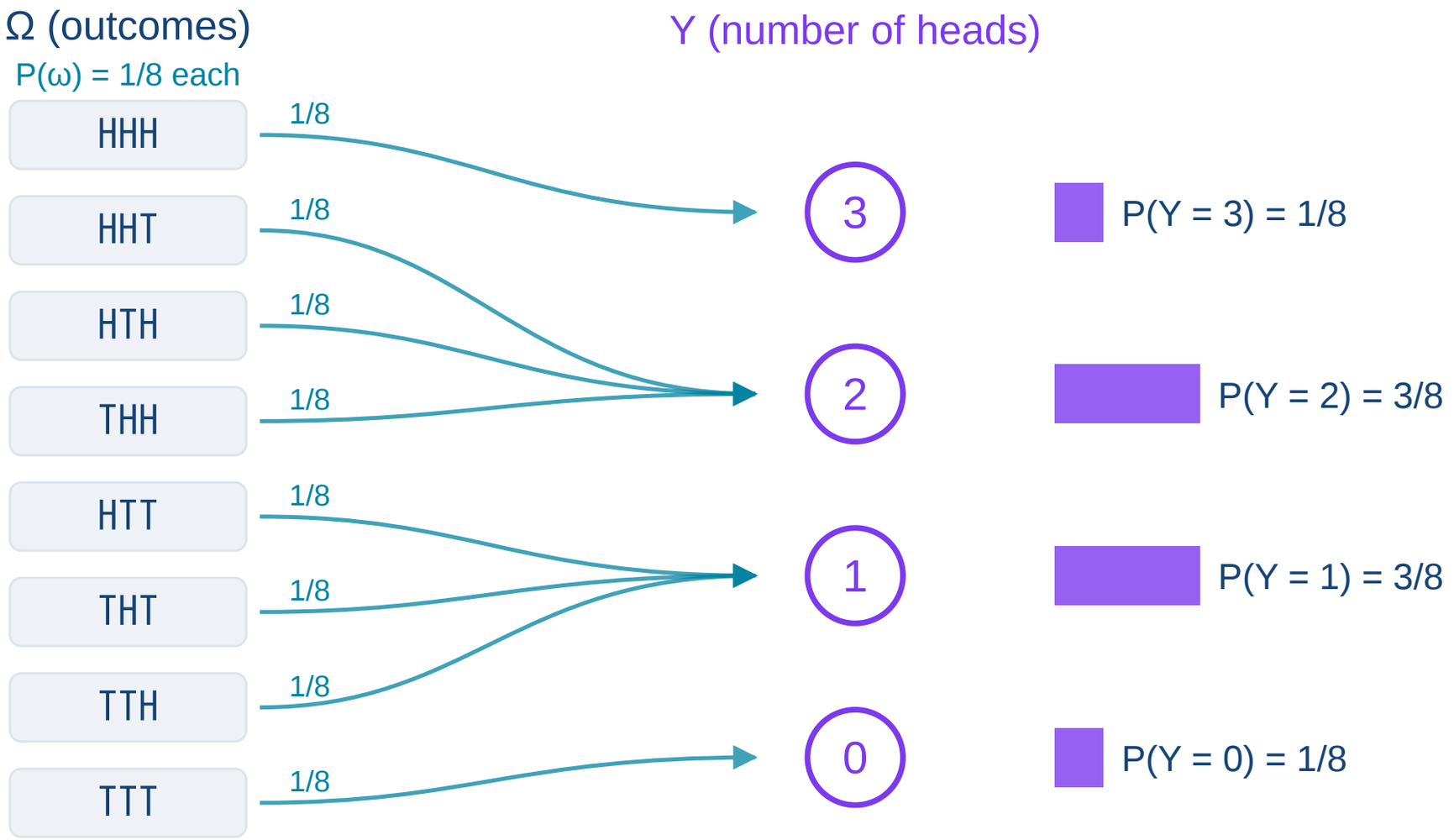
$X$ is a deterministic function: the same outcome $\omega$ always gives the same value $X(\omega)$. The randomness is in which $\omega$ occurs. The notation $P(X \in S)$ abbreviates the probability of the event $X^{-1}(S) = \{\omega : X(\omega) \in S\}$, the preimage of $S$; for a single value, $P(X = x) = P(\{\omega : X(\omega) = x\})$.

"Measurable" means that every such preimage is an event, so that $P$ is defined on it. On a finite $\Omega$ with the power set as event space every function is measurable. The convention is capital letters for random variables and lower case for their values: $P(X = x)$.

The resulting assignment $S \mapsto P(X \in S)$ is itself a probability measure on $E$, the distribution of $X$.

# The Function, Drawn

Three arrows arrive at 1 and at 2: 3/8 each — the whole shape



Speaker notes

The steps show the eight outcomes of three coin tosses; one arrow from each outcome to its number of heads; the four values with the count of arrows arriving at each (1, 3, 3, 1); and the resulting probability mass function $P(Y = k)$.

Every outcome has exactly one outgoing arrow, which is what makes $Y$ a function. A value may receive several arrows (the function need not be injective) or none (it need not be surjective onto $E$; a value with no preimage has probability 0). The values 1 and 2 are three times as probable as 0 and 3 solely because three outcomes map to each of them; no probability is assigned to $Y$ directly, it is inherited from the measure on $\Omega$.

# Examples

| experiment | sample space $\Omega$ | random variable $X(\omega)$ | values |
|---|---|---|---|
| two coins | $\{HH, HT, TH, TT\}$ | number of heads | $0, 1, 2$ |
| two dice | pairs $\langle n_1, n_2 \rangle$ | sum $n_1 + n_2$ | $2, \ldots, 12$ |
| draw a person | a population | height of the person | an interval of $\mathbb{R}$ |

The same outcome can carry many random variables: one per question asked of it.

Speaker notes

Two coins: $X$ takes values 0, 1, 2 with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Two dice: the sum takes values 2 to 12; $P(X = 7) = 6/36$ is the most probable value, $P(X = 2) = P(X = 12) = 1/36$.

People: $\Omega$ is a population and the experiment is drawing one person at random. Height is then a random variable, and its distribution is the distribution of heights in the population. A column of a dataset is read the same way: a function from records to values, whose distribution is that of the attribute in the population the records were drawn from. Several columns on the same records are several random variables on the same $\Omega$, which is what makes a joint distribution of them meaningful.

# Discrete and Continuous

**Discrete random variables**

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

**Continuous random variables**

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — with a mass function or with a density.

Speaker notes

A discrete random variable takes countably many values (finitely many or indexed by $\mathbb{N}$); its distribution is fully described by the probability mass function $p(x) = P(X = x)$, and probabilities of sets are sums. A continuous random variable has a density $f$ with $P(a \le X \le b) = \int_a^b f(x)\,dx$; every single value then has probability 0, and a density value $f(x)$ is not a probability and can exceed 1.

Real-valued range alone is not what makes a variable continuous: it is the existence of a density. Mixed cases occur, e.g. daily rainfall, which is exactly 0 with positive probability and continuous otherwise. Recorded data are always discrete because of finite measurement precision; a continuous model is an idealisation chosen for convenience, as with height.

# A Random Variable With Infinitely Many Values

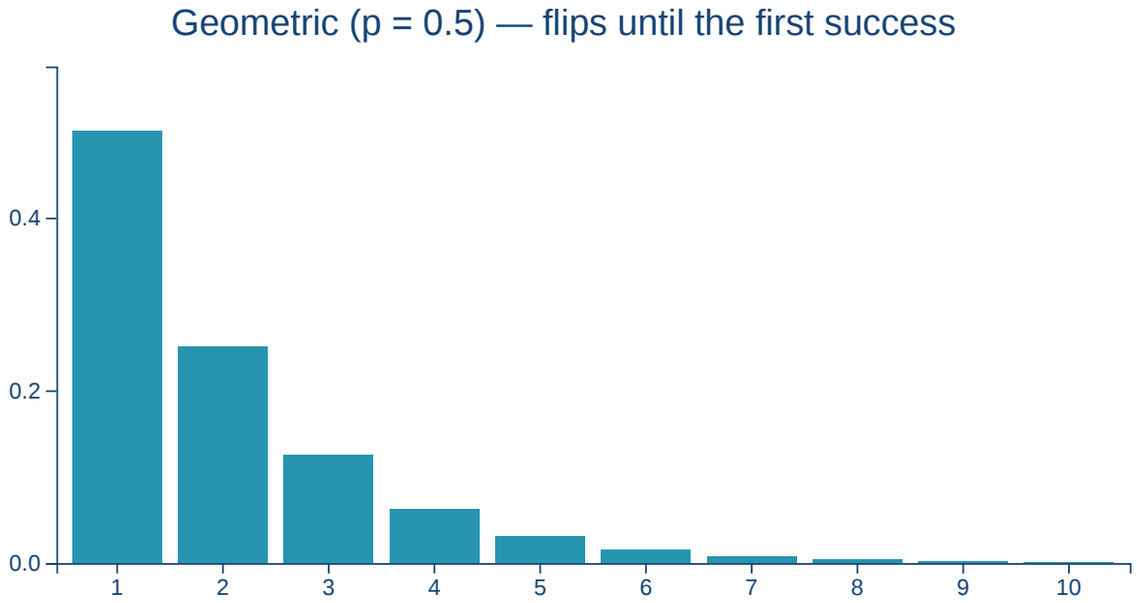
Flip a biased coin ($P(H) = p$) until the first head; $X$ is the number of flips needed.

**Geometric distribution**, $X \sim \mathrm{Geom}(p)$:

$$P(X = 1) = p$$

$$P(X = 2) = (1 - p)p$$

$$P(X = n) = (1 - p)^{n-1}p$$




Countably infinite range, and the probabilities still sum to 1.

Speaker notes

The figure shows the geometric distribution for $p = \frac{1}{2}$ up to $n = 10$. $X = n$ means $n - 1$ tails followed by a head; with independent flips the probability of this sequence is the product $(1 - p)^{n-1}p$.

The probabilities sum to 1 by the geometric series, for $0 < p \leq 1$: $\sum_{n=1}^{\infty}(1 - p)^{n-1}p = p \cdot \frac{1}{1-(1-p)} = 1$. Countable additivity is what allows this infinite sum to be used. The expected number of flips is $E[X] = 1/p$, e.g. 2 for a fair coin. The distribution is memoryless: $P(X > m + n \mid X > m) = P(X > n)$, so past tails do not make a head more likely — the formal reason the gambler's fallacy is a fallacy for independent trials.

# Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

# Probability Mass Function

**Definition (PMF).** For a discrete random variable $X$, the **probability mass function** is

$$p(a) = P(X = a)$$

**INTUITION**

$p(a)$ is how much of the total probability sits exactly on the value $a$.

What follows from the axioms, with nothing new assumed:

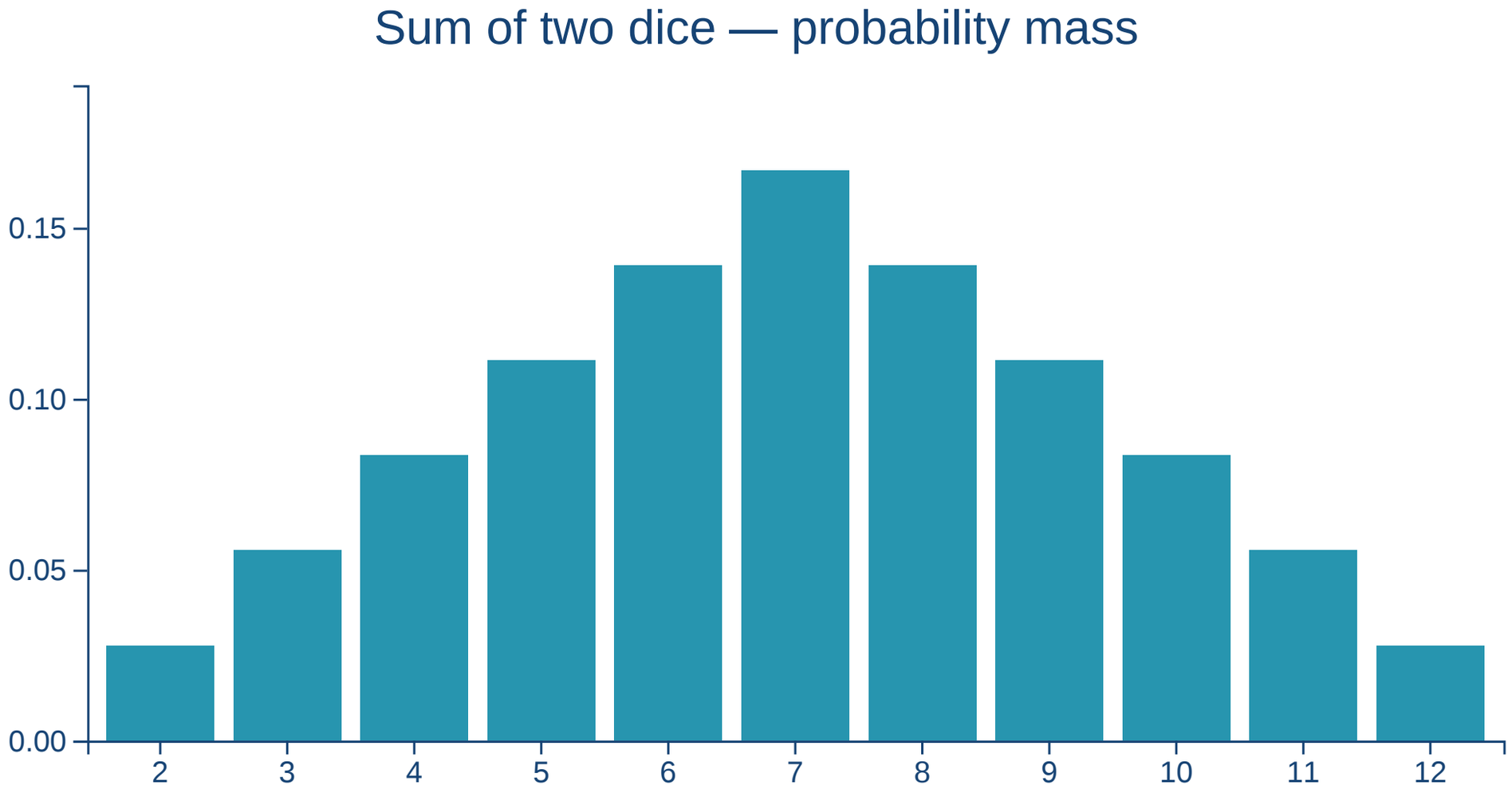
- $p(a) > 0$ for at most countably many values $a$
- $\sum_i p(a_i) = 1$ — the masses add up to the whole
- $p$ defines a probability measure on the values: the **distribution** of $X$

Speaker notes

The PMF is the probability measure transferred from outcomes to values: $p(a) = P(\{\omega : X(\omega) = a\})$. Its properties are consequences of the axioms, not new assumptions. Non-negativity is inherited directly. The events $\{X = a_i\}$ are pairwise disjoint and together cover $\Omega$, so countable additivity gives $\sum_i p(a_i) = 1$.

Strictly, $p$ is a function on values; the probability measure it defines is $P_X(A) = \sum_{a_i \in A} p(a_i)$ for any set $A$ of values, the **distribution** of $X$. A value outside the range of $X$ has $p(a) = 0$. The cumulative distribution function $F(a) = P(X \leq a) = \sum_{a_i \leq a} p(a_i)$ carries the same information as the PMF and is a step function for a discrete variable.

# Example: Sum of Two Dice




The triangular shape is counting, drawn: six of the 36 outcomes give 7, only one gives 2.

Speaker notes

The sample space has 36 equally likely **ordered** pairs $(i, j)$. The number of pairs with sum $s$ is $6 - |s - 7|$ for $s = 2, \ldots, 12$, so $p(s) = (6 - |s - 7|)/36$. The shape is symmetric about 7 because the map $(i, j) \mapsto (7 - i, 7 - j)$ is a bijection that sends sum $s$ to $14 - s$; it peaks at 7 because middling sums can be formed in the most ways.

Treating the pairs as unordered (21 outcomes, e.g. $\{1, 2\}$ counted once) is a frequent error: the unordered outcomes are not equally likely, since $\{1, 2\}$ arises from two ordered pairs and $\{1, 1\}$ from only one.

The sum has $E = 7$ and variance $2 \cdot 35/12 = 35/6 \approx 5.83$. For the sum of $n$ dice the standard deviation grows like $\sqrt{n}$ while the range grows like $n$, so the sum concentrates relative to its range — the first instance of the behaviour described by the law of large numbers and the central limit theorem.

# Expectation

**Definition (Expected value).**

$$E[X] = \sum_i a_i \cdot p(a_i)$$

INTUITION

The **average outcome** if the experiment were repeated infinitely often — the balance point of the bar chart of $p$.

| | |
|---|---|
| function of $X$ | $E[g(X)] = \sum_i g(a_i)p(a_i)$ |
| linear transformation | $E[aX + b] = aE[X] + b$ |
| first moment | $\mu = E[X]$; the first *central* moment $E[X - \mu] = 0$ |

Speaker notes

The frequentist reading is justified by the law of large numbers: the average of $n$ independent repetitions converges to $E[X]$. The expectation exists only if $\sum_i |a_i| p(a_i) < \infty$; for heavy-tailed variables the sum can diverge.

$E[g(X)] = \sum_i g(a_i)p(a_i)$ computes the expectation of $g(X)$ without first deriving the distribution of $g(X)$. Linearity, $E[aX + b] = aE[X] + b$, extends to sums: $E[X + Y] = E[X] + E[Y]$ holds for any two random variables, independent or not.

In general $E[g(X)] \neq g(E[X])$. With $X$ uniform on $\{-1, 1\}$ and $g(x) = x^2$: $E[X^2] = 1$, but $(E[X])^2 = 0$. For convex $g$, Jensen's inequality gives $E[g(X)] \geq g(E[X])$. The first central moment vanishes because $E[X - \mu] = E[X] - \mu = 0$.

# Variance

**Definition (Variance).**

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$$

## Spread

The second central moment: the expected squared distance from the mean. Larger variance, larger typical deviation.

## Scale and unit

$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$

$\sigma = \sqrt{\mathrm{Var}(X)}$, the **standard deviation**, has the unit of $X$.

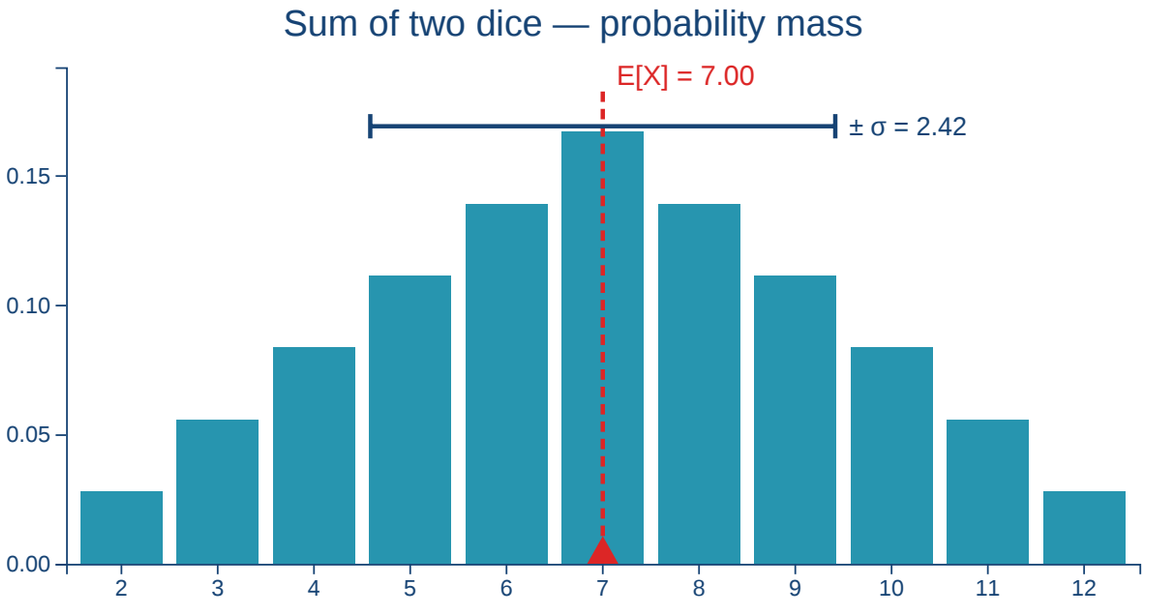
Speaker notes

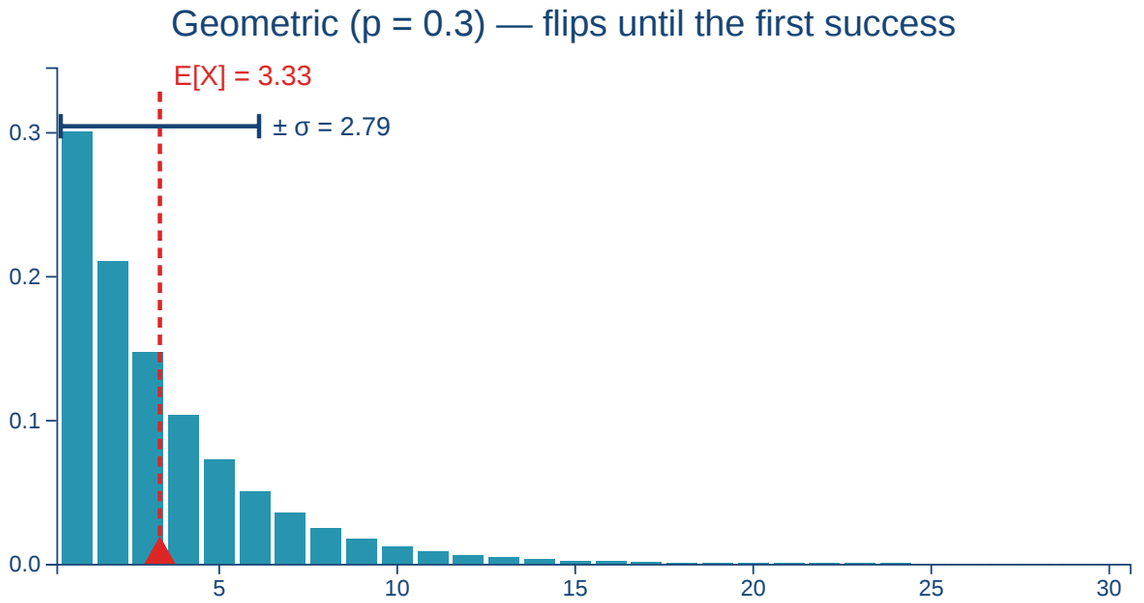
The computational identity follows from linearity: $E[(X-\mu)^2] = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$. Since the variance is non-negative, $E[X^2] \geq (E[X])^2$, with equality exactly when $X$ is constant with probability 1. In floating-point arithmetic the identity can lose precision through cancellation when the mean is large relative to the spread; numerical code therefore subtracts the mean first.

$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ because $(aX+b) - (a\mu+b) = a(X-\mu)$: a shift moves the distribution without changing its spread, and a scaling by $a$ scales $\sigma$ by $|a|$. The standard deviation is reported because it has the unit of $X$; the variance has the squared unit (cm² for heights in cm).

Fair die: $E[X] = 3.5$, $E[X^2] = 91/6$, so $\mathrm{Var}(X) = 91/6 - 12.25 = 35/12 \approx 2.92$ and $\sigma \approx 1.71$.

# Expectation and Variance, Drawn




The red triangle marks $E[X]$: read the bars as weights on a beam, and the beam would rest level on it.

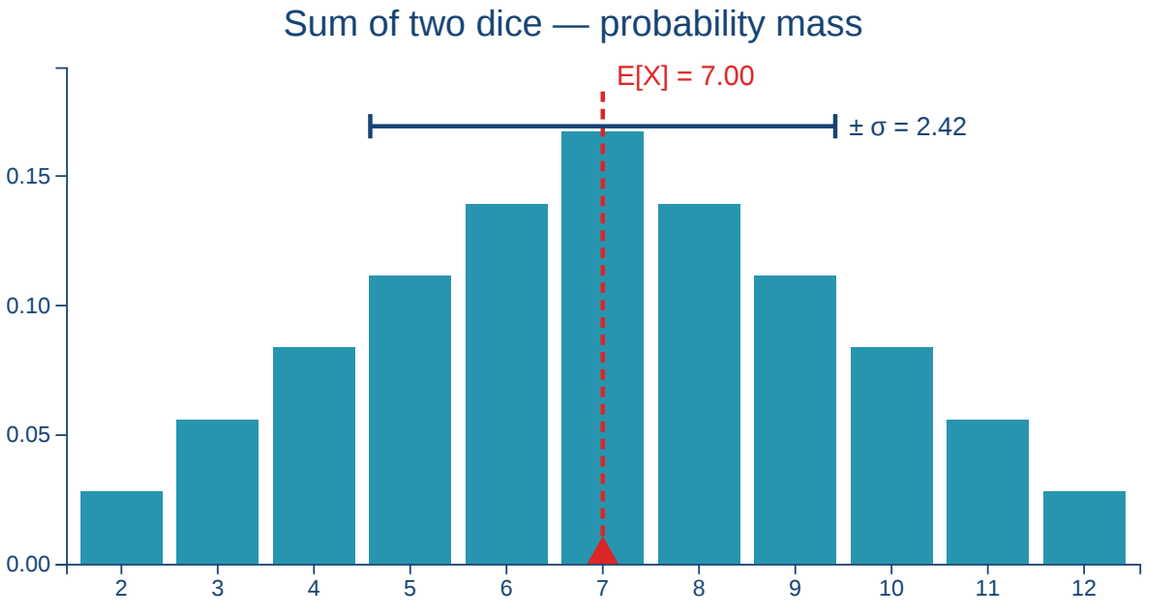
Does $E[X]$ sit under the **highest bar**?

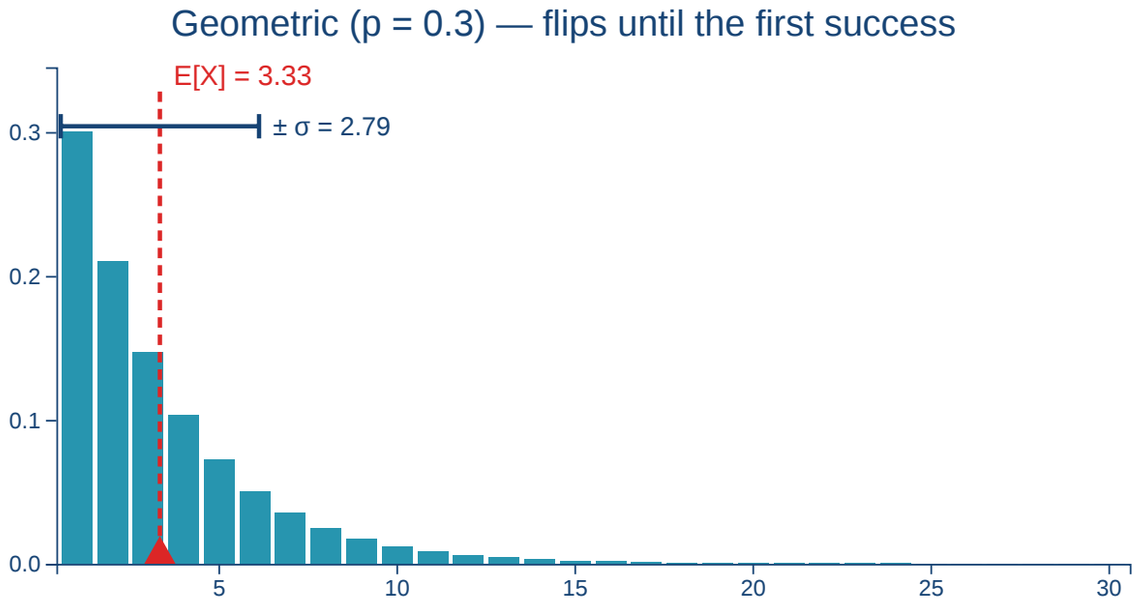
Speaker notes

The red fulcrum marks $E[X]$, the point at which the bars, read as weights $p(a_i)$ placed at positions $a_i$, balance: $\sum_i (a_i - \mu)\, p(a_i) = 0$. The bracket spans $\mu \pm \sigma$. Sum of two dice: $E[X] = 7$, $\mathrm{Var}(X) = 35/6$, $\sigma \approx 2.42$; the distribution is symmetric about 7, so expectation, peak (mode) and median coincide. Geometric with $p = 0.3$ (number of trials up to the first success): $E[X] = 1/p \approx 3.33$, $\mathrm{Var}(X) = (1-p)/p^2 \approx 7.78$, $\sigma \approx 2.79$, mode 1, median 2. The figure truncates the support at 30, which carries probability $1 - 0.7^{30} > 0.9999$.

For a right-skewed distribution the order is typically mode < median < mean. Far values pull the expectation in proportion to their distance, the variance in proportion to their squared distance. The next slide turns the asymmetry itself into a number.

# Expectation and Variance, Drawn




The red triangle marks $E[X]$: read the bars as weights on a beam, and the beam would rest level on it.

Does $E[X]$ sit under the **highest bar**?

Only for the symmetric one. The geometric's highest bar is at 1, but $E[X] = 1/p \approx 3.33$: its **long right tail** pulls the expectation away from the peak, and widens $\sigma$.

# Skewness

**Definition (Skewness).** With $\mu = E[X]$ and $\sigma^2 = \mathrm{Var}(X)$,

$$\mathrm{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{E\big[(X-\mu)^3\big]}{\sigma^3}$$

Cubing **keeps the sign** and lets far values dominate: a long right tail gives $\mathrm{Skew} > 0$, a long left tail $\mathrm{Skew} < 0$.

| distribution | skewness | shape |
|---|---|---|
| sum of two dice | $0$ | symmetric about 7 |
| geometric, $p = 0.3$ | $\frac{2-p}{\sqrt{1-p}} \approx 2.03$ | long right tail |

Speaker notes

Skewness is the standardised third central moment; the third moment $E[(X-\mu)^3]$ alone has the cubed unit of $X$, and dividing by $\sigma^3$ makes it unit-free. It is invariant under $aX+b$ with $a>0$ and changes sign for $a<0$: $\mathrm{Skew}(-X)=-\mathrm{Skew}(X)$. A distribution symmetric about $\mu$ has skewness 0, because every $(a-\mu)^3$ is cancelled by its mirror image; the converse fails, and asymmetric distributions with skewness 0 exist. Skewness exists only if $E[|X|^3]<\infty$.

Values: for the geometric distribution $\mathrm{Skew}=(2-p)/\sqrt{1-p}=1.7/\sqrt{0.7}\approx 2.03$, positive for every $p<1$. For the Bernoulli distribution $\mathrm{Skew}=(1-2p)/\sqrt{p(1-p)}$: $p=0.9$ gives $-0.8/0.3\approx -2.67$, $p=0.5$ gives 0. The binomial has $(1-2p)/\sqrt{np(1-p)}$, which tends to 0 as $n$ grows, one sign of its approach to the normal distribution. The fourth standardised moment, the kurtosis $E[((X-\mu)/\sigma)^4]$, measures the weight of the tails and is 3 for every normal distribution. The sample version of the skewness is in Lecture 02.

# Takeaway

A discrete distribution is a PMF; expectation and variance compress it into **where it sits** and **how much it spreads**.

# Bernoulli and Binomial

| | models | PMF | $E[X]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|---|
| **Bernoulli**$(p)$ | one trial: $X = 1$ success, $X = 0$ failure | $p(x) = p^x(1-p)^{1-x}$ | $p$ | $p(1-p)$ |
| **Binomial** $B(n, p)$ | successes in $n$ independent trials | $p(k) = \binom{n}{k} p^k (1-p)^{n-k}$ | $np$ | $np(1-p)$ |

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the sequences with $k$ successes; the PMF multiplies it by the probability of one such sequence — combinatorics and independence in a single line.

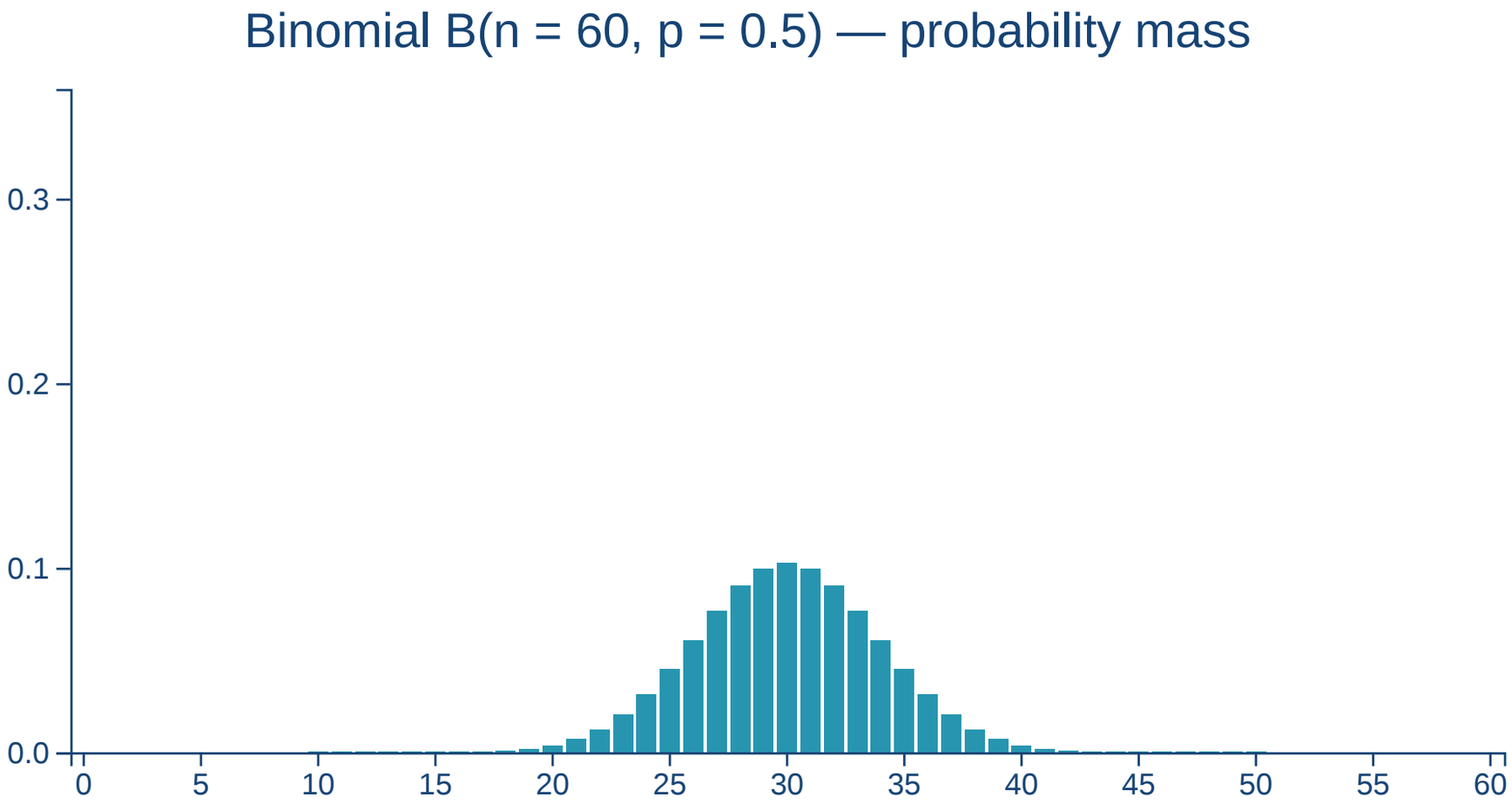
Speaker notes

Bernoulli moments: $E[X] = 1 \cdot p + 0 \cdot (1-p) = p$; since $X^2 = X$, $E[X^2] = p$ and $\operatorname{Var}(X) = p - p^2 = p(1-p)$, which is largest ($\frac{1}{4}$) at $p = \frac{1}{2}$ and 0 at $p \in \{0, 1\}$.

The binomial count is a sum of $n$ independent Bernoulli$(p)$ variables. A particular sequence with $k$ successes has probability $p^k(1-p)^{n-k}$ by independence, and $\binom{n}{k}$ such sequences exist. The PMF sums to 1 by the binomial theorem, $\sum_k \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n$. The mean $np$ follows from linearity alone; the variance $np(1-p)$ uses independence, because only for uncorrelated summands do the variances add.

The model requires independent trials with the same success probability $p$. If $p$ varies between trials or the trials influence one another, the count is not binomial.

# How the Shape Depends on the Parameters




**Predict before each click:** as $n$ grows, does the distribution get wider or narrower — and what happens to its shape?

Speaker notes

With $p = 0.5$ the mean is $n/2$ and the standard deviation $\sqrt{np(1-p)} = \sqrt{n}/2$. For the five values of $n$ shown: $n = 5$: $\sigma \approx 1.12$; $n = 10$: $1.58$; $n = 20$: $2.24$; $n = 40$: $3.16$; $n = 60$: $3.87$. Each step shows the next value of $n$.

Two effects occur together. In absolute terms the distribution gets wider, growing like $\sqrt{n}$. Relative to $n$ it gets narrower: the proportion of successes $k/n$ has standard deviation $\sqrt{p(1-p)/n}$, falling from $0.22$ at $n = 5$ to $0.065$ at $n = 60$. The shape also becomes bell-like, because a binomial is a sum of independent Bernoulli variables and the central limit theorem applies. For $p = 0.5$ the PMF is symmetric for every $n$; for other $p$ it is skewed at small $n$ and becomes symmetric only as $n$ grows.

# Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

Speaker notes

Model: $X \sim B(10, 0.25)$, so $E[X] = 2.5$ and $\mathrm{Var}(X) = 10 \cdot 0.25 \cdot 0.75 = 1.875$, $\sigma \approx 1.37$. Five correct answers lie about $1.8\sigma$ above the mean. The exact tail is $P(X \geq 5) = \sum_{k=5}^{10} \binom{10}{k} 0.25^k 0.75^{10-k} \approx 0.078$, a sum of six PMF values (or $1 - P(X \leq 4)$).

The binomial model assumes that the ten guesses are independent and each succeeds with the same probability $0.25$. It fails for a real exam taker who partly knows the material (the success probability then differs between questions), who can rule out some options (raising $p$ above $0.25$), or when questions build on each other (dependence).

# Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

$B(10, 0.25)$: on average $10 \cdot 0.25 = 2.5$ correct. Five or more is twice the expectation — unlikely (about 8%).

# Takeaway

A yes/no attribute is a **Bernoulli** variable with one parameter $p$; counting the yeses in $n$ independent trials gives a **binomial**. Both are counting in probabilistic clothing — and $p$ is the number data will have to tell us.

# From Mass to Density

**Definition (PDF).** A continuous random variable $X$ has a **probability density function** $f$ with

$$P(b \le X \le c) = \int_b^c f(a)\,da, \qquad \int_{-\infty}^{\infty} f(a)\,da = 1$$

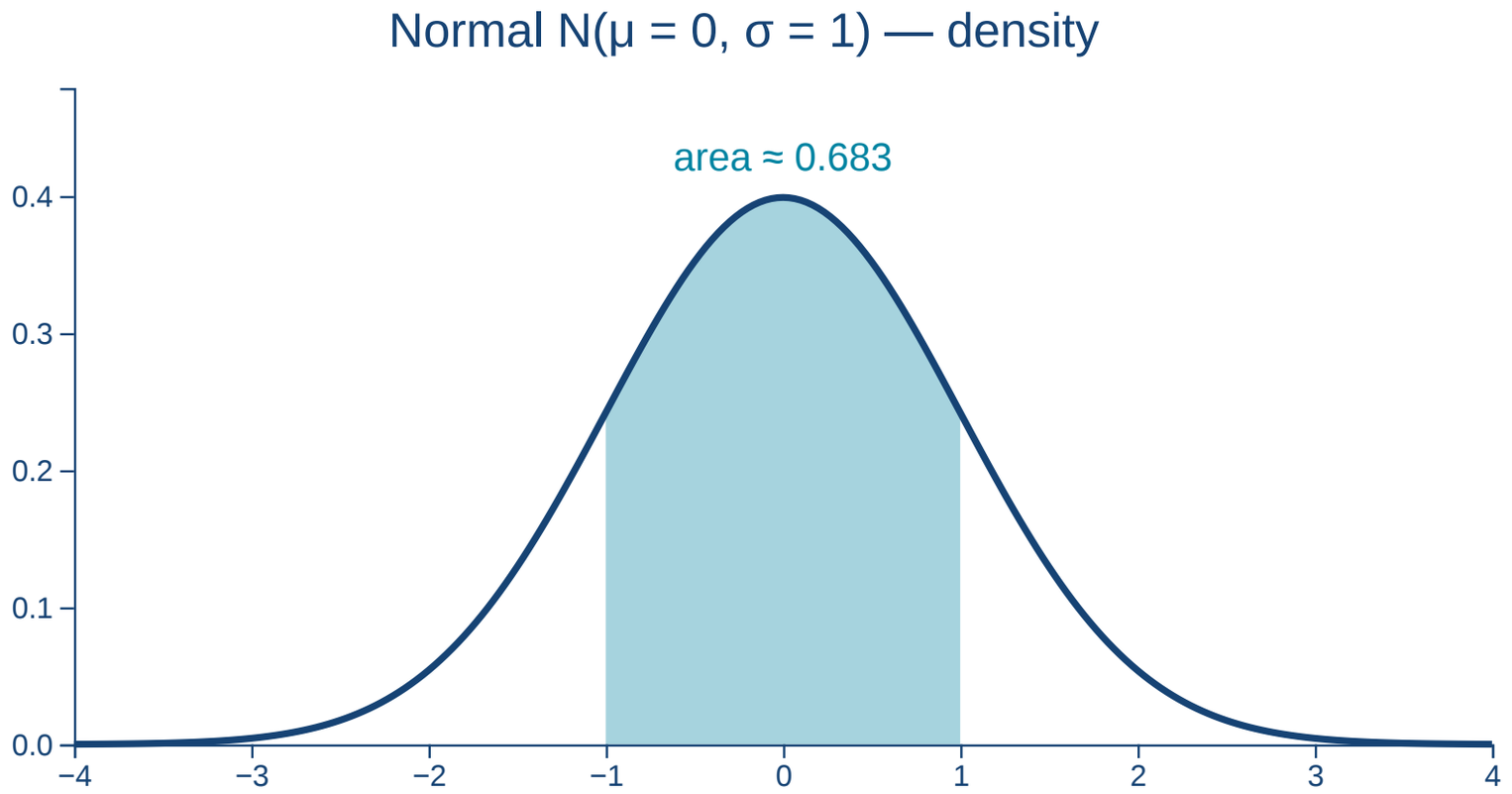
$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value $a$.

Speaker notes

For small $h$, $P(a \le X \le a+h) \approx f(a)h$, so $f$ has the unit "probability per unit of $X$". A density can therefore exceed 1: the uniform distribution on $[0, 0.5]$ has $f = 2$ on that interval. The probability of a single value is an integral over an interval of length zero, so $P(X = a) = 0$; a person being exactly 180.000… cm tall has probability zero. As a consequence, $\le$ and $<$ give the same interval probabilities. Probability zero does not mean impossible: some value is always observed.

The cumulative distribution function $F(a) = P(X \le a) = \int_{-\infty}^{a} f(t)dt$ links the two views; where $f$ is continuous, $f = F'$. Not every random variable is either discrete or continuous — daily rainfall, for instance, has a point mass at 0 and a density above it.

# Probability Is Area Under the Curve




**One value: probability 0.** $P(X = a) = \int_a^a f = 0$ for every real $a$ — a line has no area. The curve's height $f(a)$ is not a probability.

At 0 the curve is 0.40 high, yet $P(X = 0) = 0$.

The shaded region is $P(-1 \le X \le 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Speaker notes

In terms of the CDF, $P(b \le X \le c) = F(c) - F(b)$. For the standard normal the shaded area is $P(-1 \le X \le 1) \approx 0.683$; within two standard deviations it is $\approx 0.954$, within three $\approx 0.997$. The normal CDF has no closed form and is computed numerically (via the error function or tables).

The total area under any density is 1. This is what fixes the constant in the normal density: $\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)} dx = \sigma\sqrt{2\pi}$ (the Gaussian integral), so the factor $1/(\sigma\sqrt{2\pi})$ is exactly the normalisation.

# Expectation for Continuous Variables

| | discrete — mass $p$ | continuous — density $f$ |
|---|---|---|
| total | $\sum_i p(a_i) = 1$ | $\int_{-\infty}^{\infty} f(a)\,da = 1$ |
| expectation | $E[X] = \sum_i a_i p(a_i)$ | $E[X] = \int_{-\infty}^{\infty} a f(a)\,da$ |
| function of $X$ | $E[g(X)] = \sum_i g(a_i) p(a_i)$ | $E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a)\,da$ |

Only the sum turns into an integral. The rules are unchanged: $E[aX + b] = aE[X] + b$ and $\mathrm{Var}(X) = E[(X - \mu)^2]$.

Speaker notes

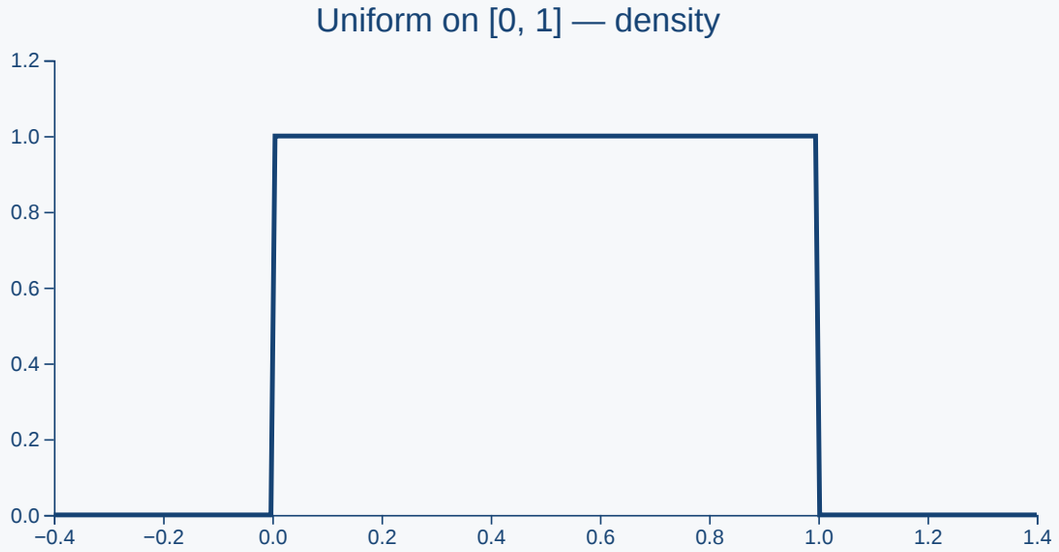
Only the summation becomes an integral; linearity of expectation, the variance identity $\operatorname{Var}(X) = E[X^2] - (E[X])^2$, and $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$ carry over with the same proofs. In practice the integral runs over the support of $f$, the region where $f > 0$.

Existence is again a condition: the expectation is defined only if $\int |a| f(a)\, da < \infty$. The Cauchy distribution, $f(x) = 1/(\pi(1 + x^2))$, is a standard example of a density without a mean (and without a variance), even though it is symmetric about 0.

# Two Basic Continuous Distributions

## Uniform on $[a, b]$

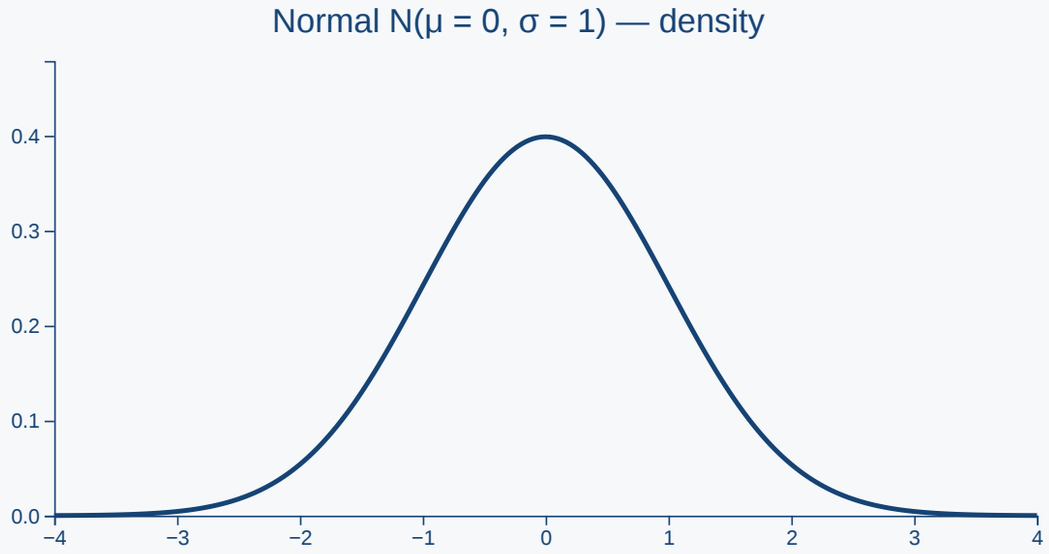
$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$




Equal-length intervals are equally likely — the Laplace assumption, continuous.

## Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



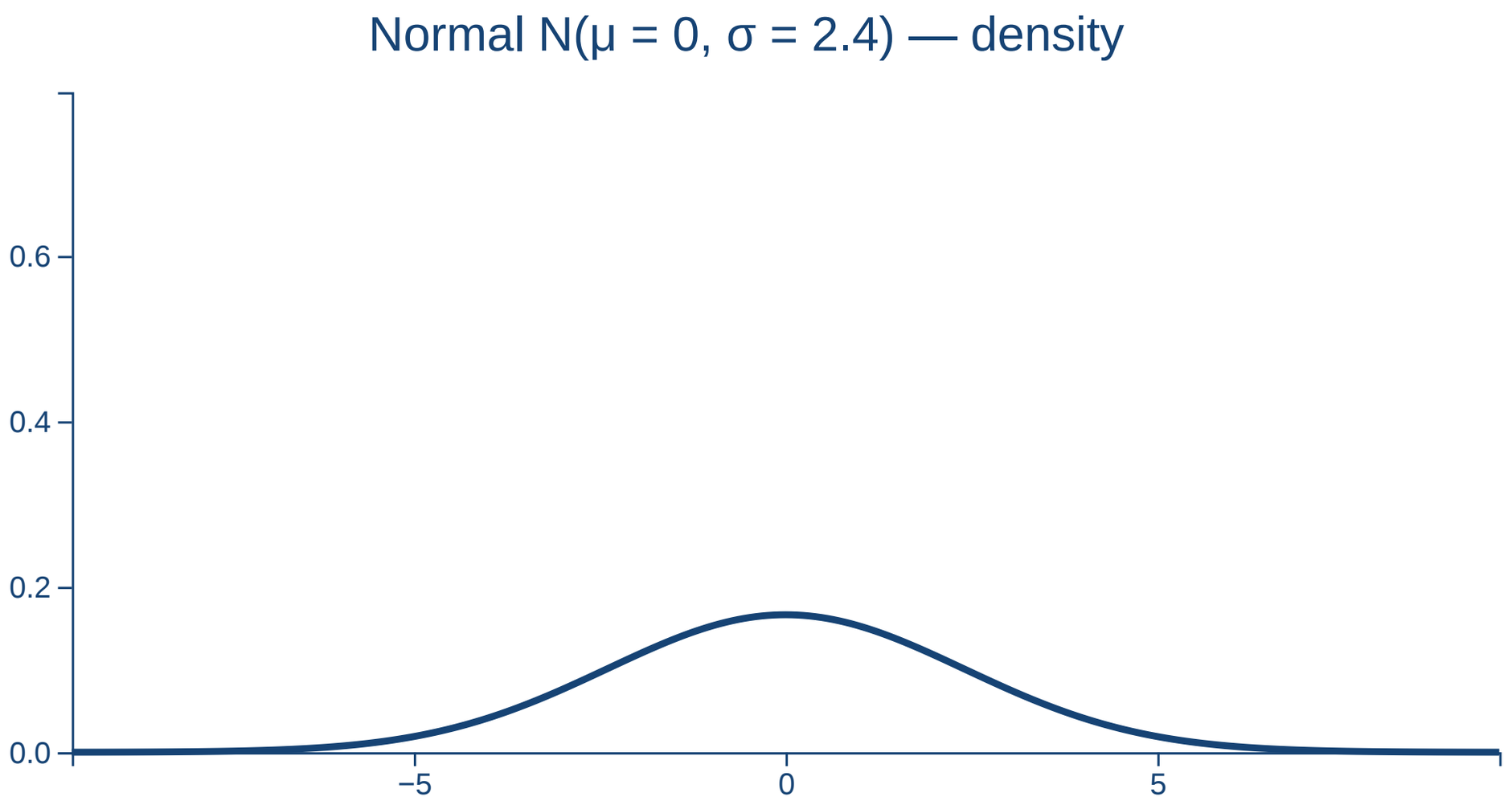

$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

Speaker notes

The figures show $[a, b] = [0, 1]$ and $N(0, 1)$. The uniform density is $1/(b - a)$ on $[a, b]$ and 0 outside, with $E[X] = (a + b)/2$ and $\mathrm{Var}(X) = (b - a)^2/12$. It is the continuous counterpart of the Laplace assumption: equal-length subintervals receive equal probability.

The normal density is symmetric about $\mu$, has its maximum $1/(\sigma\sqrt{2\pi})$ at $\mu$ and its inflection points at $\mu \pm \sigma$. Linear transformations preserve normality: if $X \sim N(\mu, \sigma^2)$ then $aX + b \sim N(a\mu + b, a^2\sigma^2)$. In particular the standardisation $Z = (X - \mu)/\sigma$ is $N(0, 1)$, which is why a single table of the standard normal suffices. The notation $N(\mu, \sigma^2)$ uses the variance, not the standard deviation, as second parameter.

# What σ Does to the Shape




The peak falls as the curve widens — the area is always 1.

Speaker notes

The peak height of $N(\mu, \sigma^2)$ is $1/(\sigma\sqrt{2\pi})$, inversely proportional to $\sigma$. For the four values shown: $\sigma = 0.6$ gives $\approx 0.665$, $\sigma = 1$ gives $\approx 0.399$, $\sigma = 1.6$ gives $\approx 0.249$, $\sigma = 2.4$ gives $\approx 0.166$. Widening by a factor $k$ lowers the peak by the same factor, so the area stays 1.

Independently of $\sigma$, about 68% of the mass lies within $\mu \pm \sigma$: changing $\sigma$ only rescales the horizontal axis. For $\sigma < 1/\sqrt{2\pi} \approx 0.40$ the peak exceeds 1, another instance of a density larger than 1.

# Why the Normal Is Everywhere

> The **central limit theorem**: under mild conditions, the average of $n$ independent observations is approximately normal for large $n$ — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown — and why a sum of many independent coin flips, the binomial, looks bell-shaped.

Speaker notes

Precise statement: for independent, identically distributed $X_1, \ldots, X_n$ with mean $\mu$ and finite variance $\sigma^2$, the standardised mean $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges in distribution to $N(0, 1)$. The "mild conditions" are independence and finite variance; versions exist that relax identical distribution. Without finite variance the theorem fails: the mean of independent Cauchy variables is again Cauchy, however large $n$.

The mean $\bar{X}_n$ has variance $\sigma^2/n$, so its standard deviation, the **standard error**, is $\sigma/\sqrt{n}$. This is the basis of error bars and confidence intervals computed from a sample.

A binomial $\mathrm{Bin}(n, p)$ is a sum of $n$ independent Bernoulli variables and is therefore approximately $N(np, np(1-p))$ for large $n$. The approximation is poor when $p$ is close to 0 or 1; rules of thumb require both $np$ and $n(1-p)$ to be reasonably large (commonly at least 5 or 10).

# Other Distributions You Will Meet

| distribution | values | typically models |
|---|---|---|
| **Exponential** | $[0, \infty)$ | time until an event occurs (e.g. the next earthquake) |
| **Gamma** | $[0, \infty)$ | time until $n$ events have occurred |
| **Beta** | $[0, 1]$ | an unknown probability |
| **Dirichlet** | probability vectors | distributions *over* probabilities |

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

Speaker notes

The exponential distribution with rate $\lambda$ has density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and mean $1/\lambda$. It is memoryless, $P(X > s + t \mid X > s) = P(X > t)$, which encodes a constant event rate. The gamma distribution with integer shape $n$ and rate $\lambda$ is the distribution of a sum of $n$ independent exponential waiting times, with mean $n/\lambda$.

The Beta$(\alpha, \beta)$ distribution lives on $[0, 1]$ with mean $\alpha/(\alpha + \beta)$; $\mathrm{Beta}(1, 1)$ is the uniform distribution. It is the standard prior for an unknown probability, because a Beta prior combined with Bernoulli or binomial data yields a Beta posterior. The Dirichlet distribution generalises the Beta to vectors of probabilities that sum to 1 and plays the same role for categorical data. None of these four is needed for the rest of this unit.

# A Table or a Formula?

How many numbers does it take to write down each distribution completely?

Speaker notes

A table lists one probability per value, so a variable with $k$ values needs $k-1$ numbers (the last follows because they sum to 1). A formula computes every probability from a few parameters: the binomial $P(X=k)=\binom{n}{k}\theta^k(1-\theta)^{n-k}$ fixes all eleven probabilities through $\theta$ alone, and the normal density fixes a whole continuum through $\mu$ and $\sigma$. The saving exists only when the formula has fewer parameters than the table has cells. For the loaded die there is no such formula, and the table is itself the parametric family "categorical with 6 values", with 5 parameters.

The notation $p(x \mid \theta)$ reads "the probability of $x$ for the parameter value $\theta$". Once $\theta$ is treated as unknown and given a distribution, the bar becomes an ordinary conditional — the step taken in Lecture 03. A model that does not fix the number of parameters in advance (a histogram whose bins grow with the data, a kernel density estimate) is called **non-parametric**.

# A Table or a Formula?

How many numbers does it take to write down each distribution completely?

| random variable | as a table | as a formula |
|---|---|---|
| heads in 10 flips, $0, \ldots, 10$ | 10 numbers | binomial$(10, \theta)$: **1** number |
| height in cm, 20 bins of 2.5 cm | 19 numbers | normal$(\mu, \sigma^2)$: **2** numbers |
| a loaded die | 5 numbers | no shorter formula: still 5 |

# A Table or a Formula?

How many numbers does it take to write down each distribution completely?

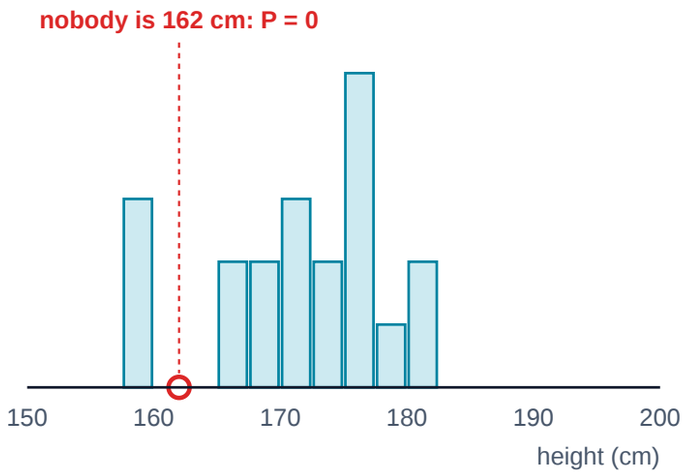
| random variable | as a table | as a formula |
|---|---|---|
| heads in 10 flips, $0, \ldots, 10$ | 10 numbers | binomial$(10, \theta)$: **1** number |
| height in cm, 20 bins of 2.5 cm | 19 numbers | normal$(\mu, \sigma^2)$: **2** numbers |
| a loaded die | 5 numbers | no shorter formula: still 5 |

A **parametric family** is one formula $p(x \mid \theta)$ for a whole set of distributions. The **parameter** $\theta$ picks one member: $\theta$ for the binomial, $(\mu, \sigma)$ for the normal.

# What the Formula Buys — and What It Costs

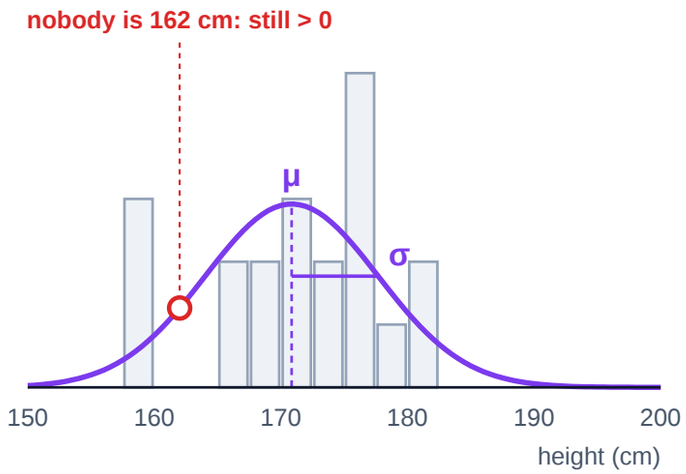
**A table: one number per bin**

20 bins → 19 free numbers




**A formula: the normal family**

μ and σ → 2 numbers




- ✓ **Fewer numbers to estimate:** 20 heights for 2 numbers, not for 19.
- ✓ **Unseen values get probability:** the curve fills the gaps.

Speaker notes

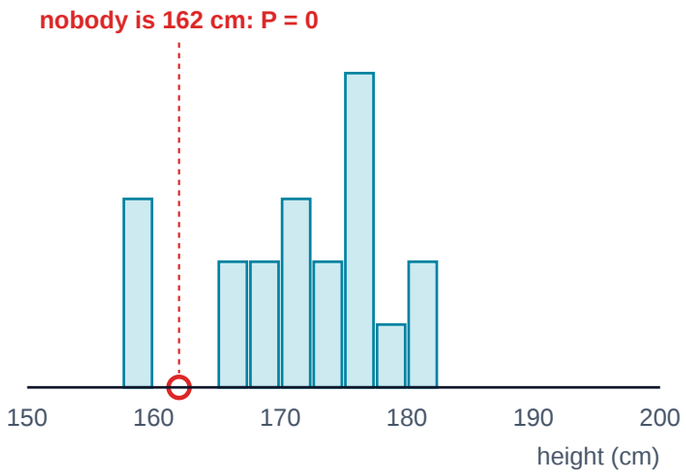
Twenty heights, drawn from a normal distribution with mean 172 cm and standard deviation 8 cm. The table needs one number per bin; with 20 observations spread over 19 free numbers, most bins hold zero or one person, and an empty bin — no one is 162 cm tall — gets probability 0, although that height is clearly possible. The normal family estimates only $\mu$ and $\sigma$, from the same 20 values, and assigns every height a positive density. Each parameter is estimated from all observations, so the estimates are far less noisy.

The price is an assumption about the shape. With enough data a histogram approaches any distribution; a normal curve stays a normal curve, so if the true distribution is skewed or has two peaks, the fitted model is wrong however many observations are added. This is the systematic-error pattern again: more data reduces the noise of the estimate, not the error of the assumption. The trade-off between few parameters (stable, possibly wrong) and many parameters (flexible, data-hungry) returns as the bias–variance trade-off.

# What the Formula Buys — and What It Costs

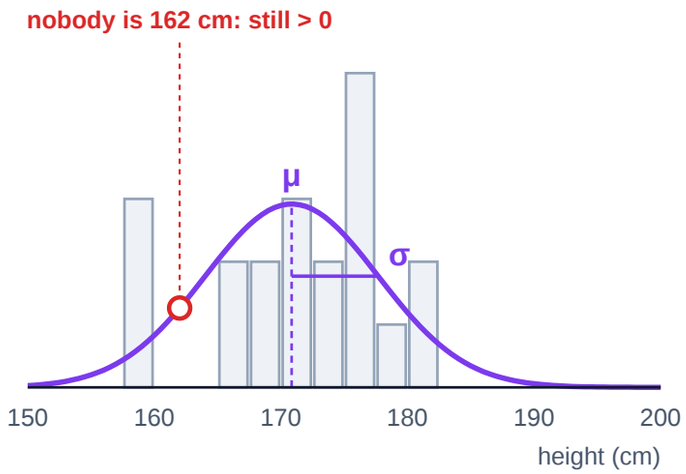
**A table: one number per bin**

20 bins → 19 free numbers




**A formula: the normal family**

μ and σ → 2 numbers




✓ **Fewer numbers to estimate:** 20 heights for 2 numbers, not for 19.

✓ **Unseen values get probability:** the curve fills the gaps.

✗ **The shape is an assumption.** If the family is wrong, more data does not fix it.

# Mini-Exercise

A class of **children and their parents** is measured, and a normal distribution is fitted to all heights.

Where does the fitted curve have its peak, and how many people stand there?

Speaker notes

With children around 130 cm and parents around 172 cm, the fitted mean lies near 150 cm, in the gap between the groups, and the fitted standard deviation is large enough to cover both. More measurements make $\hat{\mu}$ and $\hat{\sigma}$ more precise but leave the peak in the gap: the family, not the sample, is the problem. A mixture of two normals, $\pi\mathcal{N}(\mu_1, \sigma_1^2) + (1-\pi)\mathcal{N}(\mu_2, \sigma_2^2)$, has five parameters and fits; mixture models and how to fit them are treated later in the course.

# Mini-Exercise

A class of **children and their parents** is measured, and a normal distribution is fitted to all heights.

Where does the fitted curve have its peak, and how many people stand there?

> Between the two groups, where **almost nobody** is. The data has two peaks; the normal family has only one. A mixture of two normals fits.

# Takeaway

A **parametric family** describes a distribution by a formula with a few parameters instead of a table with one number per value. It needs far less data and gives unseen values probability — but its **shape is an assumption** that no amount of data corrects.

# Joint Distributions

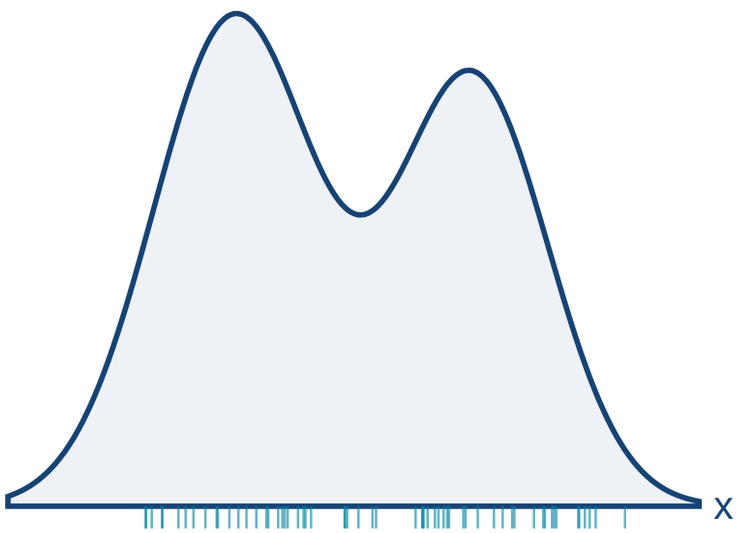
> ***Guiding Question:*** *Never one quantity alone. How do we describe several random variables at once — the joint the three moves start from?*

Speaker notes

For several random variables on the same probability space, the joint distribution $P(X, Y)$ assigns probabilities to combinations of values: the count table of the introductory example, generalised. Marginals follow from the joint by the sum rule and conditionals by the product rule, but the joint cannot in general be recovered from the marginals — that is the reason modelling starts from the joint.

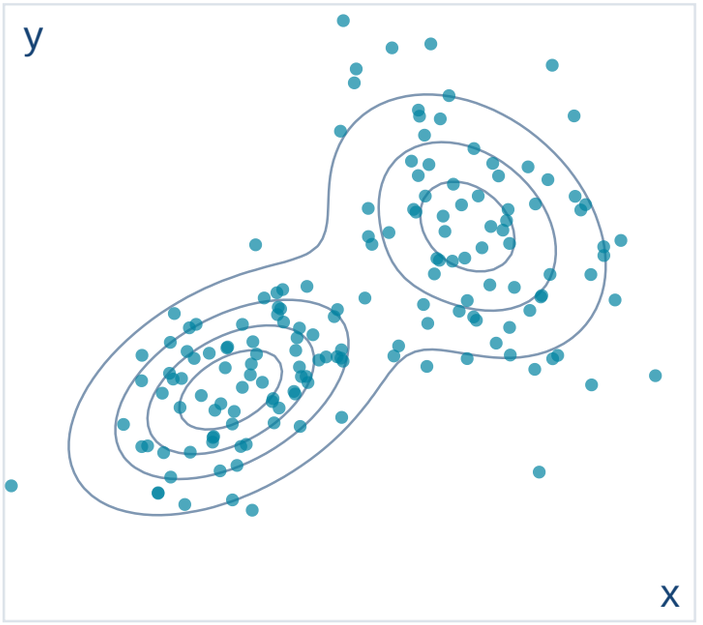
# More Than One Random Variable

## Univariate



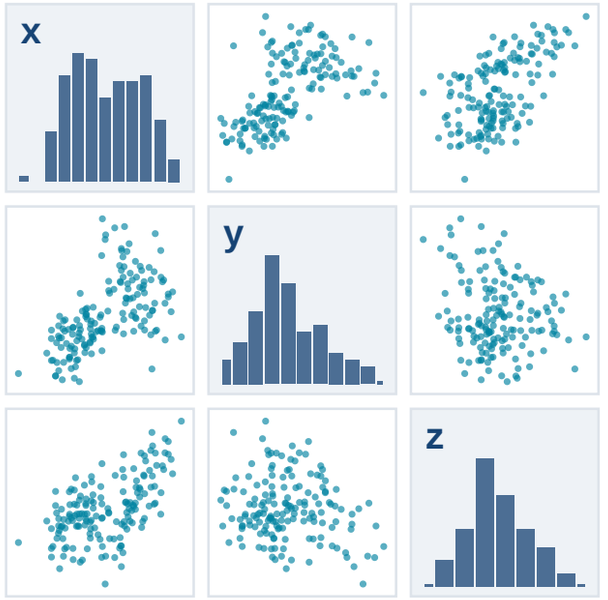

p(x) — one column

## Bivariate




p(x, y) — two columns

## Multivariate




p(x, y, z, …) — three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Speaker notes

The figure shows one variable as a density curve, two as a point cloud with the contour lines of their joint density, and three as a scatter matrix: every pair of columns as a cloud, every single column as a histogram on the diagonal. A scatter matrix shows only the two-dimensional projections; the joint of three or more variables can no longer be drawn directly.

A table with columns age, income and churn is a sample from a trivariate joint distribution: each row is one realisation of the vector $(X_{\text{age}}, X_{\text{income}}, X_{\text{churn}})$, and the joint distribution assigns probability to combinations of values.

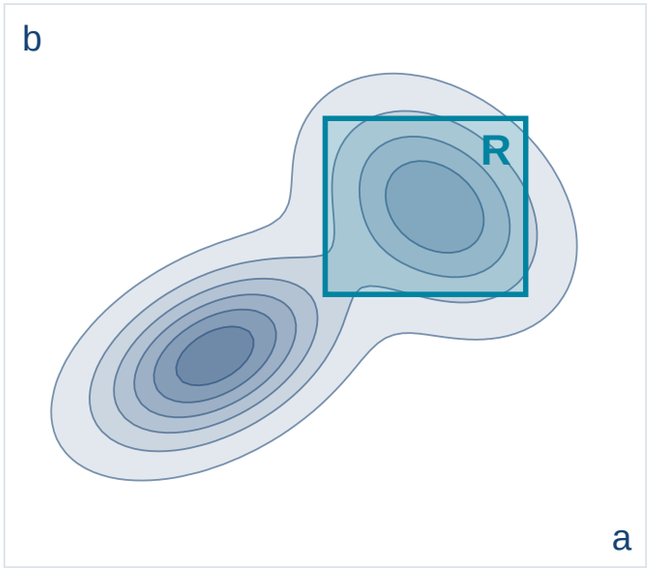
The joint determines every marginal and every conditional distribution. The converse fails: the marginals alone do not determine the joint, except under independence, where the joint is the product of the marginals. The size of a joint grows exponentially: $d$ binary variables need $2^d - 1$ free probabilities ($d = 30$ already gives more than $10^9$). Any practical model therefore imposes structure, such as independence or conditional independence assumptions, to reduce this number.

# Joint Mass and Density

| joint PMF (discrete) | joint PDF (continuous) |
|---|---|
| $p_{X,Y}(a,b) = P(X=a, Y=b)$ | $P((X,Y)\in R) = \iint_R f_{X,Y}(a,b)\,da\,db$ |

| | G=0 | G=1 | G=2 | G=3 |
|---|---|---|---|---|
| B=0 | .15 | .1 | .0875 | .0375 |
| B=1 | .1 | .175 | .1125 | 0 |
| B=2 | .0875 | .1125 | 0 | 0 |
| B=3 | .0375 | 0 | 0 | 0 |

one probability per cell · all sum to 1




darker = denser · volume over R ≈ 0.33

**From here on** we write $p(a, b, c, \ldots)$ for the joint of $X_1, X_2, X_3, \ldots$ — a mass or a density, as the variables require.

Speaker notes

Left: the joint PMF of the number of boys and girls in a family (the example that follows), one probability per cell. Right: a joint density drawn as contour lines; the probability of the region $R$ is the volume under the density surface above $R$, here computed numerically.

Both joint forms are normalised: $\sum_a \sum_b p_{X,Y}(a,b) = 1$ and $\iint f_{X,Y}(a,b)\,da\,db = 1$. A joint may also mix the two kinds, e.g. a class label and a measurement; then one sums over the discrete and integrates over the continuous variable.

The convention $p(a,b,c,\ldots)$ is safe because marginalisation, conditioning and factorisation follow the same rules in both cases; only summing versus integrating differs. Two caveats apply. For continuous variables $p$ is a density, so its values are not probabilities and can exceed 1; reading $p(a,b,c)$ as "the probability of state $a,b,c$" is then shorthand for a density value. And $p$ is overloaded: $p(a)$ and $p(b)$ denote different functions, distinguished only by which variable the argument refers to.

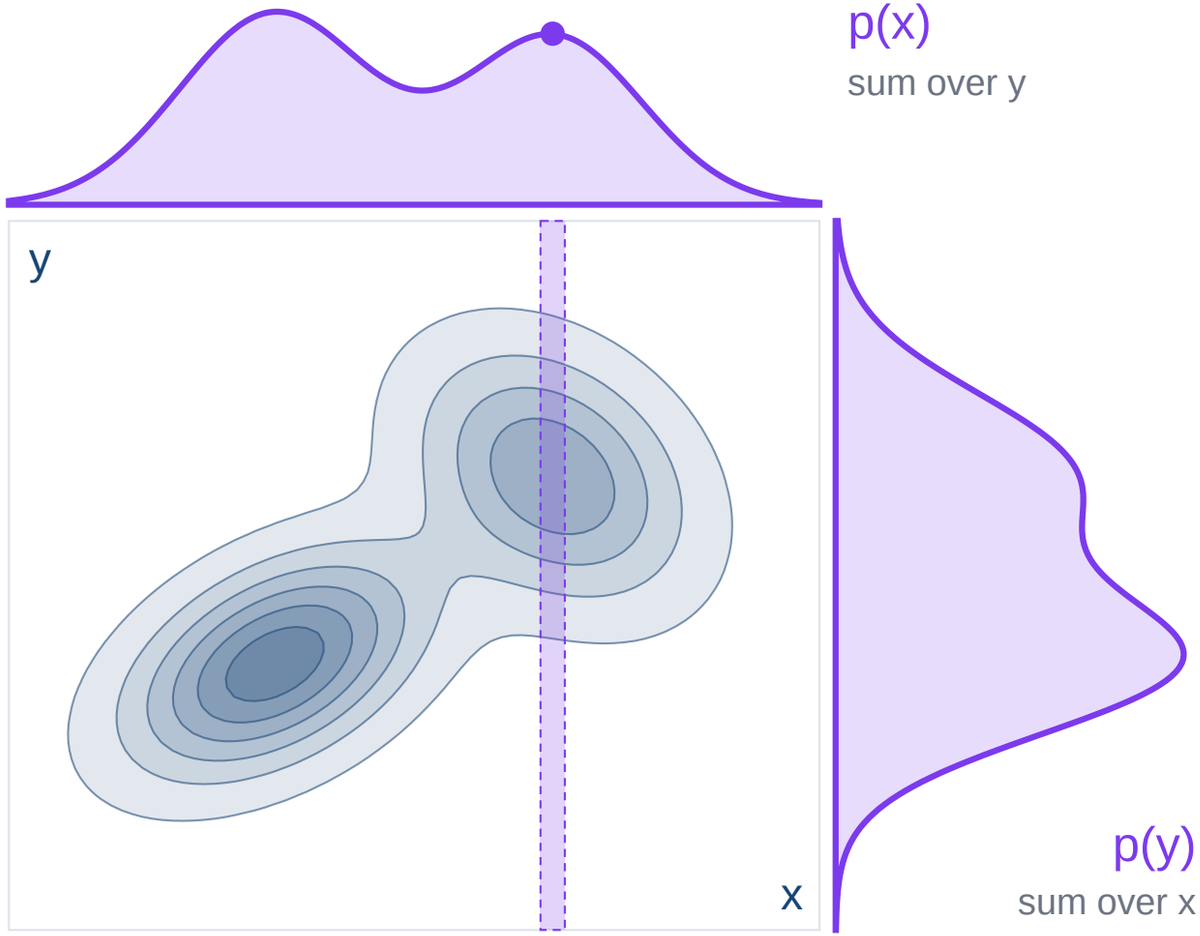
# Marginalisation

**Marginalisation** eliminates a random variable by summing over all its states. For $p(x_1, \ldots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \ldots, x_n)$$

**INTUITION**

Forget about a variable, but keep its influence correctly accounted for: collapse the joint onto the axis of the variable you keep.



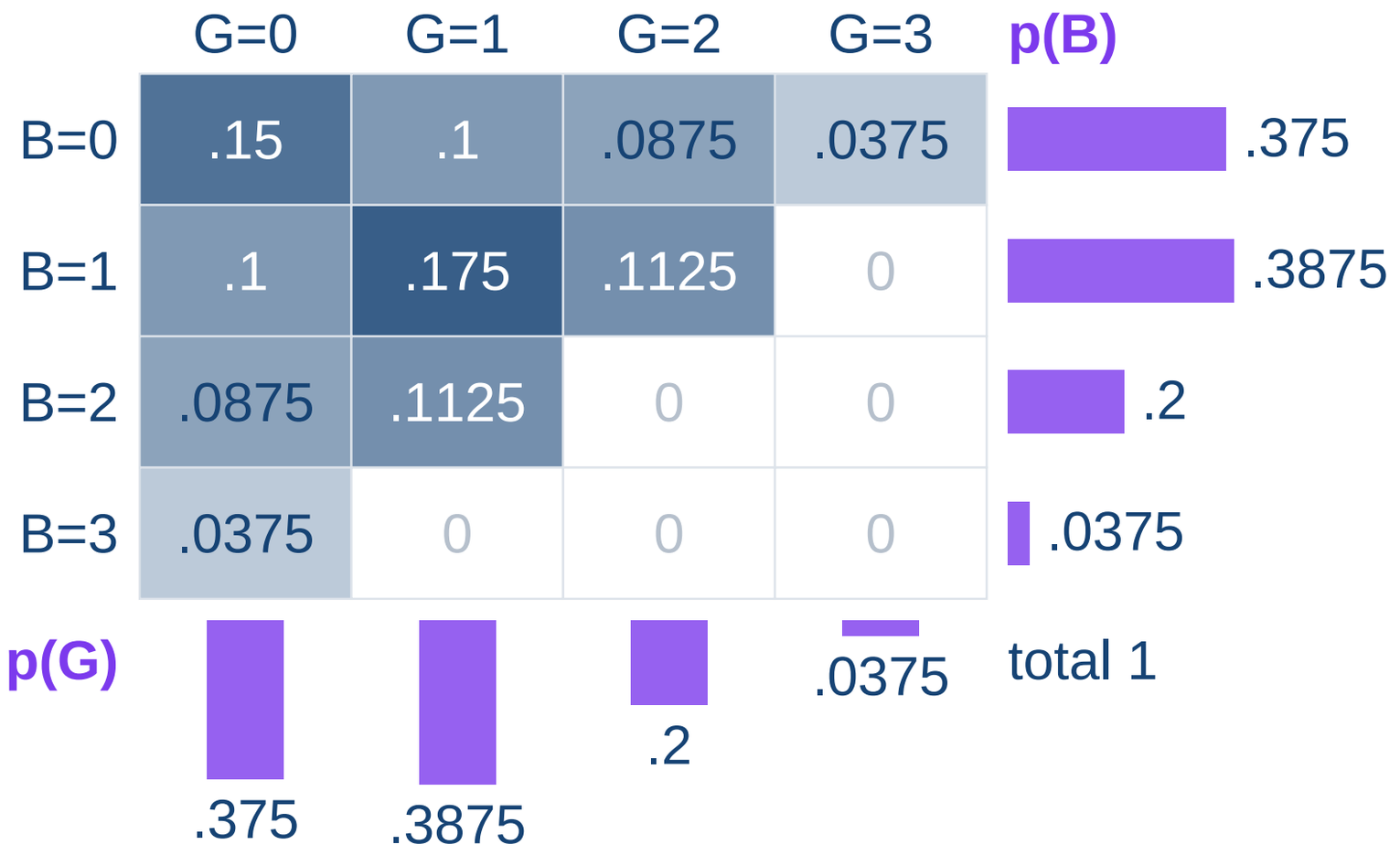
Speaker notes

The steps project the joint density from before onto its axes: (1) summing (integrating) every vertical strip gives one point of $p(x)$, (2) summing every horizontal strip gives $p(y)$. The marginals are computed from the joint, not assumed; the two bumps of the joint reappear in the marginals, as two peaks in $p(x)$ and as a peak with a shoulder in $p(y)$.

Marginalisation is the law of total probability applied to random variables: the events $\{X_2 = x_2\}$ for all states $x_2$ partition the sample space, so $p(x_1) = \sum_{x_2} p(x_1, x_2) = \sum_{x_2} p(x_1 \mid x_2)p(x_2)$. The second form shows how the eliminated variable keeps its influence: each conditional is weighted by how probable that state is. For continuous variables the sum becomes an integral.

In general, the marginal of any subset of variables is obtained by summing over all the remaining ones. Done naively, marginalising $n - 1$ variables with $k$ states each takes $k^{n-1}$ terms; factorised models allow the sums to be rearranged and computed far more cheaply.

# Example: Boys and Girls




Families: 15% have no children, 20% one, 35% two, 30% three. Each child is a boy or a girl with probability $\frac{1}{2}$, independently.

$B$ boys, $G$ girls.

> The margins of the table *are* the marginals — hence the name.

Speaker notes

The steps sum each row into $p(B)$ on the right and each column into $p(G)$ below; both sum to 1.

Each cell follows from family size times a binomial probability. One child (20%): one boy or one girl, 10% each. Two children (35%): BB and GG with probability $\frac{1}{4}$ each ($.0875$), one of each with $\frac{1}{2}$ ($.175$). Three children (30%): 3 boys or 3 girls with $\frac{1}{8}$ each ($.0375$), 2+1 in either direction with $\frac{3}{8}$ each ($.1125$). The cell $(B = 1, G = 2) = .1125$ reads: 11.25% of families have one boy and two girls. Rows and columns sum to the marginals, which are identical for $B$ and $G$ by symmetry.

$B$ and $G$ are dependent, because family size couples them: $p(B = 3, G = 3) = 0$, whereas $p(B = 3)p(G = 3) = .0375^2 \approx .0014$. The covariance is negative: $E[B] = E[G] = 0.9$, $E[BG] = .175 + 2 \cdot .1125 + 2 \cdot .1125 = .625$, so $\text{cov}(B, G) = .625 - .81 = -.185$.

# Covariance

$$\operatorname{cov}(A, B) = E\big[(A - E[A])(B - E[B])\big]$$

$\operatorname{cov} > 0$
$A$ large when $B$ large — positively related

$\operatorname{cov} < 0$
$A$ large when $B$ small — negatively related

$\operatorname{cov}(A, A) = \operatorname{Var}(A)$
variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *linear* association.

Speaker notes

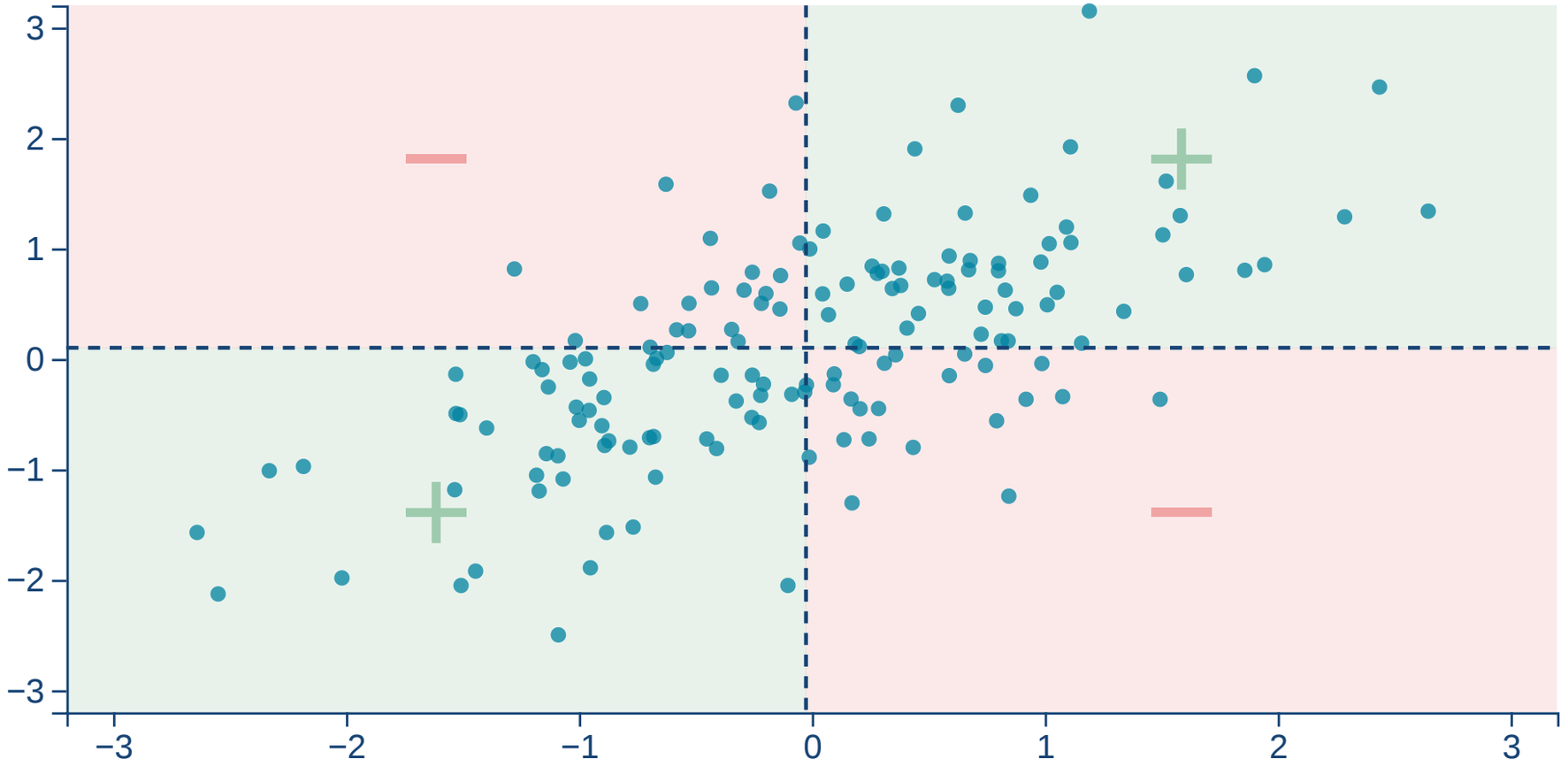
Expanding the definition gives the computational form $\operatorname{cov}(A, B) = E[AB] - E[A]E[B]$. For independent variables $E[AB] = E[A]E[B]$, hence zero covariance. The converse fails: $X$ uniform on $[-1, 1]$ and $Y = X^2$ have $E[XY] = E[X^3] = 0$ and $E[X] = 0$, so $\operatorname{cov}(X, Y) = 0$, although $Y$ is a function of $X$. Methods that rely on covariance or correlation alone can therefore miss real dependence. An important exception: for jointly normal variables, zero covariance does imply independence.

Covariance measures linear association. Its size depends on the units; the correlation $\rho = \operatorname{cov}(A, B)/(\sigma_A \sigma_B)$ removes them and lies in $[-1, 1]$, with $|\rho| = 1$ exactly when $B$ is a linear function of $A$. Covariance also governs the variance of a sum: $\operatorname{Var}(A + B) = \operatorname{Var}(A) + \operatorname{Var}(B) + 2\operatorname{cov}(A, B)$.

# Covariance Sees Only One Kind of Structure

Slide ρ, then switch the relation. Predict the covariance first.

cov(X, Y) = 0.62




**Predict first:** with $Y = X^2 - 1$, every point is fixed by its $x$. The covariance?

Speaker notes

The steps show (1) a cloud of points with correlation $\rho = 0.7$, (2) the two means as crosshairs, from which deviations are measured, and (3) the four quadrants: the product $(x - \bar{x})(y - \bar{y})$ is positive in the upper-right and lower-left quadrants and negative in the other two. The covariance is the average of these products, so it is large and positive when most points fall in the positive quadrants. The final step allows changing $\rho$ and switching to the relation $Y = X^2 - 1$.

For $Y = X^2 - 1$ with $X$ symmetric about 0, each point at $x$ has a mirror point at $-x$ with the same $y$ and an opposite-signed product, so the products cancel: $\operatorname{cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$. The sample value is close to, not exactly, zero. The dependence is complete — $Y$ is fixed by $X$ — yet the covariance is about 0. This is why "uncorrelated" and "independent" are different properties.

# Takeaway

The joint distribution holds everything; **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

# Data as a Sample

> ***Guiding Question:*** *A table is all we ever get. Which joint distribution could have produced it — and what else went into the recorded values?*

Speaker notes

The module reads a dataset from the outside in. Its shape first: record, graph, grid or spatio-temporal, which decides whether rows can be treated as independent at all. The simplest shape, a table, is then read as a sample: columns are random variables, and $n$ rows are $n$ draws from a joint distribution over them. The columns' scales (nominal, ordinal, interval, ratio) decide which statistics are meaningful and which parametric family suits each column; the rows are usually assumed i.i.d. Last, the single recorded value: noise, artefacts, outliers and missing values, each a statement about how it was generated. Three mini-exercises and the takeaway close the module.

# Four Shapes of Data

### Record data

Objects × attributes — the familiar table. Rows are *treated as* independent.

### Graph data

Objects carry **links** to other objects: web pages, social networks, citations.

### Fields / grids

Objects sit in a **spatial** arrangement: images, sensor lattices, simulations.

### Spatio-temporal

Position **and** time: monthly land temperatures, traffic movement, trajectories.

The shape decides which methods even apply. A table whose rows are treated as independent can be shuffled without loss; a grid or a time series cannot — the arrangement *is* information. Independence of rows is an assumption, not a property of the file.

Speaker notes

Every shape can be flattened into a table, but not without loss: a graph becomes an edge list whose rows share nodes, an image becomes one row per pixel whose neighbours are correlated, a time series becomes rows whose order carries the dynamics. In each case the rows are no longer independent, so methods that assume independent rows give wrong standard errors and over-optimistic validation (a random train/test split leaks information between neighbouring rows). Network science, spatial statistics and time-series analysis are the methods that model the arrangement instead.

# A Table Is a Sample From a Joint Distribution

| | payment | satisfaction | born | income (k€) | returned? |
|---|---|---|---|---|---|
| **column** | $X_1$ · nominal | $X_2$ · ordinal | $X_3$ · interval | $X_4$ · ratio | $X_5$ · nominal |
| **draw** $\mathbf{x}_1$ | card | 4 | 1987 | 52 | yes |
| **draw** $\mathbf{x}_2$ | invoice | 2 | 1995 | 31 | no |

Each **column** is a random variable **with a measurement scale**; each **row** is one draw $\mathbf{x}_i$ from **one joint distribution** $p(x_1, \ldots, x_5)$ that nobody has written down.

A is 180 cm long, B 90 cm. Is A "twice as long"?

Speaker notes

The second header row is not data: it names the random variable behind each column and the scale its values are measured on. A random variable is a function from the population (customers) to values; the scale states which relations between those values carry information. A row is one customer drawn from the population with all attributes recorded at once, a realisation of the vector $(X_1, \ldots, X_5)$. The table as a whole is a sample of $n$ such draws.

A dataset is not the distribution but a sample from it. Every quantity computed from the table — a mean, a proportion, a correlation — is therefore an estimate of a property of $p(x_1, \ldots, x_5)$. A different sample from the same distribution would give a different value, and this sample-to-sample variation is the estimate's error.

The length example keeps the object fixed and changes only the recording. A ratio scale has a true zero, so ratios survive every change of unit ($x \mapsto ax$, $a > 0$): 180/90 = 70.9/35.4 = 2. Measuring from an arbitrary reference point, here 100 cm, is an interval scale: the map $x \mapsto x - 100$ preserves differences (90 cm) but not ratios (80 / −10 = −8). Size classes keep only the order, and any strictly increasing recoding of 1, 2, 3 is equally valid. A label keeps only identity. The same number 3, or the same pair of values, therefore supports different statements depending on the scale; Lecture 02 defines the four scales and the statistics each permits (Stevens 1946).

For the probabilistic reading the scale matters twice: it decides which summaries of a sample are meaningful, and which distribution family is a sensible model for the column, taken up in "The Scale Suggests the Model".

# A Table Is a Sample From a Joint Distribution

| | payment | satisfaction | born | income (k€) | returned? |
|---|---|---|---|---|---|
| **column** | $X_1$ · nominal | $X_2$ · ordinal | $X_3$ · interval | $X_4$ · ratio | $X_5$ · nominal |
| **draw** $\mathbf{x}_1$ | card | 4 | 1987 | 52 | yes |
| **draw** $\mathbf{x}_2$ | invoice | 2 | 1995 | 31 | no |

Each **column** is a random variable **with a measurement scale**; each **row** is one draw $\mathbf{x}_i$ from **one joint distribution** $p(x_1, \ldots, x_5)$ that nobody has written down.

A is 180 cm long, B 90 cm. Is A "twice as long"?

| length recorded as | A | B | what the values support |
|---|---|---|---|
| cm — **ratio** | 180 | 90 | "twice as long" — in inches 70.9 vs 35.4, still ×2 |
| cm above 100 cm — **interval** | 80 | −10 | "90 cm longer"; the ratio −8 means nothing |
| size class 1–3 — **ordinal** | 3 | 1 | "longer"; "three times" means nothing |
| size label — **nominal** | L | S | "different" — nothing more |

# Four Attribute Types — What the Values Allow

| | type | the values… | meaningful | statistics | examples |
|---|---|---|---|---|---|
| categorical | **nominal** | are names that only distinguish objects | $=, \neq$ | mode, entropy, contingency table, $\chi^2$ test | ZIP code, employee ID, eye colour |
| | **ordinal** | carry an order | $<, >$ | median, percentiles, rank correlation, sign test | grades, rankings, {good, better, best} |
| numeric | **interval** | have meaningful differences; the zero is arbitrary | $+, -$ | mean, standard deviation, Pearson correlation, $t$-test | temperature in °C or °F, calendar dates |
| | **ratio** | have a true zero, so ratios are meaningful | $\times, \div$ | geometric mean, harmonic mean, coefficient of variation | length, mass, counts, age, temperature in K |

Each type permits everything the one above it does, plus more.

Speaker notes

The four scales go back to S. S. Stevens (1946). Nominal and ordinal attributes are called categorical (qualitative), interval and ratio attributes numeric (quantitative). The scales form a hierarchy: each permits the comparisons and statistics of the scales above it in the table, plus its own. So the mode is meaningful on every scale, the median from ordinal on, the mean from interval on, and ratios, the geometric mean and the coefficient of variation only on a ratio scale.

The type is a property of the measurement, not of the storage format: a ZIP code stored as an integer is still nominal, and a street number is ordinal at best. Temperature in kelvin, unlike °C, is a ratio scale.

# Four Attribute Types — What You May Transform

| | type | permissible transformation | example |
|---|---|---|---|
| categorical | **nominal** | any one-to-one relabelling | reassigning employee IDs changes nothing |
| | **ordinal** | any strictly increasing $f$: $x \mapsto f(x)$ | {good, better, best} is as well coded by {1, 2, 3} as by {1, 5, 9} |
| numeric | **interval** | $x \mapsto ax + b$, $a > 0$ | °C and °F differ in zero point and unit |
| | **ratio** | $x \mapsto ax$, $a > 0$ | length in metres or in feet |

**INTUITION**

A statement is **meaningful** on a scale if no permissible transformation can change its truth. "Twice as long" survives metres → feet; "twice as warm" does not survive °C → °F.

Speaker notes

Each scale is characterised by the transformations that preserve its information: any one-to-one relabelling for nominal, any strictly increasing function for ordinal, positive affine maps $x \mapsto ax + b$ ($a > 0$) for interval (°C to °F), and positive scalings $x \mapsto ax$ for ratio (metres to feet). The smaller the set of permissible transformations, the more structure the values carry, and the more statements are meaningful.

The criterion explains the statistics column of the previous table. The mean of an ordinal attribute is not meaningful because a strictly increasing recoding (1, 2, 3 to 1, 5, 9) can change which of two groups has the larger mean; the median's position is preserved by every such recoding. A ratio of two temperatures in °C changes under the affine map to °F (20 °C / 10 °C = 2, but 68 °F / 50 °F = 1.36); a ratio of two lengths is the same in metres and feet.

# Why the Type Decides the Method

- ✗ **The classic error:** averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.
- ✗ **The subtle one:** 20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

> The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every method in this course depends on it.

# The Scale Suggests the Model

Which distribution would you write down for each column — and how many numbers does it need?

> The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

Speaker notes

Each entry is a parametric family: a formula with a few parameters instead of one probability per value. The column "numbers" counts the parameters of the family: what has to be estimated from the data, instead of one probability per value. The categorical distribution assigns $P(X = j) = p_j$ to each of $k$ values, with $k - 1$ free parameters because $\sum_j p_j = 1$; the Bernoulli is the case $k = 2$. Naive Bayes classifiers, for example, commonly model each nominal attribute this way, with one categorical distribution per class.

Ordinal attributes are the hard case. A categorical model ignores the order; a numeric model (e.g. normal) assumes equal gaps between levels, which the ordinal scale does not justify. Dedicated ordinal models, such as cumulative-logit models, use the order without assuming equal gaps. For height a normal model is only an approximation, since its support is the whole real line, including negative values that carry negligible probability. Income is positive and right-skewed, so a normal model would put probability on negative incomes; a log-normal model means that $\log$ income is normal. ZIP codes are nominal despite consisting of digits.

The Poisson distribution, $P(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$, models the number of events in a fixed period at a constant rate $\lambda$, which is also its mean and its variance; real counts often vary more than that (overdispersion), and a negative binomial adds a second parameter for it. The exponential distribution models the waiting time between such events. A share in $[0, 1]$ is modelled by a Beta distribution, the same family that serves as the standard prior for an unknown probability.



# The Scale Suggests the Model

Which distribution would you write down for each column — and how many numbers does it need?

| attribute | values | scale | a first model | numbers |
|---|---|---|---|---|
| returned? | yes / no | nominal | **Bernoulli**$(p)$ | 1 |
| payment | card, invoice, … ($k$) | nominal | **categorical**$(p_1, \ldots, p_k)$ | $k - 1$ |
| satisfaction | 1 – 5 stars | ordinal | categorical (order unused) | 4 |
| orders last month | 0, 1, 2, … | ratio, count | **Poisson**$(\lambda)$ | 1 |
| height | around 170 cm | ratio | **normal**$(\mu, \sigma^2)$ | 2 |
| income | $> 0$, skewed | ratio | **log-normal**: normal on $\log$ income | 2 |
| share of orders returned | $[0, 1]$ | ratio | **Beta**$(\alpha, \beta)$ | 2 |

> The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

# What "i.i.d." Claims

The rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **independent and identically distributed** (i.i.d.) if

| | | |
|---|---|---|
| **independent** | $p(\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_i p_i(\mathbf{x}_i)$ | *one customer's row tells nothing about another's* |
| **identically distributed** | $p_i = p$ for every row $i$ | *every row comes from the **same** joint distribution* |

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

Speaker notes

Independence is a statement about rows, not columns: the attributes within a row may be strongly dependent (income and age, say). Under i.i.d. the probability of the whole dataset factorises, $p(\mathcal{D} \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$, which is the form every likelihood-based fitting method starts from.

Answer to the question. Daily sales of one shop are autocorrelated (one day predicts the next) and follow trends and seasons, so they are neither independent nor identically distributed. Two visits by the same customer are dependent because they share the person. Neighbouring pixels of one image are strongly correlated. A survey sent to a random sample yields approximately independent draws from the population (exactly so for sampling with replacement; without replacement the dependence is negligible when the population is large), though non-response can still make it unrepresentative. Record data is the setting where i.i.d. is at least plausible; graphs, grids and time series are the settings where it is not. A frequent violation of "identically distributed" in practice is distribution shift between the data a model is fitted on and the data it is applied to.

# What "i.i.d." Claims

The rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **independent and identically distributed** (i.i.d.) if

| | | |
|---|---|---|
| **independent** | $p(\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_i p_i(\mathbf{x}_i)$ | *one customer's row tells nothing about another's* |
| **identically distributed** | $p_i = p$ for every row $i$ | *every row comes from the **same** joint distribution* |

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

Only the random survey is plausibly i.i.d. Time series, repeated measurements and spatial data all violate it — the arrangement *is* information.

# Where Does a Recorded Value Come From?

A sensor records 22.4 °C. Besides the temperature, what went into that number?

$$x = s + b + \varepsilon$$

$s$: the true value

$b$: artefact — deterministic, the same every time

$\varepsilon$: noise — random, mean 0

Speaker notes

The equation is a measurement model: the recorded value $x$ is the quantity of interest $s$ plus a systematic error $b$ plus a random error $\varepsilon$ with $E[\varepsilon] = 0$. Noise is aleatoric: repeating the measurement gives a different $\varepsilon$ each time. An artefact is introduced by the measurement process itself (a sensor in the sun, a miscalibrated scale, a logging bug, a resampled image) and gives the same $b$ for every reading taken through that process. More generally an artefact is any deterministic error $x = g(s)$: a decimal point shifted (1.85 m recorded as 18.5 m), °F stored as °C, a wrong conversion in the pipeline. The same $s$ always gives the same wrong $x$, so repeating the measurement does not reveal it, and it may affect every row or only those from one source. A third case is a reading that does not follow the model at all, an outlier. The same decomposition underlies regression later in the course, where $s$ becomes a function $f(\mathbf{x})$ and $\varepsilon$ the observation noise.

# Where Does a Recorded Value Come From?

A sensor records 22.4 °C. Besides the temperature, what went into that number?

$$x = s + b + \varepsilon$$

$s$: the true value

$b$: artefact — deterministic, the same every time

$\varepsilon$: noise — random, mean 0

A dataset records $x$, never $s$. Every data problem is a claim about the terms that were added on the way.

# Three Problems on One Axis




Which of the three problems does collecting more data fix?

Speaker notes

Twelve readings of a true value $s = 20$ °C in each row; the triangle marks the sample mean. Noise: the readings scatter around $s$ and their mean is close to it; with $n$ readings the mean's random error shrinks like $1/\sqrt{n}$ (made precise as the standard error in the statistics module). Artefact: the same scatter shifted by $b = 2.5$; the mean converges to $s + b$, not to $s$, so more data makes the estimate more precise but not more correct. Outlier: one reading of 35 °C (a mistyped value, say) raises the mean from about 20 to about 21.3, while the median stays near 20.

# Three Problems on One Axis




Which of the three problems does collecting more data fix?

Only **noise**. An artefact shifts every new reading too; an outlier comes back in the same proportion.

# Each Problem Is a Model Statement

| problem | in the model | more data | what helps |
|---|---|---|---|
| **noise** | $\varepsilon \sim p(\varepsilon)$, $E[\varepsilon] = 0$ | averages it out | model it; average |
| **artefact** | $x = g(s)$, deterministic | does **not** help | calibration, a second instrument, domain knowledge |
| **outlier** | a draw from another $q$ | a share $\pi$ remains | robust statistics, or model $q$: $(1-\pi)\,p + \pi\,q$ |
| **missing** | $M$ = missing, $P(M \mid x)$ | depends on why | ask for the mechanism first |

None of these can be read off the data alone. Calling a value noise, an artefact or an outlier is an assumption about how it was generated.

Speaker notes

Noise is modelled by a distribution for $\varepsilon$, most often $\mathcal{N}(0, \sigma^2)$; heavier-tailed choices (Laplace, Student-t) allow occasional large errors without calling them outliers. An artefact is a bias and is invisible from within one dataset, because every reading carries it; it shows only against an external reference (a calibrated instrument, a second data source, known physical limits: no adult is 18.5 m tall), or once the source of each value (the scale, the sensor, the clerk) is recorded as an attribute and the groups can be compared. Outliers are modelled as a mixture: with probability $1 - \pi$ a reading comes from the model $p$, with probability $\pi$ from a broad contamination distribution $q$. Fitting the mixture yields, for each reading, the posterior probability that it is an outlier — Bayes' theorem with two hypotheses. Missing values are described by a missingness indicator $M$: missing completely at random (MCAR) if $P(M = 1)$ is a constant, missing at random (MAR) if it depends only on observed attributes, missing not at random (MNAR) if it depends on the missing value itself, as for a sensor that fails in the cold. Only under MCAR is deleting incomplete rows unbiased.

# Mini-Exercise: Stars and Birth Years

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

Speaker notes

Satisfaction in stars is ordinal. The mean treats the gap from one to two stars as equal to the gap from four to five, which the scale does not guarantee; the median and the mode use only the order and the labels. Means of rating scales are nonetheless widely reported in practice, which amounts to treating the scale as interval.

Year of birth is interval: year 0 is a convention, so "born twice as late" is meaningless, but the mean year and differences between years are meaningful. A difference of two interval values, such as an age computed from years of birth, is on a ratio scale. Payment method is nominal (only counts, shares and the mode are meaningful); income is ratio, with a true zero, so "twice the income" is a valid statement.

# Mini-Exercise: Stars and Birth Years

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

Satisfaction is **ordinal**: its mean assumes equal gaps between the stars — report the median. Year of birth is **interval**: the mean year is fine, but ratios are not ("born twice as late" means nothing); differences such as age are fine.

# Mini-Exercise: Three Scales

Patients are weighed on three scales, a third on each. Two record $X = S + \varepsilon$; the third is miscalibrated and adds exactly 3 kg every time: $X = S + 3 + \varepsilon$. Does weighing ten times as many patients bring the mean of $X$ closer to $E[S]$?

Speaker notes

$S$ is the true weight, $\varepsilon$ the random noise with $E[\varepsilon] = 0$, the same for all three scales. By linearity $E[X] = E[S] + \frac{1}{3} \cdot 3 + \frac{2}{3} \cdot 0 = E[S] + 1$ kg. The sample mean converges to $E[X]$, so a larger sample makes the estimate more precise but not more correct. The artefact is detectable once the scale is recorded as an attribute: the mean per scale then differs by about 3 kg, more than the noise explains.

# Mini-Exercise: Three Scales

Patients are weighed on three scales, a third on each. Two record $X = S + \varepsilon$; the third is miscalibrated and adds exactly 3 kg every time: $X = S + 3 + \varepsilon$. Does weighing ten times as many patients bring the mean of $X$ closer to $E[S]$?

> No. $E[X] = E[S] + \frac{1}{3} \cdot 3 = E[S] + 1$ kg: the mean converges, but to a value 1 kg too high — an **artefact**. More data shrinks only the noise $\varepsilon$.

# Mini-Exercise: A Sensor That Fails in the Cold

Temperature $T$, missingness indicator $M$ (1 = reading missing). The sensor fails mostly in the cold: $P(M = 1 \mid T)$ is large for low $T$. After deleting the incomplete rows, does the mean estimate $E[T]$ — or what?

Speaker notes

The rows that remain are draws from the conditional distribution $p(t \mid M = 0)$, not from $p(t)$. By Bayes' theorem, $p(t \mid M = 0) \propto P(M = 0 \mid t)\, p(t)$: low temperatures are down-weighted because they are the ones likely to be missing, so the mean of the remaining rows overestimates $E[T]$, and more data does not remove the bias. This is missing not at random (MNAR). If $P(M = 1 \mid t)$ is a constant, $M$ and $T$ are independent, the weighting is flat and $p(t \mid M = 0) = p(t)$: missing completely at random (MCAR). The two cases produce files that look alike; telling them apart requires knowledge of the mechanism, such as a second thermometer or the sensor's specification.

# Mini-Exercise: A Sensor That Fails in the Cold

Temperature $T$, missingness indicator $M$ (1 = reading missing). The sensor fails mostly in the cold: $P(M = 1 \mid T)$ is large for low $T$. After deleting the incomplete rows, does the mean estimate $E[T]$ — or what?

> It estimates $E[T \mid M = 0]$, which is **larger** than $E[T]$: the coldest days are gone. Only if $M$ is **independent** of $T$ is $E[T \mid M = 0] = E[T]$ and deletion harmless.

# Takeaway

> A dataset is $n$ **draws from an unknown joint distribution** — and every way of reading it is an assumption about how it was generated.

1. **Shape:** only record data can plausibly be read as independent rows.
2. **Table:** columns are random variables, rows are draws from $p(x_1, \ldots, x_d)$.
3. **Scale:** licenses a column's statistics and suggests its model.
4. **i.i.d.:** lets the rows factorise — to be checked, not presumed.
5. **Recorded value:** truth + artefact + noise. More data removes **noise only**; artefacts, outliers and missing values need a model of how they arose.

# Statistics and Estimation

> ***Guiding Question:*** *A model has a parameter nobody told us, and all we have is a sample. What does the sample say about the parameter — and how sure may we be?*

Speaker notes

What is computed from a sample: a statistic is a function of the sample, and as an estimator of a population quantity it has a sampling distribution whose standard deviation is the standard error; a confidence interval turns it into a range. The Bayesian answer — a posterior over the parameter, in the Beta–Bernoulli model — follows in Lecture 03, where naive Bayes' smoothing needs it.

# We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

| | what it is | its numbers | do we see them? |
|---|---|---|---|
| **Population** | the entire group — all customers, all transactions, all documents | **parameters** $\mu$, $\sigma^2$ | no — the truth we want |
| **Sample** | the observations we actually have | **statistics** $\overline{x}$, $s^2$ | yes — and we estimate the parameters from them |

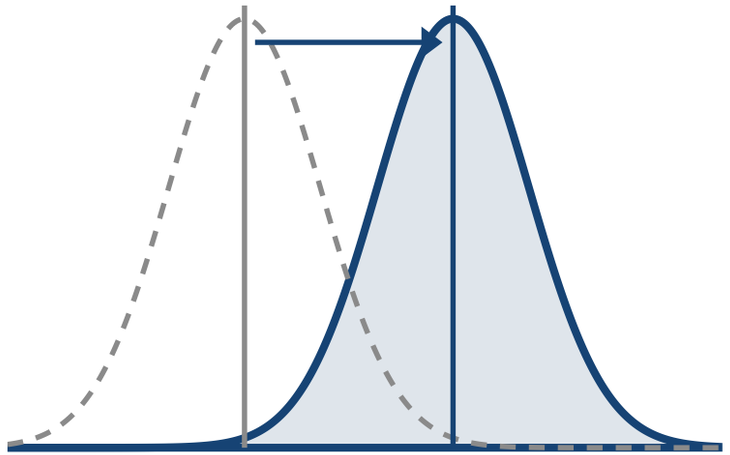
**Descriptive statistics** summarise a sample, usually while assuming something about the underlying distribution.

Speaker notes

Descriptive statistics summarise the observed data; inferential statistics draw conclusions about the population from them, with quantified uncertainty. The two vocabularies match: a **parameter** ($\mu$, $\sigma^2$) is a fixed but unknown property of the population; a **statistic** ($\overline{x}$, $s^2$) is a function of the sample. Because the sample is random, a statistic is itself a random variable, with a distribution of its own across possible samples.

The population can be a finite set (all customers of a shop) or a conceptual one (all future transactions, the process that generates documents). In the latter case the population parameters are properties of the generating distribution, e.g. $\mu = E[X]$.

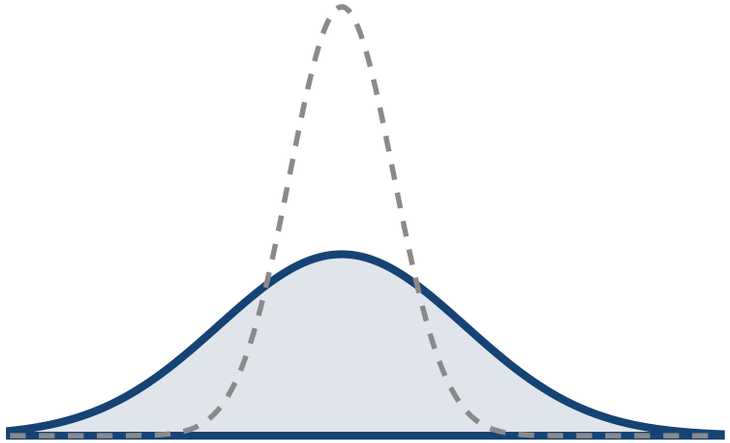
# What We Summarise



## Centre

1st moment

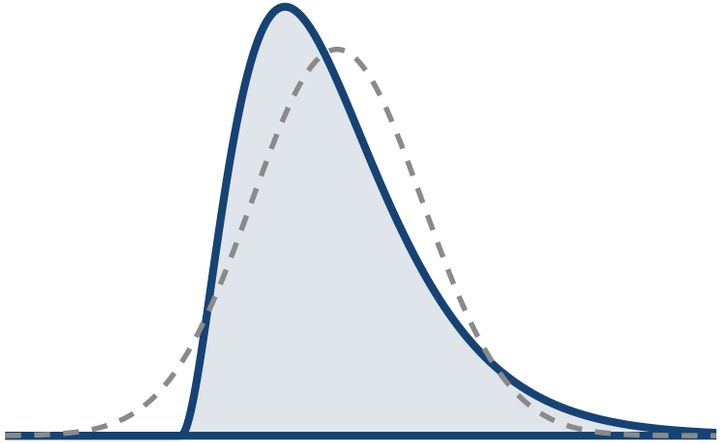
mean, median



## Dispersion

2nd central moment

variance, std. dev., IQR



## Shape

3rd central moment

skewness, quantiles

Univariate statistics describe one random variable; **multivariate** statistics describe $X_1, \ldots, X_d$ together.

Speaker notes

Each panel changes one aspect of a distribution against the dashed grey reference: the centre shifts, the spread widens, the shape becomes lopsided with a long right tail. The $k$-th moment of $X$ is $E[X^k]$; the $k$-th central moment is $E[(X-\mu)^k]$. The mean is the first moment, the variance the second central moment. Skewness is the standardised third central moment, $E[(X-\mu)^3]/\sigma^3$: positive for a long right tail (incomes), negative for a long left tail, zero for symmetric distributions. The standardised fourth moment (kurtosis) describes the weight of the tails.

Moment-based summaries are not robust: a single extreme value can move them arbitrarily far. Quantile-based summaries are: the median for location, the interquartile range $Q_3 - Q_1$ (spread of the middle 50%) for dispersion. The multivariate counterparts of mean and variance are the mean vector and the covariance matrix.

# Means

| mean | formula | use it for | example |
|---|---|---|---|
| **sample** | $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ | values that add up | average salary |
| **weighted** | $\overline{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$ | values that count unequally | grade over courses of different credits |
| **geometric** | $\left(\prod_{i=1}^{n} x_i\right)^{1/n}$ | factors that multiply | +50% then −50%: factor 0.87 per period |
| **harmonic** | $\frac{n}{\sum_i 1/x_i}$ | rates over equal amounts | 60 km/h out, 40 back: 48 km/h |
| **trimmed** | average after dropping the $k$ smallest and $k$ largest | robustness on the cheap | judges' scores |

Speaker notes

The sample mean estimates $E[X]$; it is also the value $c$ that minimises the sum of squared deviations $\sum_i (x_i - c)^2$. Which mean is appropriate depends on how the quantities combine.

Speeds over equal distances average harmonically: 60 km/h out and 40 km/h back give $2/(1/60 + 1/40) = 48$ km/h, not 50, because more time is spent at the lower speed. Growth factors average geometrically: +50% followed by −50% gives the factor $1.5 \cdot 0.5 = 0.75$, so the average factor is $\sqrt{0.75} \approx 0.866$ (−13.4% per period), while the arithmetic mean of the rates, 0%, is wrong. For positive values, harmonic $\le$ geometric $\le$ arithmetic mean, with equality only when all values are equal; the geometric and harmonic means require positive values. A trimmed mean with $k$ values dropped at each end tolerates up to $k$ outliers per side; judged sports use it when the highest and lowest score are discarded. A grade average weighted by course credits is a weighted mean with the credits as weights.

# Median

The **median** is the middle value of the sorted sample: for odd $n$ the $\frac{n+1}{2}$-th value, for even $n$ the average of the two middle values.

**INTUITION**

Half of the values lie below it, half above. Only their **order** counts, not how far out they lie.

✓ **Robust** against outliers — one extreme value moves the median by at most one position

✗ **Costlier** to compute — it needs sorting or a selection algorithm, not one pass with a running sum

*For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.*

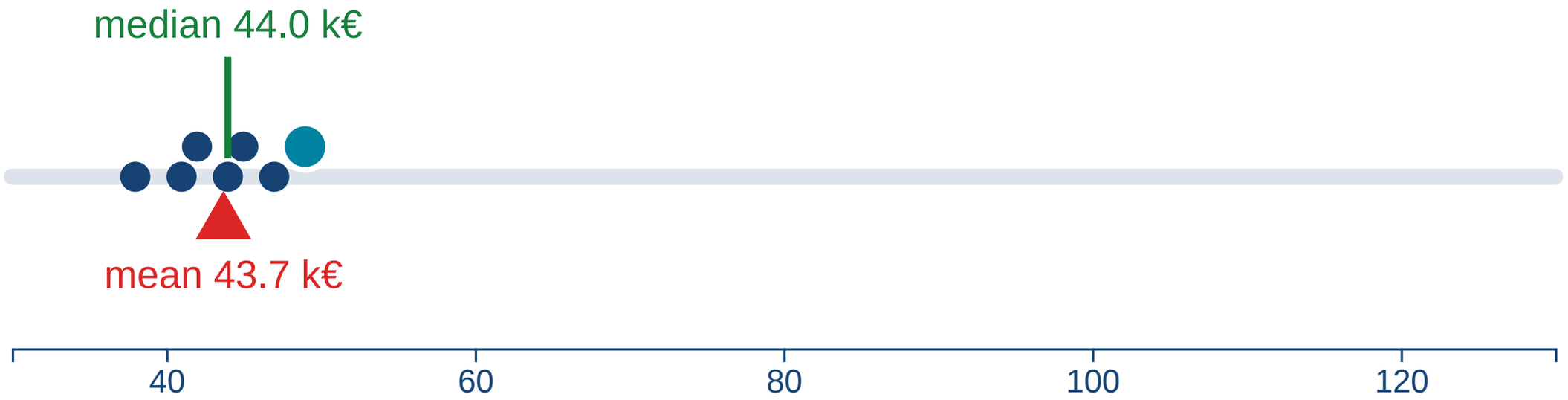
Speaker notes

The median minimises the sum of absolute deviations $\sum_i |x_i - c|$, as the mean minimises the sum of squares. Its robustness can be quantified by the breakdown point, the fraction of values that can be made arbitrarily large before the statistic follows: 50% for the median, 0% for the mean.

Full sorting ($O(n \log n)$) is sufficient but not necessary: selection algorithms such as quickselect find the median in expected $O(n)$ time. What the median lacks is the one-pass, constant-memory computation of the mean; streaming systems therefore use approximate quantiles. For binned data the median is interpolated as $L + \frac{n/2 - F}{f} \cdot w$, where $L$ is the lower border of the interval containing the median, $F$ the cumulative count below it, $f$ the count in it, and $w$ its width; this assumes the values are spread evenly within the interval.

# One Value, Two Very Different Reactions

Drag the light point — which marker follows it?




**Commit first:** drag the top salary to the far right. Which marker follows?

Speaker notes

The steps show (1) the mean as the point where the beam balances, (2) the median as a position in the sorted order, and (3) the top salary made draggable. Moving one of $n$ values by $\Delta$ moves the mean by $\Delta/n$, so the mean follows the dragged value at a seventh of its speed. The median does not move at all while the dragged value stays above the middle value: it remains 44 k€.

With the top salary far to the right, the mean lies above six of the seven salaries, a "typical" value almost nobody earns. Which summary is quoted is therefore not neutral: an employer advertising pay has an incentive to quote the mean, an employee representative the median.

# Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

# Mini-Exercise — Solution

Speaker notes

Mean: $(38+41+42+44+45+47+260)/7 = 517/7 \approx 73.9$. Median: the 4th of the 7 sorted values, $44$. Without the top salary the six remaining values have mean $257/6 \approx 42.8$ and median $43$, so the two summaries nearly agree once the outlier is gone.

Neither number is wrong; each answers a different question. The mean is payroll divided by headcount and is the right figure for total cost ($7 \times 73.9 \approx 517$ k€). The median is the right figure for the salary of a typical employee. Quoting the mean in a recruitment ad suggests a salary only one employee earns.

# Mini-Exercise — Solution

Mean $\approx 73.9$, median $= 44$.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee; the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

# Dispersion

**Sample variance** and **standard deviation** of the observed data:

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2, \qquad s = \sqrt{s^2}$$

INTUITION

The average squared distance from the centre: how stretched or squeezed the data is. $s$ brings it back to the unit of the data.

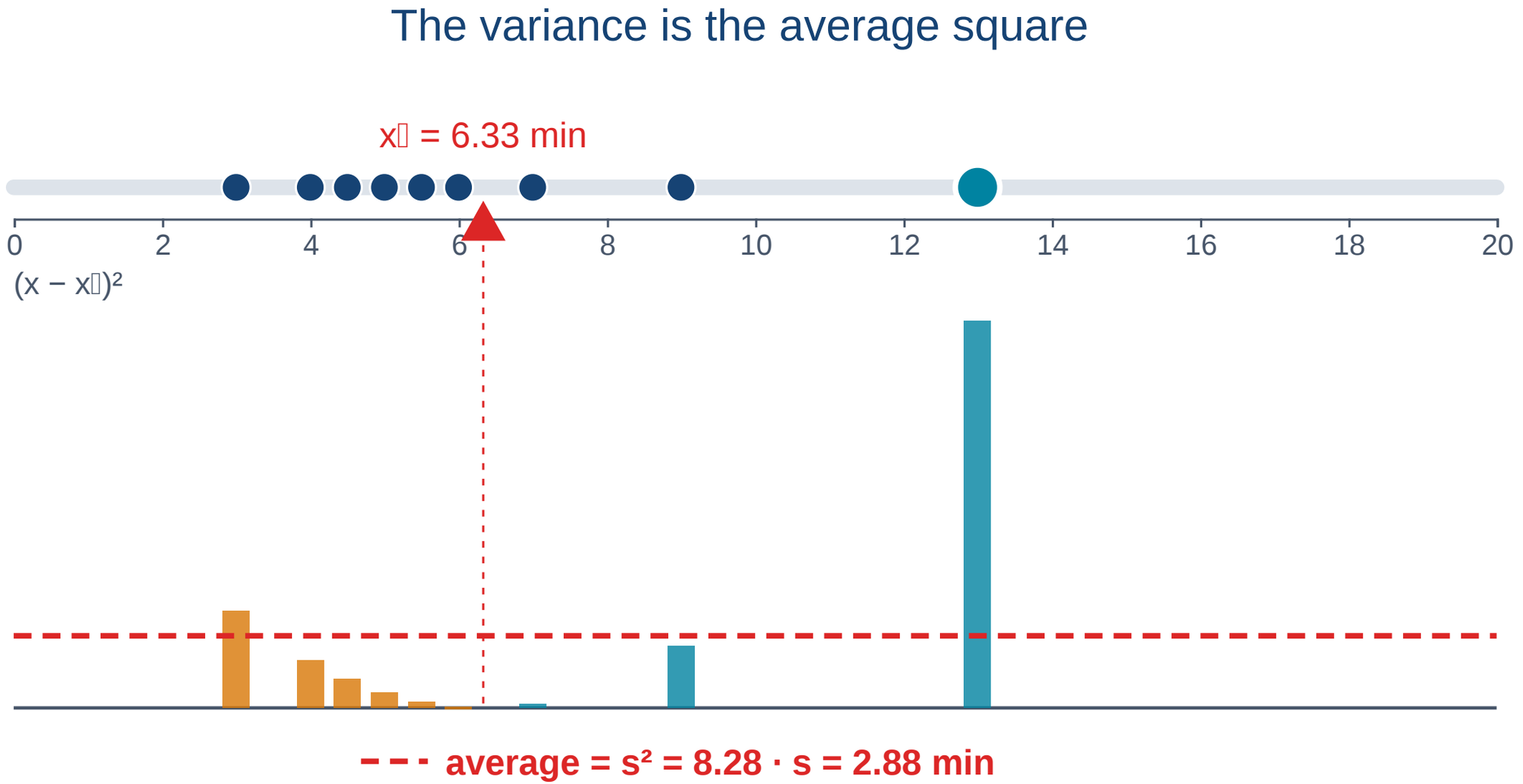
Computable in **one pass**, $O(n)$, from the running totals $n$, $\sum x_i$ and $\sum x_i^2$ — which matters when the data does not fit in memory.

Speaker notes

Expanding the square gives $s^2 = \frac{1}{n}\sum_i x_i^2 - \overline{x}^2$, so the counts $n$, $\sum x_i$ and $\sum x_i^2$ suffice. These three totals can be accumulated in one pass and merged across partitions by simple addition, which is why streaming and distributed systems can compute variances without holding the data.

The formula is numerically unstable in floating point when the mean is large relative to the spread: two large, almost equal numbers are subtracted and the significant digits cancel (values around $10^9$ with standard deviation 1 lose essentially all precision in double arithmetic). Welford's algorithm is the standard numerically stable one-pass alternative; it updates the running mean and the running sum of squared deviations with each new value.

# The Variance, Drawn



Speaker notes

The bars are the squared deviations $(x_i - \overline{x})^2$; their average is $s^2$, the dashed line. The unsquared deviations would be useless: they always sum to 0, $\sum_i (x_i - \overline{x}) = \sum_i x_i - n\overline{x} = 0$, the balance-point property of the mean. Squaring removes the sign, so both sides contribute, and it weights a deviation by its own size: the point at 13 minutes, 6.67 from the mean, contributes 44.4 of the total 74.5, i.e. 60% of $s^2$ on its own. Here $s^2 = 74.5/9 \approx 8.28$ min² and $s \approx 2.88$ min.

Moving that point to 20 minutes raises $s^2$ to about 23.5 min², almost three times as much, and $s$ to 4.85 min; the point then contributes 79% of the total. Moving it to 7 minutes gives $s^2 \approx 2.94$ min². The variance is therefore as sensitive to single far values as the mean, and more: a value at distance $d$ moves the mean by $d/n$ but contributes $d^2/n$ to the variance. Robust alternatives are the interquartile range and the median absolute deviation.

The steps show the mean, the squared deviations as bars, and their average. The light point also moves with the slider below the figure, in steps of 0.5; Reset returns it to 13.

# Why the Denominator Changes

The **population variance** $\sigma^2$ is the variance over the whole population. With the true mean $\mu$ unknown, it is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

**INTUITION**

$\overline{x}$ is computed from the same data, so it sits closer to the data than $\mu$ does: the deviations come out **systematically too small**. Dividing by $n-1$ instead of $n$ compensates — one degree of freedom was spent on the mean.

*Naming is not uniform: many textbooks, and pandas, call the $n-1$ version $s^2$; NumPy's `var` divides by $n$ unless told otherwise.*

Speaker notes

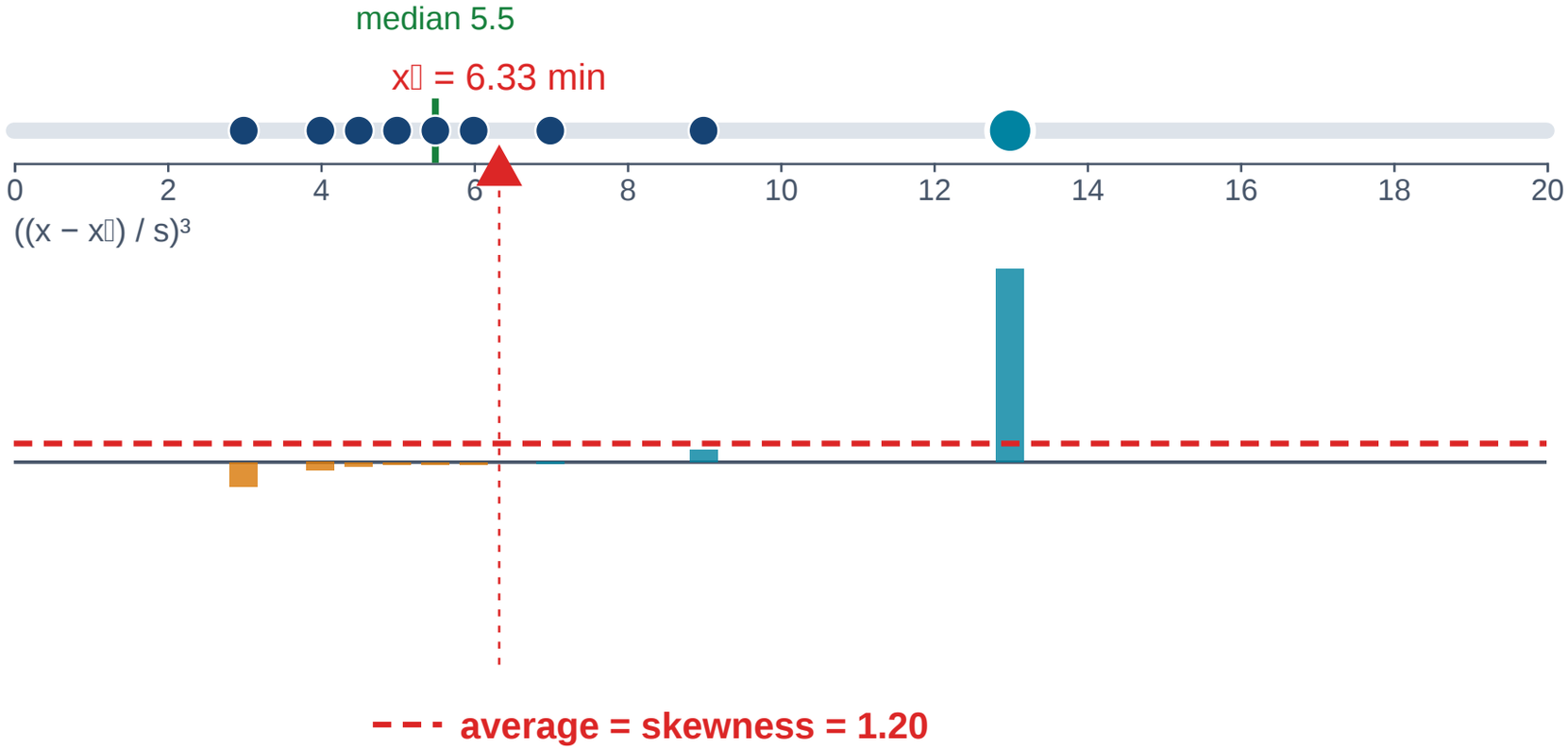
Derivation of the correction for i.i.d. data with mean $\mu$ and variance $\sigma^2$. Algebraically, $\sum_i (x_i - \overline{x})^2 = \sum_i (x_i - \mu)^2 - n(\overline{x} - \mu)^2$. Taking expectations, $E[\sum_i (x_i - \mu)^2] = n\sigma^2$ and $E[(\overline{x} - \mu)^2] = \mathrm{Var}(\overline{x}) = \sigma^2/n$, so $E[\sum_i (x_i - \overline{x})^2] = (n-1)\sigma^2$. Dividing by $n-1$ makes the estimator unbiased. The deviations are too small on average because $\overline{x}$ is, by construction, the value closest to the data in squared distance.

With $n = 1$ the sample variance is 0 (the point equals its own mean) and the corrected formula is $0/0$, undefined: one observation carries no information about spread. If $\mu$ is known, $\frac{1}{n}\sum_i (x_i - \mu)^2$ is already unbiased. Unbiasedness does not carry over to the square root: $\hat{\sigma}$ underestimates $\sigma$ slightly on average. For large $n$ the two denominators differ negligibly. In pandas, `var` uses `ddof=1` by default; in NumPy, `ddof=0`.

# Sample Skewness: Which Tail Is Longer?

$$\widehat{\text{Skew}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \overline{x}}{s}\right)^3 \qquad \begin{cases} > 0 & \text{long right tail} \\ < 0 & \text{long left tail} \end{cases}$$

Right side outweighs left: skewed right




Drag the light point left. Where is the skewness 0 — and why there?

Speaker notes

The bars are the cubed standardised deviations $z_i^3$ with $z_i = (x_i - \overline{x})/s$; their average is the (moment) skewness. Cubing keeps the sign, so left and right deviations compete, and it weights a deviation by the cube of its size, so the farthest point dominates: at 13 minutes $z^3 \approx 12.4$, against $-2.5$ for all points left of the mean together. The skewness is $1.20$; the mean (6.33) lies to the right of the median (5.5). Dividing by $s$ makes the measure unit-free: it is unchanged when minutes are converted to seconds.

Answer: at 1 minute. The sample 1, 3, 4, 4.5, 5, 5.5, 6, 7, 9 is symmetric about 5 (1 and 9, 3 and 7, 4 and 6, 4.5 and 5.5 mirror each other), so every cube is cancelled by its mirror image, mean and median are both 5, and the skewness is exactly 0. At 0 minutes it turns negative ($-0.36$). Skewness 0 does not imply symmetry in general, but symmetry implies skewness 0.

The formula replaces $E$, $\mu$ and $\sigma$ in the definition of $\mathrm{Skew}(X)$ by the sample average, $\overline{x}$ and $s$. Right-skewed data are common wherever values are bounded below but not above: waiting times, incomes, file sizes, counts. A log transformation often makes them nearly symmetric, which is why such attributes are modelled with a log-normal distribution. Statistical software often reports a bias-corrected sample skewness, which differs from this formula by a factor that tends to 1 as $n$ grows.

# Takeaway

Variance and skewness average the deviations from the mean raised to the power **2 and 3**: squares add up, cubes compete by sign. $s^2$ describes **the sample**; $\hat{\sigma}^2$ estimates **the population** — the $n - 1$ is the price of having estimated the mean from the same data.

# You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean $\mu$ is exactly 73.9 k€?

Speaker notes

The question has no useful answer. If uncertainty about $\mu$ is described by a continuous distribution, the probability of one exact value is exactly zero, as for any single value under a density; read as "73.9 to the nearest 0.1 k€", it is the probability of an interval of width 0.1 k€, small but positive. In the frequentist view $\mu$ is a fixed number and carries no probability at all (it is either 73.9 or not). In both readings, $\overline{x} = \mu$ exactly is not to be expected: a different sample of seven people would give a different mean.

The well-posed question concerns the size of the error $\overline{x} - \mu$. Its answer needs the distribution of $\overline{x}$ across samples, which is the subject of this unit. With seven values, one of them an outlier, the error can be large.

# You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean $\mu$ is exactly 73.9 k€?

Close to zero. $\overline{x}$ is not $\mu$ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is* $\mu$ but **how far from** $\mu$ **can this number be?**

# Estimator and Estimate

**Definition.** An **estimator** is a function of the sample, $\hat{\theta} = g(X_1, \ldots, X_n)$, used to approximate an unknown quantity $\theta$ of the population. The number it takes on for one particular sample is an **estimate**.

INTUITION

The estimator is the **recipe**, $\overline{X} = \frac{1}{n}\sum_i X_i$; the estimate is what one batch of data cooks from it, $\overline{x} = 73.9$.

The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

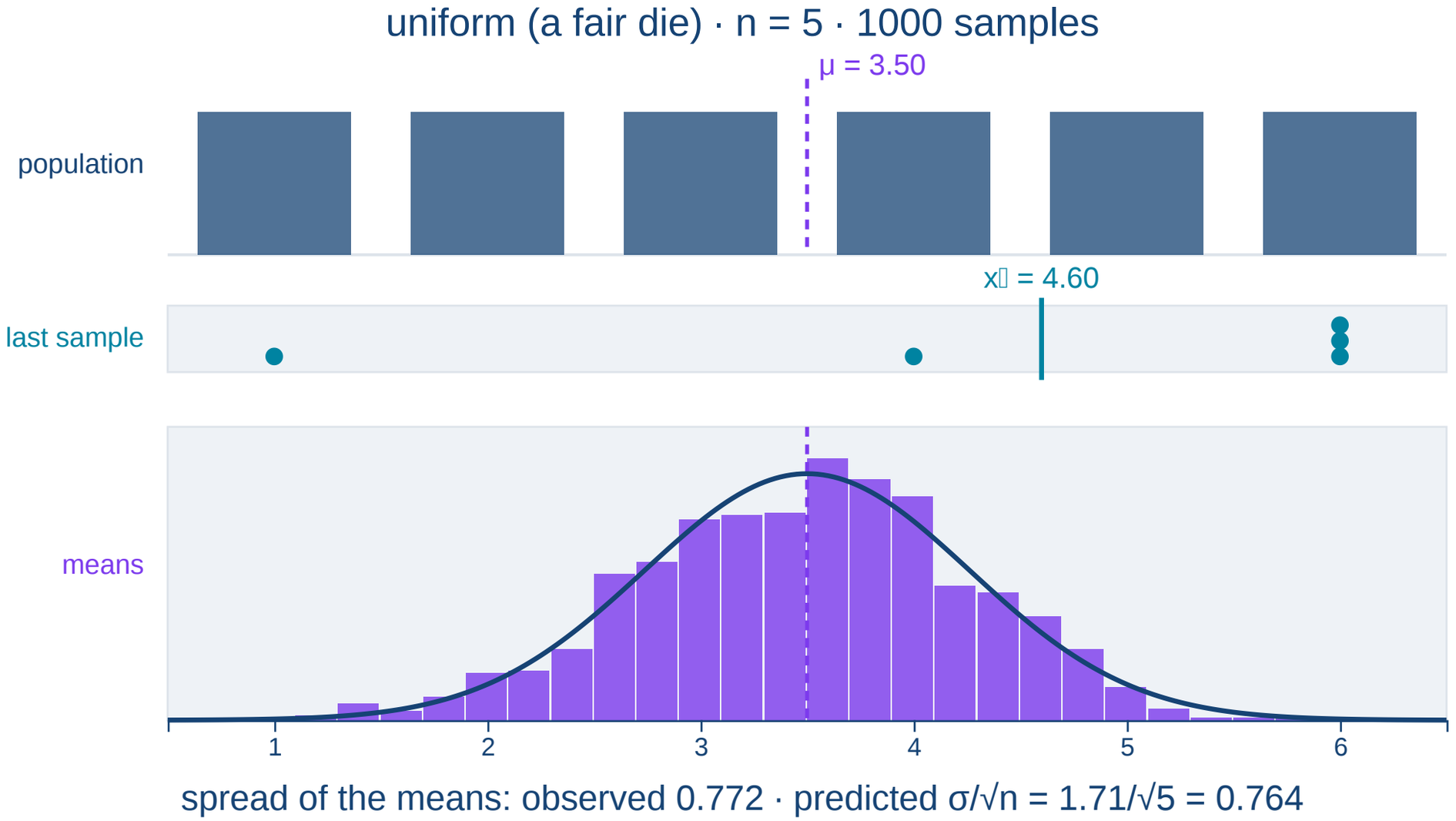
*$\overline{x}$, $s^2$ and $\hat{\sigma}^2$ from descriptive statistics are all estimators — used so far without saying what kind of object they are.*

Speaker notes

The distinction is the one between a rule and its output. The estimator $\overline{X} = \frac{1}{n}\sum_i X_i$ is written with capital letters because it is a function of the random variables $X_1, \ldots, X_n$; the estimate $\overline{x}$ is the number obtained by plugging in the observed values. A statistic inherits randomness from the sample, so properties such as bias and spread belong to the estimator, not to a single estimate.

Several estimators may target the same $\theta$: the sample mean, the median and a trimmed mean all estimate the centre of a symmetric population. They differ in their sampling distributions, and comparing those distributions is how one estimator is judged better than another.

# An Estimator Has a Distribution



Speaker notes

The steps show (1) one sample of $n = 5$ draws from the population in the upper panel, (2) its mean recorded as a single point in the lower panel, (3) the means of fifty samples, and (4) the means of a thousand samples with a normal curve and the standard error overlaid. Both panels share the horizontal axis, so their widths can be compared directly. Afterwards the population shape and $n$ can be changed.

The populations live on the values 1 to 6. The uniform one is a fair die, with $\mu = 3.5$ and $\sigma = \sqrt{35/12} \approx 1.71$. For $n = 5$ the means have standard deviation $1.71/\sqrt{5} \approx 0.76$, less than half the population's. A frequent wrong prediction is that the means are spread as widely as the observations.

# Three Things the Picture Shows

- → The sample means **centre on** $\mu$ — the estimator is not systematically off
- → They **scatter less than single draws**: standard deviation 0.76 instead of 1.71 — averaging cancels noise
- → Their histogram is **bell-shaped**, even though the population was not

> **Definition (Sampling distribution).** The distribution of an estimator $\hat{\theta}$ over all possible samples of size $n$. $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Speaker notes

The three observations are unbiasedness ($E[\overline{X}] = \mu$, by linearity), the standard error ($\mathrm{Var}(\overline{X}) = \sigma^2/n$, by independence) and the central limit theorem (approximately normal shape). The first two hold exactly for every $n$; the third is an approximation that improves with $n$.

For the sample median the picture differs. In a symmetric population the median is centred on $\mu$; in a skewed population it targets the population median, which differs from $\mu$, and is in general not exactly unbiased even for that. Its spread also shrinks like $1/\sqrt{n}$, and for large $n$ it is approximately normal, but this follows from a separate asymptotic result, not from the CLT for sums.

# Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

Speaker notes

Unbiasedness and the standard error $\sigma/\sqrt{n}$ do not depend on the population's shape at all; they need only independence and a finite variance. The bell shape is the approximate part. The bimodal population is the clearest case: two separate humps, yet the means of five draws already form a single hump.

How fast the bell shape appears depends on the population. For a symmetric population, a handful of draws suffices; for a strongly skewed one, the sampling distribution of the mean stays visibly skewed at small $n$, its skewness falling like $1/\sqrt{n}$. At $n = 1$ the sampling distribution is the population itself, and at $n = 2$ the population shape is still clearly visible. The CLT is a limit statement; "large enough $n$" depends on the population.

# Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as $n$ is more than a handful — which is why one formula can cover cases we never inspect.

# Standard Error

**Definition (Standard error).** The standard error of an **estimator** $\hat{\theta}$ is the standard deviation of its **sampling distribution**: $\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$. For the sample mean of $n$ independent draws, $\mathrm{SE}(\overline{X}) = \sigma/\sqrt{n}$.

Two distributions, two spreads — the die from the widget, $n = 5$:

| | the population | the estimator $\overline{X}$ |
|---|---|---|
| distribution of | one draw $X$ | the mean of $n$ draws, sample after sample |
| centre | $\mu = 3.5$ | $\mu = 3.5$ (unbiased) |
| spread | $\sigma = 1.71$, the **standard deviation** | $\sigma/\sqrt{n} = 0.76$, the **standard error** |
| more data | unchanged | shrinks like $1/\sqrt{n}$ |

Speaker notes

$\sigma/\sqrt{n}$ follows from independence: $\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_i \mathrm{Var}(X_i) = \sigma^2/n$. For dependent observations (e.g. a time series with positive correlation) the true standard error is larger, and the formula is overconfident. When sampling without replacement from a finite population of size $N$, the standard error is multiplied by $\sqrt{(N-n)/(N-1)}$, which is negligible for $n \ll N$ and gives 0 when the whole population is sampled.

The standard error is a property of the estimator, i.e. of the procedure "take $n$ draws and average them", not of a single estimate: one number $\overline{x}$ has no spread. It describes how far a typical estimate lies from $\mu$, in the unit of the data. In practice $\sigma$ is unknown and replaced by the sample estimate $\hat{\sigma}$, giving $\hat{\sigma}/\sqrt{n}$, which is itself an estimate of the standard error; this is the "SE" printed by statistical software. A weather forecast's "±3 °C" generally mixes both kinds of uncertainty: the inherent variability of the weather and the forecaster's limited knowledge.

# Precision Costs Data

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}} \quad \Longrightarrow \quad \frac{\sigma}{\sqrt{kn}} = \frac{1}{\sqrt{k}} \cdot \frac{\sigma}{\sqrt{n}}$$

Multiply the data by $k$, and the standard error is multiplied by $1/\sqrt{k}$:

| | | | |
|---|---|---|---|
| **0.71** | → | ½ | → | ⅕ | → | 1⁄10 |
| 2 × the data | | 4 × the data | | 25 × the data | | 100 × the data |

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Speaker notes

The table entries are $1/\sqrt{k}$ for $k$ times the data: $1/\sqrt{2} \approx 0.71$, $1/\sqrt{4} = \frac{1}{2}$, $1/\sqrt{25} = \frac{1}{5}$, $1/\sqrt{100} = \frac{1}{10}$. Equivalently, the sample size needed for a target standard error $e$ is $n = (\sigma/e)^2$.

Example: an A/B test comparing two conversion rates of about 5% that should detect a difference of 0.1 percentage points (at 5% significance and 80% power) needs roughly $2 \cdot 0.05 \cdot 0.95 \cdot (2.8/0.001)^2 \approx 750{,}000$ users per variant; the factor $2.8 \approx 1.96 + 0.84$ combines the two normal quantiles. Since precision bought with data is expensive, reducing $\sigma$ itself — through better measurement or a model that explains part of the variation — is often the cheaper route.

# Two Theorems We Have Already Seen Happen

For i.i.d. observations with mean $\mu$ and finite variance $\sigma^2$:

| | | |
|---|---|---|
| **Law of large numbers** | $\overline{X}_n \to \mu \text{ as } n \to \infty$ | *the pile keeps narrowing onto the truth* |
| **Central limit theorem** | $\overline{X}_n \approx N(\mu, \sigma^2/n) \text{ for large } n$ | *and we know the shape of what is left* |

The first says the error goes away; the second says how much of it is left after $n$ observations. Only the second lets us put a number on our ignorance.

Speaker notes

Law of large numbers: for i.i.d. observations with finite mean, $\overline{X}_n$ converges to $\mu$. It states that the error vanishes but not how quickly. The central limit theorem adds the rate and the shape: the error $\overline{X}_n - \mu$ is approximately $N(0, \sigma^2/n)$. Only the second allows probability statements such as "within $\pm 2\,\mathrm{SE}$ in about 95% of samples".

A frequent error is to invoke the LLN where the CLT is needed, e.g. to justify an error bar. Another is to read the LLN as a balancing law ("after many heads, tails is due"): the mean converges because the influence of early deviations is diluted by later observations, not because they are compensated.

# From a Number to an Interval

If $\overline{X} \approx N(\mu, \sigma^2/n)$, then for 95% of samples $\overline{X}$ lands within $1.96\,\mathrm{SE}$ of $\mu$ — and therefore $\mu$ lands within $1.96\,\mathrm{SE}$ of $\overline{X}$.

$$\overline{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

the estimate — our best single guess

how confident we insist on being

how noisy the estimate is

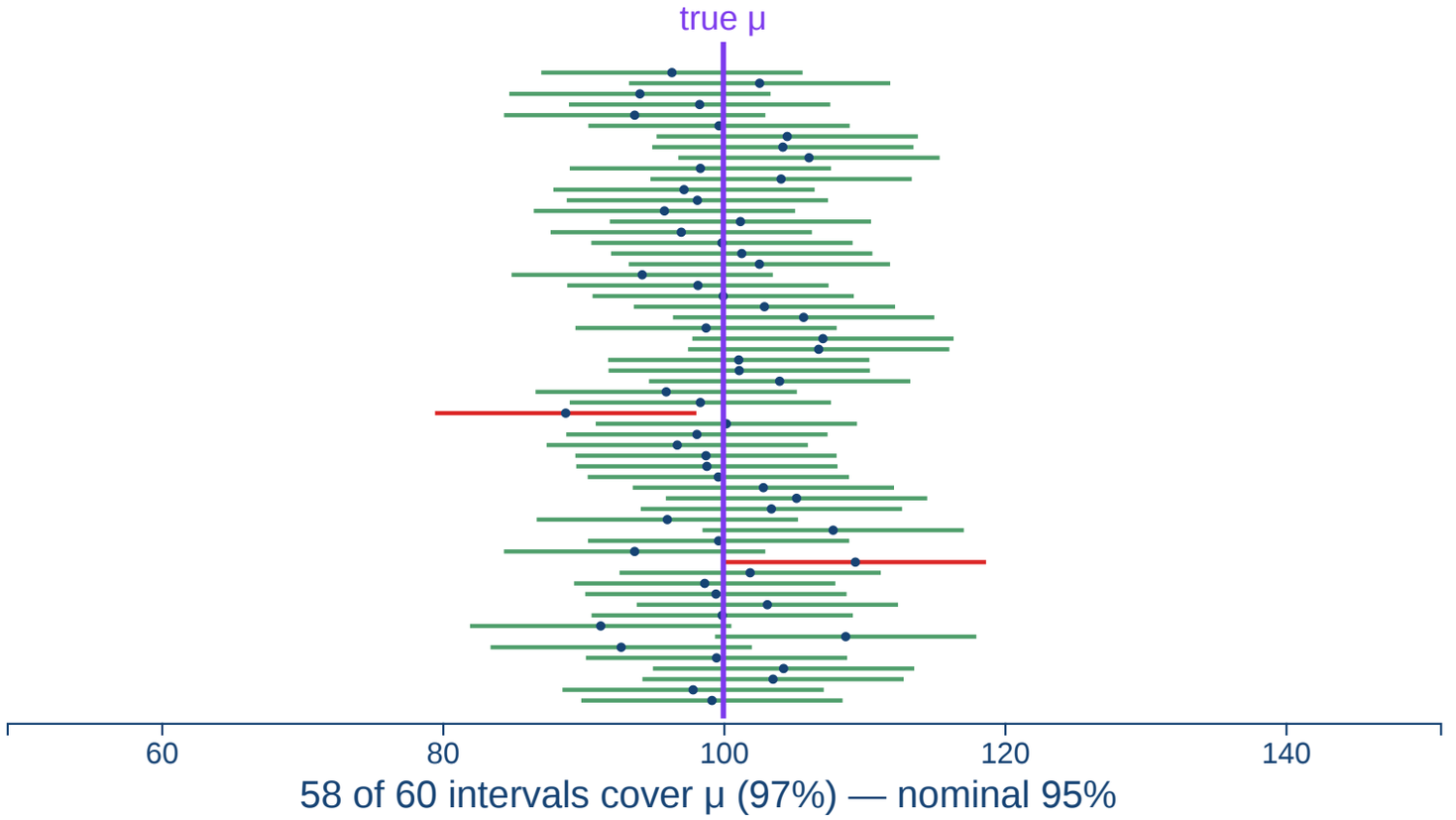
Speaker notes

The interval follows from the normal approximation: $P(|\overline{X} - \mu| \leq 1.96\,\mathrm{SE}) \approx 0.95$, and the event $|\overline{X} - \mu| \leq 1.96\,\mathrm{SE}$ is the same as $\mu \in [\overline{X} - 1.96\,\mathrm{SE}, \overline{X} + 1.96\,\mathrm{SE}]$. The value 1.96 is the 97.5% quantile of $N(0, 1)$; 90% intervals use 1.645, 99% intervals 2.576.

The interval widens with a higher confidence level, with larger $\sigma$, and with smaller $n$. The formula treats $\sigma$ as known. When $\sigma$ is replaced by $\hat{\sigma}$, the quantile 1.96 is replaced by a quantile of the $t$ distribution with $n - 1$ degrees of freedom, which is larger for small samples (2.262 for $n = 10$) and approaches 1.96 as $n$ grows.

# What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?




**How to read it**

- **Row:** one sample, $n = 10$, from $\mu = 100$, $\sigma = 15$
- **Dot:** its mean $\overline{x}$
- **Bar:** $\overline{x} \pm 1.96 \cdot 15/\sqrt{10} = \overline{x} \pm 9.3$, equal in every row
- **Purple:** the true $\mu$, unknown in practice
- **Red:** a bar that misses $\mu$
- **confidence** sets the multiplier: 80 / 90 / 95 / 99 % → 1.28 / 1.64 / 1.96 / 2.58

Speaker notes

The steps show (1) one sample's interval against the fixed true mean, (2) twenty intervals, (3) all sixty with the coverage count, misses drawn in red, and (4) controls for the confidence level and $n$. With $\mu = 100$, $\sigma = 15$ and $n = 10$, the standard error is $15/\sqrt{10} \approx 4.74$ and each 95% interval has half-width $\approx 9.3$; all intervals have the same width because $\sigma$ is known, and only their centres move.

Of sixty 95% intervals, on average 3 miss $\mu$; the count is itself binomial, $B(60, 0.05)$, so anything from 0 to about 6 is unremarkable. Changing the level changes the width and the miss rate; changing $n$ changes the width but not the miss rate. A missing interval cannot be recognised from the data: it looks exactly like one that covers $\mu$.

# The Sentence Almost Everyone Gets Wrong

**WHAT IT DOES NOT MEAN**

"There is a 95% probability that $\mu$ lies in **this** interval."

$\mu$ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

**WHAT IT MEANS**

"95% of the intervals produced **by this procedure** contain $\mu$."

The probability is a property of the recipe, evaluated before the data is seen.

Speaker notes

Before the data are seen, the interval's end points are random variables, and $P(\mu \in [\overline{X} \pm 1.96\,\mathrm{SE}]) = 0.95$ is a genuine probability statement. After the data are plugged in, both end points and $\mu$ are fixed numbers, and the statement is either true or false. The 95% is therefore the coverage rate of the procedure under repeated sampling.

A Bayesian credible interval does support the first sentence: it is an interval to which the posterior distribution of $\mu$, given the observed data, assigns 95% probability. This requires a prior distribution for $\mu$. For the mean of normal data with a flat prior, the 95% credible interval coincides numerically with the 95% confidence interval, although the two statements mean different things.

# Where This Leaves Us

Two kinds of uncertainty meet in one formula. We can now measure one of them.

- → $\sigma$ — how variable the world is. **Aleatoric.** Unchanged by data.
- → $\mathrm{SE} = \sigma/\sqrt{n}$ — how badly we know our own estimate. **Epistemic.** Shrinks with data.

"More data helps" was a slogan.
It is now a rate: $1/\sqrt{n}$.

Speaker notes

Aleatoric uncertainty is the variability inherent in the data-generating process; it is described by $\sigma$ and does not shrink with more data. Epistemic uncertainty is uncertainty due to limited knowledge; for the sample mean it is the standard error, which shrinks like $1/\sqrt{n}$. A prediction of a single new observation carries both: its variance is approximately $\sigma^2 + \sigma^2/n$, so for large $n$ the epistemic part vanishes and the aleatoric part remains.

The claim that a model is uncertain "because it has not seen enough data" therefore refers to epistemic uncertainty, and the $1/\sqrt{n}$ rate gives its order of magnitude.

# Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

Speaker notes

$\mathrm{SE} = 60/\sqrt{100} = 6$ s. The 95% interval is $180 \pm 1.96 \cdot 6 = 180 \pm 11.8$, i.e. about $[168, 192]$ s. Halving the SE requires doubling $\sqrt{n}$, so $n = 400$. In general, a target standard error $e$ requires $n = (\sigma/e)^2$; for $e = 3$ s this gives $n = 400$.

If $\sigma$ had to be estimated from the same 100 customers, the interval would use $\hat{\sigma}$ and a $t$ quantile with 99 degrees of freedom ($\approx 1.98$), practically the same as 1.96 at this sample size. The calculation also assumes independent customers; customers served in the same queue at the same time can be correlated, which would make the true standard error larger.

# Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

$\mathrm{SE} = 60/\sqrt{100} = 6$ s, so the 95% interval is about $180 \pm 12$ s. Halving the SE needs twice the $\sqrt{n}$ — **four times the data**, $n = 400$. $\sigma = 60$ s does not change: customers stay as variable as they were; only our knowledge of their average improves.

# Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.

Speaker notes

The three points are: a statistic computed from a sample is a random variable; its standard error shrinks like $1/\sqrt{n}$; and the confidence level of an interval is a property of the procedure. They apply to every reported measurement with an error bar, including an accuracy or error rate estimated on a test set: an accuracy measured on 1,000 test cases at around 90% has a standard error of $\sqrt{0.9 \cdot 0.1/1000} \approx 0.95$ percentage points. A Bayesian treatment answers the same question with a posterior distribution over the unknown quantity instead of a sampling distribution.

# Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.